

[MS-MQDSSM]:

Message Queuing (MSMQ): Directory Service Schema Mapping

Intellectual Property Rights Notice for Open Specifications Documentation

- **Technical Documentation.** Microsoft publishes Open Specifications documentation ("this documentation") for protocols, file formats, data portability, computer languages, and standards support. Additionally, overview documents cover inter-protocol relationships and interactions.
- **Copyrights.** This documentation is covered by Microsoft copyrights. Regardless of any other terms that are contained in the terms of use for the Microsoft website that hosts this documentation, you can make copies of it in order to develop implementations of the technologies that are described in this documentation and can distribute portions of it in your implementations that use these technologies or in your documentation as necessary to properly document the implementation. You can also distribute in your implementation, with or without modification, any schemas, IDLs, or code samples that are included in the documentation. This permission also applies to any documents that are referenced in the Open Specifications documentation.
- **No Trade Secrets.** Microsoft does not claim any trade secret rights in this documentation.
- **Patents.** Microsoft has patents that might cover your implementations of the technologies described in the Open Specifications documentation. Neither this notice nor Microsoft's delivery of this documentation grants any licenses under those patents or any other Microsoft patents. However, a given Open Specifications document might be covered by the Microsoft [Open Specifications Promise](#) or the [Microsoft Community Promise](#). If you would prefer a written license, or if the technologies described in this documentation are not covered by the Open Specifications Promise or Community Promise, as applicable, patent licenses are available by contacting iplg@microsoft.com.
- **License Programs.** To see all of the protocols in scope under a specific license program and the associated patents, visit the [Patent Map](#).
- **Trademarks.** The names of companies and products contained in this documentation might be covered by trademarks or similar intellectual property rights. This notice does not grant any licenses under those rights. For a list of Microsoft trademarks, visit www.microsoft.com/trademarks.
- **Fictitious Names.** The example companies, organizations, products, domain names, email addresses, logos, people, places, and events that are depicted in this documentation are fictitious. No association with any real company, organization, product, domain name, email address, logo, person, place, or event is intended or should be inferred.

Reservation of Rights. All other rights are reserved, and this notice does not grant any rights other than as specifically described above, whether by implication, estoppel, or otherwise.

Tools. The Open Specifications documentation does not require the use of Microsoft programming tools or programming environments in order for you to develop an implementation. If you have access to Microsoft programming tools and environments, you are free to take advantage of them. Certain Open Specifications documents are intended for use in conjunction with publicly available standards specifications and network programming art and, as such, assume that the reader either is familiar with the aforementioned material or has immediate access to it.

Support. For questions and support, please contact dochelp@microsoft.com.

Revision Summary

Date	Revision History	Revision Class	Comments
7/25/2008	0.1	Editorial	Initial Availability.
8/29/2008	1.0	Major	Added section 2.3.
10/24/2008	2.0	Major	Updated and revised the technical content.
12/5/2008	2.0.1	Editorial	Changed language and formatting in the technical content.
1/16/2009	2.0.2	Editorial	Changed language and formatting in the technical content.
2/27/2009	2.1	Minor	Clarified the meaning of the technical content.
4/10/2009	2.2	Minor	Clarified the meaning of the technical content.
5/22/2009	2.3	Minor	Clarified the meaning of the technical content.
7/2/2009	2.3.1	Editorial	Changed language and formatting in the technical content.
8/14/2009	2.3.2	Editorial	Changed language and formatting in the technical content.
9/25/2009	3.0	Major	Updated and revised the technical content.
11/6/2009	3.0.1	Editorial	Changed language and formatting in the technical content.
12/18/2009	4.0	Major	Updated and revised the technical content.
1/29/2010	5.0	Major	Updated and revised the technical content.
3/12/2010	6.0	Major	Updated and revised the technical content.
4/23/2010	6.0.1	Editorial	Changed language and formatting in the technical content.
6/4/2010	7.0	Major	Updated and revised the technical content.
7/16/2010	8.0	Major	Updated and revised the technical content.
8/27/2010	9.0	Major	Updated and revised the technical content.
10/8/2010	10.0	Major	Updated and revised the technical content.
11/19/2010	11.0	Major	Updated and revised the technical content.
1/7/2011	12.0	Major	Updated and revised the technical content.
2/11/2011	13.0	Major	Updated and revised the technical content.
3/25/2011	14.0	Major	Updated and revised the technical content.
5/6/2011	15.0	Major	Updated and revised the technical content.
6/17/2011	15.1	Minor	Clarified the meaning of the technical content.
9/23/2011	15.1	None	No changes to the meaning, language, or formatting of the technical content.
12/16/2011	16.0	Major	Updated and revised the technical content.
3/30/2012	16.1	Minor	Clarified the meaning of the technical content.
7/12/2012	16.2	Minor	Clarified the meaning of the technical content.

Date	Revision History	Revision Class	Comments
10/25/2012	17.0	Major	Updated and revised the technical content.
1/31/2013	17.0	None	No changes to the meaning, language, or formatting of the technical content.
8/8/2013	18.0	Major	Updated and revised the technical content.
11/14/2013	18.0	None	No changes to the meaning, language, or formatting of the technical content.
2/13/2014	18.0	None	No changes to the meaning, language, or formatting of the technical content.
5/15/2014	18.0	None	No changes to the meaning, language, or formatting of the technical content.
6/30/2015	19.0	Major	Significantly changed the technical content.
10/16/2015	19.0	None	No changes to the meaning, language, or formatting of the technical content.
7/14/2016	19.0	None	No changes to the meaning, language, or formatting of the technical content.
6/1/2017	19.0	None	No changes to the meaning, language, or formatting of the technical content.
9/15/2017	20.0	Major	Significantly changed the technical content.
9/12/2018	21.0	Major	Significantly changed the technical content.
4/7/2021	22.0	Major	Significantly changed the technical content.
6/25/2021	23.0	Major	Significantly changed the technical content.
4/23/2024	24.0	Major	Significantly changed the technical content.

Table of Contents

1	Introduction	9
1.1	Glossary	9
1.2	References	10
1.2.1	Normative References	10
1.2.2	Informative References	11
1.3	Overview	11
1.4	Relationship to Other Protocols	11
1.5	Prerequisites/Preconditions	11
1.6	Applicability Statement	12
1.7	Versioning and Capability Negotiation	12
1.8	Vendor-Extensible Fields	12
1.9	Standards Assignments.....	12
2	Messages.....	13
2.1	Transport	13
2.2	Common Data Types	13
2.2.1	LDAP Distinguished Names of Directory Objects	13
2.2.2	Attributes of Directory Objects	14
2.2.3	mSMQEnterpriseSettings Object	16
2.2.4	Unused Active Directory Properties	16
2.2.5	Hash String Calculation From Queue Name	16
2.2.6	LDAP Result Code to DirectoryOperationResult Mapping	25
2.3	Queue Alias.....	25
2.4	Directory Service Schema Elements	25
3	Protocol Details.....	28
3.1	Algorithm Details	28
3.1.1	Abstract Data Model.....	28
3.1.1.1	Externally Defined Data Elements	28
3.1.1.2	ReadDirectoryIteratorState Data Element	28
3.1.1.3	ReadDirectoryIteratorStateCollection Data Element	29
3.1.1.4	User Data Element	29
3.1.1.5	CachedConfigurationNamingContext	29
3.1.1.6	CachedLDAPConnection	29
3.1.2	Timers	29
3.1.3	Initialization.....	29
3.1.4	Message Processing Events and Sequencing Rules	30
3.1.5	Timer Events.....	30
3.1.6	Other Local Events.....	31
3.1.6.1	Create Directory Object	31
3.1.6.1.1	QueueManager	31
3.1.6.1.1.1	Preconditions.....	31
3.1.6.1.1.2	Creation	31
3.1.6.1.1.3	Postprocessing	33
3.1.6.1.1.4	mSMQOSType	35
3.1.6.1.1.5	mSMQServiceType	35
3.1.6.1.1.6	mSMQOutRoutingServers.....	36
3.1.6.1.1.7	mSMQInRoutingServers.....	36
3.1.6.1.1.8	nTSecurityDescriptor	37
3.1.6.1.1.9	mSMQSignCertificates and mSMQDigests.....	38
3.1.6.1.2	Queue.....	38
3.1.6.1.2.1	Preconditions.....	38
3.1.6.1.2.2	Creation	38
3.1.6.1.2.3	Postprocessing	40
3.1.6.1.2.4	mSMQPrivacyLevel	40

3.1.6.1.2.5	Name	40
3.1.6.1.3	Site	40
3.1.6.1.3.1	Preconditions.....	40
3.1.6.1.3.2	Creation	41
3.1.6.1.3.3	Postprocessing	42
3.1.6.1.3.4	nTSecurityDescriptor	42
3.1.6.1.4	RoutingLink	43
3.1.6.1.4.1	Preconditions.....	43
3.1.6.1.4.2	Creation	43
3.1.6.1.4.3	Postprocessing	44
3.1.6.1.4.4	mSMQSite1	44
3.1.6.1.4.5	mSMQSite2	44
3.1.6.1.4.6	mSMQSiteGates.....	45
3.1.6.2	Delete Directory Object.....	45
3.1.6.2.1	QueueManager.....	46
3.1.6.2.1.1	Preconditions.....	46
3.1.6.2.1.2	Delete	46
3.1.6.2.1.3	Postprocessing	46
3.1.6.2.2	Queue.....	47
3.1.6.2.2.1	Preconditions.....	47
3.1.6.2.2.2	Delete	47
3.1.6.2.2.3	Postprocessing	48
3.1.6.2.3	Site	48
3.1.6.2.3.1	Preconditions.....	48
3.1.6.2.3.2	Delete	48
3.1.6.2.3.3	Postprocessing	48
3.1.6.2.4	RoutingLink	48
3.1.6.2.4.1	Preconditions.....	49
3.1.6.2.4.2	Delete	49
3.1.6.2.4.3	Postprocessing	49
3.1.6.3	Read Directory.....	49
3.1.6.3.1	Search For One Object	50
3.1.6.3.2	QueueManager	51
3.1.6.3.2.1	Preconditions.....	51
3.1.6.3.2.2	Read	51
3.1.6.3.3	Queue.....	52
3.1.6.3.3.1	Preconditions.....	52
3.1.6.3.3.2	Read	52
3.1.6.3.4	Enterprise.....	53
3.1.6.3.4.1	Preconditions.....	53
3.1.6.3.4.2	Read	53
3.1.6.3.5	Site	53
3.1.6.3.5.1	Preconditions.....	53
3.1.6.3.5.2	Read	54
3.1.6.3.6	RoutingLink	54
3.1.6.3.6.1	Preconditions.....	54
3.1.6.3.6.2	Read	55
3.1.6.3.7	User	55
3.1.6.3.7.1	Preconditions.....	55
3.1.6.3.7.2	Read	55
3.1.6.4	Read Directory Begin.....	56
3.1.6.4.1	QueueManager	57
3.1.6.4.1.1	Preconditions.....	57
3.1.6.4.1.2	Read Begin	58
3.1.6.4.1.3	Postprocessing	59
3.1.6.4.2	Queue.....	60
3.1.6.4.2.1	Preconditions.....	60
3.1.6.4.2.2	Read Begin	60

3.1.6.4.2.3	Postprocessing	61
3.1.6.4.3	Enterprise.....	61
3.1.6.4.3.1	Preconditions.....	62
3.1.6.4.3.2	Read Begin	62
3.1.6.4.3.3	Postprocessing	63
3.1.6.4.3.4	WeakenedSecurity	63
3.1.6.4.3.5	NonLDAPCapableQueueManagerNotification	63
3.1.6.4.4	Site	63
3.1.6.4.4.1	Preconditions.....	63
3.1.6.4.4.2	Read Begin	64
3.1.6.4.4.3	Postprocessing	65
3.1.6.4.5	RoutingLink	65
3.1.6.4.5.1	Preconditions.....	65
3.1.6.4.5.2	Read Begin	65
3.1.6.4.5.3	Postprocessing	66
3.1.6.4.5.4	Site1Identifier Filtering	67
3.1.6.4.5.5	Site2Identifier Filtering	67
3.1.6.4.6	User	67
3.1.6.4.6.1	Preconditions.....	67
3.1.6.4.6.2	Read Begin	68
3.1.6.4.6.3	Postprocessing	69
3.1.6.5	Read Directory Next	69
3.1.6.6	Read Directory End	70
3.1.6.7	Write Directory	70
3.1.6.7.1	QueueManager.....	71
3.1.6.7.1.1	Preconditions.....	71
3.1.6.7.1.2	Write.....	71
3.1.6.7.1.3	Postprocessing	72
3.1.6.7.1.4	mSMQOSType	73
3.1.6.7.1.5	mSMQServiceType	73
3.1.6.7.1.6	mSMQOutRoutingServers.....	74
3.1.6.7.1.7	mSMQInRoutingServers.....	74
3.1.6.7.1.8	mSMQSignCertificates and mSMQDigests.....	75
3.1.6.7.1.9	mSMQSettings Objects	75
3.1.6.7.1.10	PublicSigningKeyList.....	78
3.1.6.7.2	Queue.....	78
3.1.6.7.2.1	Preconditions.....	78
3.1.6.7.2.2	Write.....	78
3.1.6.7.2.3	Postprocessing	80
3.1.6.7.2.4	PrivacyLevel	80
3.1.6.7.2.5	<queue name>	80
3.1.6.7.3	Enterprise.....	80
3.1.6.7.3.1	Preconditions.....	81
3.1.6.7.3.2	Write.....	81
3.1.6.7.3.3	Postprocessing	81
3.1.6.7.3.4	mSMQCSPName	82
3.1.6.7.4	Site	82
3.1.6.7.4.1	Preconditions.....	82
3.1.6.7.4.2	Write.....	82
3.1.6.7.4.3	Postprocessing	83
3.1.6.7.5	RoutingLink	84
3.1.6.7.5.1	Preconditions.....	84
3.1.6.7.5.2	Write.....	84
3.1.6.7.5.3	Postprocessing	85
3.1.6.7.5.4	mSMQSite1	85
3.1.6.7.5.5	mSMQSite2.....	85
3.1.6.7.5.6	mSMQSiteGates.....	86
3.1.6.7.6	User	86

3.1.6.7.6.1	Preconditions.....	86
3.1.6.7.6.2	Write.....	86
3.1.6.7.6.3	Postprocessing	87
3.1.6.7.6.4	Note on mSMQSignCertificates and mSMQDigests.....	87
3.1.6.8	Resolve Queue Alias	88
3.1.6.9	Resolve Distribution List.....	88
3.1.6.10	Create LDAP Attribute List.....	90
3.1.6.10.1	QueueManager	90
3.1.6.10.2	Queue.....	91
3.1.6.10.3	Enterprise.....	92
3.1.6.10.4	Site	92
3.1.6.10.5	RoutingLink	93
3.1.6.10.6	User	93
3.1.6.11	Create ADM Element From LDAP Values.....	93
3.1.6.11.1	QueueManager.....	94
3.1.6.11.1.1	ComputerName	97
3.1.6.11.1.2	OperatingSystemType	97
3.1.6.11.1.3	OutRoutingServerIdentifierList.....	97
3.1.6.11.1.4	InRoutingServerIdentifierList.....	98
3.1.6.11.1.5	DirectoryServerType	98
3.1.6.11.1.6	Clustered	99
3.1.6.11.2	Queue.....	99
3.1.6.11.2.1	Pathname	101
3.1.6.11.2.2	QualifiedPathname	101
3.1.6.11.2.3	PrivacyLevel	102
3.1.6.11.3	Enterprise.....	102
3.1.6.11.3.1	Name	103
3.1.6.11.3.2	WeakenedSecurity	103
3.1.6.11.3.3	NonLDAPCapableQueueManagerNotification	103
3.1.6.11.4	Site	104
3.1.6.11.5	RoutingLink	104
3.1.6.11.5.1	Site1Identifier	105
3.1.6.11.5.2	Site2Identifier	105
3.1.6.11.5.3	SiteGateIdentifierList.....	106
3.1.6.11.6	User	106
3.1.6.12	Create Object Using LDAP	107
3.1.6.13	Delete Object Using LDAP	109
3.1.6.14	Get Object Properties Using LDAP	112
3.1.6.15	Search Using LDAP.....	113
3.1.6.16	Set Object Properties Using LDAP.....	116
3.1.6.17	Set Object Security Using LDAP	118
3.1.6.18	Prepare an LDAP Connection	120
3.1.6.19	Find Object By GUID Using LDAP	121
3.1.6.20	Data Element Directory Attribute Tables	122
3.1.6.20.1	QueueManager.....	122
3.1.6.20.2	Queue.....	124
3.1.6.20.3	Enterprise.....	124
3.1.6.20.4	Site	125
3.1.6.20.5	RoutingLink	125
3.1.6.20.6	User	126
3.1.6.21	Shut Down an LDAP Connection.....	126
4	Algorithm Examples	127
5	Security	128
5.1	Security Considerations for Implementers	128
5.1.1	QueueManager	128
5.1.2	Queue	129
5.1.3	Enterprise.....	129

5.1.4	Site.....	129
5.1.5	RoutingLink.....	130
5.1.6	User.....	130
5.1.7	Queue Alias	130
5.1.8	Distribution List.....	130
5.2	Index of Security Parameters	130
6	Appendix A: Product Behavior	131
7	Change Tracking.....	134
8	Index.....	135

1 Introduction

This document specifies the Message Queuing (MSMQ): Directory Service Schema Mapping.

[[MS-MQDMPR](#)] section 3.1.1 specifies a common abstract data model (ADM) used by all protocols in the MSMQ family. A subset of the ADM elements and ADM element attributes specified there can be stored in **Active Directory**, which provides a **Lightweight Directory Access Protocol (LDAP)** interface. The Directory Service Schema Mapping specifies an algorithm by which ADM elements are persisted as specific objects in Active Directory. It also provides a set of events that trigger LDAP operations to access those objects in Active Directory.

Sections 1.5, 1.8, 1.9, 2, and 3 of this specification are normative. All other sections and examples in this specification are informative.

1.1 Glossary

This document uses the following terms:

Active Directory: The Windows implementation of a general-purpose directory service, which uses **LDAP** as its primary access protocol. Active Directory stores information about a variety of objects in the network such as user accounts, computer accounts, groups, and all related credential information used by Kerberos [[MS-KILE](#)]. Active Directory is either deployed as Active Directory Domain Services (AD DS) or Active Directory Lightweight Directory Services (AD LDS), which are both described in [[MS-ADOD](#)]: Active Directory Protocols Overview.

Active Directory object: A set of directory objects that are used within **Active Directory** as defined in [[MS-ADTS](#)] section 3.1.1. An **Active Directory object** can be identified by a dsname. See also directory object.

Active Directory schema: Contains formal definitions of every object class that can be created in an **Active Directory** forest. The schema also contains formal definitions of every attribute that can exist in an **Active Directory object**.

directory string: A string encoded in UTF-8 as defined in [[RFC2252](#)] section 6.10.

discretionary access control list (DACL): An access control list (ACL) that is controlled by the owner of an object and that specifies the access particular users or groups can have to the object.

distinguished name (DN): In **Lightweight Directory Access Protocol (LDAP)**, an LDAP Distinguished Name, as described in [[RFC2251](#)] section 4.1.3. The DN of an object is the DN of its parent, preceded by the RDN of the object. For example: CN=David Thompson, OU=Users, DC=Microsoft, DC=COM. For definitions of CN and OU, see [[RFC2256](#)] sections 5.4 and 5.12, respectively.

globally unique identifier (GUID): A term used interchangeably with universally unique identifier (UUID) in Microsoft protocol technical documents (TDs). Interchanging the usage of these terms does not imply or require a specific algorithm or mechanism to generate the value. Specifically, the use of this term does not imply or require that the algorithms described in [[RFC4122](#)] or [[C706](#)] have to be used for generating the GUID. See also universally unique identifier (UUID).

Lightweight Directory Access Protocol (LDAP): The primary access protocol for **Active Directory**. Lightweight Directory Access Protocol (LDAP) is an industry-standard protocol, established by the Internet Engineering Task Force (IETF), which allows users to query and update information in a directory service (DS), as described in [MS-ADTS]. The Lightweight Directory Access Protocol can be either version 2 [[RFC1777](#)] or version 3 [[RFC3377](#)].

path name: The name of the receiving computer where the messages for a particular queue are stored, and an optional PRIVATE\$ key word indicating whether the queue is private, followed by the name of the queue. Path names can also refer to subqueues; for more information, see [\[MS-MQMQ\]](#) section 2.1.

queue manager (QM): A message queuing service that manages queues deployed on a computer. A queue manager can also provide asynchronous transfer of messages to queues deployed on other queue managers.

security identifier (SID): An identifier for security principals that is used to identify an account or a group. Conceptually, the **SID** is composed of an account authority portion (typically a domain) and a smaller integer representing an identity relative to the account authority, termed the relative identifier (RID). The **SID** format is specified in [\[MS-DTYP\]](#) section 2.4.2; a string representation of **SIDs** is specified in [\[MS-DTYP\]](#) section 2.4.2 and [\[MS-AZOD\]](#) section 1.1.1.2.

Unicode: A character encoding standard developed by the Unicode Consortium that represents almost all of the written languages of the world. The **Unicode** standard [\[UNICODE5.0.0/2007\]](#) provides three forms (UTF-8, UTF-16, and UTF-32) and seven schemes (UTF-8, UTF-16, UTF-16 BE, UTF-16 LE, UTF-32, UTF-32 LE, and UTF-32 BE).

MAY, SHOULD, MUST, SHOULD NOT, MUST NOT: These terms (in all caps) are used as defined in [\[RFC2119\]](#). All statements of optional behavior use either MAY, SHOULD, or SHOULD NOT.

1.2 References

Links to a document in the Microsoft Open Specifications library point to the correct section in the most recently published version of the referenced document. However, because individual documents in the library are not updated at the same time, the section numbers in the documents may not match. You can confirm the correct section numbering by checking the [Errata](#).

1.2.1 Normative References

We conduct frequent surveys of the normative references to assure their continued availability. If you have any issue with finding a normative reference, please contact dochelp@microsoft.com. We will assist you in finding the relevant information.

[MS-ADA1] Microsoft Corporation, "[Active Directory Schema Attributes A-L](#)".

[MS-ADA2] Microsoft Corporation, "[Active Directory Schema Attributes M](#)".

[MS-ADA3] Microsoft Corporation, "[Active Directory Schema Attributes N-Z](#)".

[MS-ADSC] Microsoft Corporation, "[Active Directory Schema Classes](#)".

[MS-ADTS] Microsoft Corporation, "[Active Directory Technical Specification](#)".

[MS-DTYP] Microsoft Corporation, "[Windows Data Types](#)".

[MS-MQDMPR] Microsoft Corporation, "[Message Queuing \(MSMQ\): Common Data Model and Processing Rules](#)".

[MS-MQDS] Microsoft Corporation, "[Message Queuing \(MSMQ\): Directory Service Protocol](#)".

[MS-MQMQ] Microsoft Corporation, "[Message Queuing \(MSMQ\): Data Structures](#)".

[MS-SAMR] Microsoft Corporation, "[Security Account Manager \(SAM\) Remote Protocol \(Client-to-Server\)](#)".

[RFC1321] Rivest, R., "The MD5 Message-Digest Algorithm", RFC 1321, April 1992, <https://www.rfc-editor.org/info/rfc1321>

[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>

[RFC2251] Wahl, M., Howes, T., and Kille, S., "Lightweight Directory Access Protocol (v3)", RFC 2251, December 1997, <https://www.rfc-editor.org/info/rfc2251>

1.2.2 Informative References

[LDAP] Microsoft Corporation, "About Lightweight Directory Access Protocol", <http://msdn.microsoft.com/en-us/library/aa366075.aspx>

[MS-MQOD] Microsoft Corporation, "[Message Queuing Protocols Overview](#)".

1.3 Overview

The Message Queuing (MSMQ): Directory Service Schema Mapping is used by any protocol that manipulates the subset of the ADM elements and ADM attributes specified in [\[MS-MQDMPR\]](#) section 3.1 that can be stored in a directory, in the case in which the directory service provider is **Active Directory**. This algorithm, when combined with the common ADM and an understanding of the Active Directory **LDAP** interface, as defined in [\[MS-ADTS\]](#), allows an abstract operation on ADM elements and ADM attributes to be reduced to a concrete LDAP operation on concrete **Active Directory objects** and attributes.

The algorithm provides access to stateful information, but it is up to Active Directory to maintain that state.

1.4 Relationship to Other Protocols

The MSMQ Directory Service Schema Mapping relies upon the **LDAP** interface of **Active Directory**, as specified in [\[MS-ADTS\]](#); references to the underlying specifications of [\[LDAP\]](#) itself are found in that document.

The Message Queuing (MSMQ): Directory Service Schema Mapping uses abstract data model (ADM) elements specified in Message Queuing (MSMQ): Common Data Model and Processing Rules [\[MS-MQDMPR\]](#) and data structures specified in Message Queuing (MSMQ): Data Structures [\[MS-MQMQ\]](#).

The Message Queuing (MSMQ): Directory Service Schema Mapping is used by the processing rules specified in Message Queuing (MSMQ): Common Data Model and Processing Rules [\[MS-MQDMPR\]](#), as shown in the diagram in [\[MS-MQDMPR\]](#) section 1.4. Protocols shown in the diagram can call the events specified in [\[MS-MQDMPR\]](#) sections 3.1.7.1.3.1 and 3.1.7.1.18 through 3.1.7.1.24, and the processing rules in those sections call the events specified in this algorithm.

The Message Queuing (MSMQ): Directory Service Schema Mapping is used by the processing rules specified in Message Queuing (MSMQ): Directory Service Protocol Specification [\[MS-MQDS\]](#), as shown in the diagram in [\[MS-MQDMPR\]](#) section 1.4. Protocols shown in the diagram can call the events specified in [\[MS-MQDS\]](#) sections 3.1.6.7 through 3.1.6.9, 3.1.6.11 through 3.1.6.13, and 3.1.6.15, and the processing rules in those sections call the events specified in this algorithm.

1.5 Prerequisites/Preconditions

It is assumed that MSMQ is operating in an environment in which **Active Directory** is available and in use by MSMQ. It is further assumed that any MSMQ instance has the information required to access Active Directory via LDAP in this environment.

1.6 Applicability Statement

Message Queuing (MSMQ): Directory Service Schema Mapping is applicable for implementation in an environment in which **Active Directory** is available and desired to be used.

1.7 Versioning and Capability Negotiation

None.

1.8 Vendor-Extensible Fields

None.

1.9 Standards Assignments

None.

2 Messages

2.1 Transport

None.

2.2 Common Data Types

The following table summarizes the types defined in this specification.

Type	Description
DirectoryOperationResult ([MS-MQDMPR] section 3.1.1.17)	An enumeration that specifies the result of a directory operation.

2.2.1 LDAP Distinguished Names of Directory Objects

The Directory Service Schema Mapping uses the **Active Directory** classes listed in the following table and MUST use these **distinguished names in LDAP** queries to access objects of these classes.

Object type	Distinguished name
mSMQQueue ([MS-ADSC] section 2.166)	CN=<queue name>, CN=msmq, CN=<computer name>, CN=Computers, <root>
mSMQConfiguration ([MS-ADSC] section 2.163)	CN=msmq, CN=<computer name>, CN=Computers, <root>
computer ([MS-ADSC] section 2.21)	CN=<computer name>, CN=Computers, <root>
site ([MS-ADSC] section 2.258)	CN=<site name>, CN=Sites, CN=Configuration, <root>
mSMQEnterpriseSettings ([MS-ADSC] section 2.164)	CN=MsmqServices, CN=Services, CN=Configuration, <root>
user ([MS-ADSC] section 2.269)	CN=<name>, CN=Users, <root>
mSMQSiteLink ([MS-ADSC] section 2.168)	CN=<routing link name>, CN=MsmqServices, CN=Services, CN=Configuration, <root>
mSMQSettings ([MS-ADSC] section 2.167)	CN=MSMQ Settings, CN=<computer name>, CN=Servers, CN=<site name>, CN=Sites, CN=Configuration, <root>
mSMQ-Custom-Recipient ([MS-ADSC] section 2.161)	CN=<name>, CN=Users, <root>
group ([MS-ADSC] section 2.55)	CN=<name>, CN=Users, <root>

<queue name> MUST be the "QueueName" portion of an MSMQ Queue Name as specified in [\[MS-MQMQ\]](#) section 2.1.1.

<computer name> MUST be the "Computer" portion of an MSMQ Queue Name as specified in [\[MS-MQMQ\]](#) section 2.1.1.

<site name> MUST be the **Site.Name** ADM element attribute specified in [\[MS-MQDMPR\]](#) section 3.1.1.7.

<routing link name> is specified in section [3.1.6.1.4.2](#) of this document.

<name> is a string identifier that MUST be unique among all objects of the same type in Active Directory.

In each case, <root> MUST be a common root for these entries, which is the rootDomainNamingContext as specified in [\[MS-ADTS\]](#) section 3.1.1.3.2.16.

2.2.2 Attributes of Directory Objects

This table lists the attributes used by the Directory Service Schema Mapping for each of the **Active Directory** classes listed in section [2.2.1](#).

Object	Attributes
mSMQQueue	objectGUID ([MS-ADA3] section 2.44) mSMQLabelEx ([MS-ADA2] section 2.566) whenCreated ([MS-ADA3] section 2.371) whenChanged ([MS-ADA3] section 2.370) mSMQQueueType ([MS-ADA2] section 2.581) mSMQJournal ([MS-ADA2] section 2.563) mSMQQueueQuota ([MS-ADA2] section 2.580) mSMQQueueJournalQuota ([MS-ADA2] section 2.578) mSMQAuthenticate ([MS-ADA2] section 2.546) mSMQPrivacyLevel ([MS-ADA2] section 2.576) mSMQTransactional ([MS-ADA2] section 2.599) MSMQ-MulticastAddress ([MS-ADA2] section 2.543) nTSecurityDescriptor ([MS-ADA3] section 2.37) mSMQBasePriority ([MS-ADA2] section 2.547) mSMQQueueNameExt ([MS-ADA2] section 2.579) distinguishedName ([MS-ADA1] section 2.177)
mSMQConfiguration	objectGUID ([MS-ADA3] section 2.44) whenCreated ([MS-ADA3] section 2.371) whenChanged ([MS-ADA3] section 2.370) mSMQServiceType ([MS-ADA2] section 2.586) mSMQQuota ([MS-ADA2] section 2.582) mSMQJournalQuota ([MS-ADA2] section 2.564) mSMQForeign ([MS-ADA2] section 2.559) distinguishedName ([MS-ADA1] section 2.177) mSMQRoutingServices ([MS-ADA2] section 2.584) mSMQDsServices ([MS-ADA2] section 2.585) mSMQDependentClientServices ([MS-ADA2] section 2.553) mSMQEncryptKey ([MS-ADA2] section 2.558) nTSecurityDescriptor ([MS-ADA3] section 2.37) mSMQSites ([MS-ADA2] section 2.598) mSMQOutRoutingServers ([MS-ADA2] section 2.573) mSMQInRoutingServers ([MS-ADA2] section 2.560) mSMQComputerTypeEx ([MS-ADA2] section 2.549) mSMQOSType ([MS-ADA2] section 2.572)
computer	mSMQSignCertificates ([MS-ADA2] section 2.587)

Object	Attributes
	servicePrincipalName ([MS-ADA3] section 2.253) objectSid ([MS-ADA3] section 2.45) dNSHostName ([MS-ADA1] section 2.185) operatingSystemVersion ([MS-ADA3] section 2.56) mSMQSignCertificatesMig ([MS-ADA2] section 2.588) <1> mSMQDigestsMig ([MS-ADA2] section 2.555) <2>
site	objectGUID ([MS-ADA3] section 2.44) cn ([MS-ADA1] section 2.110) mSMQInterval1 ([MS-ADA2] section 2.561) mSMQInterval2 ([MS-ADA2] section 2.562) distinguishedName ([MS-ADA1] section 2.177) mSMQSiteForeign ([MS-ADA2] section 2.592) nTSecurityDescriptor ([MS-ADA3] section 2.37) mSMQNt4Stub ([MS-ADA2] section 2.571)
mSMQEnterpriseSettings	objectGUID ([MS-ADA3] section 2.44) mSMQNameStyle ([MS-ADA2] section 2.569) mSMQCSPName ([MS-ADA2] section 2.551) mSMQLongLived ([MS-ADA2] section 2.567) mSMQVersion ([MS-ADA2] section 2.601) nTSecurityDescriptor ([MS-ADA3] section 2.37)
user	objectGUID ([MS-ADA3] section 2.44) distinguishedName ([MS-ADA1] section 2.177) objectSid ([MS-ADA3] section 2.45) mSMQSignCertificates ([MS-ADA2] section 2.587) mSMQDigests ([MS-ADA2] section 2.554) mSMQSignCertificatesMig ([MS-ADA2] section 2.588) <3> mSMQDigestsMig ([MS-ADA2] section 2.555) <4>
mSMQSiteLink	objectGUID ([MS-ADA3] section 2.44) description ([MS-ADA1] section 2.153) distinguishedName ([MS-ADA1] section 2.177) mSMQCost ([MS-ADA2] section 2.550) mSMQSite1 ([MS-ADA2] section 2.590) mSMQSite2 ([MS-ADA2] section 2.591) mSMQSiteGates ([MS-ADA2] section 2.593) mSMQSiteGatesMig ([MS-ADA2] section 2.594) <5>
mSMQSettings	mSMQQMID ([MS-ADA2] section 2.577) mSMQServices ([MS-ADA2] section 2.585) mSMQRoutingService ([MS-ADA2] section 2.583) mSMQDsService ([MS-ADA2] section 2.556) mSMQDependentClientService ([MS-ADA2] section 2.552) mSMQMigrated ([MS-ADA2] section 2.568) <6>
mSMQ-Custom-Recipient	msMQ-Recipient-FormatName ([MS-ADA2] section 2.544) objectGUID ([MS-ADA3] section 2.44)
group	objectGUID ([MS-ADA3] section 2.44)

Object	Attributes
	member ([MS-ADA2] section 2.43)

2.2.3 mSMQEnterpriseSettings Object

As specified in [\[MS-ADSC\]](#) section 2.164, there MUST NOT be more than one mSMQEnterpriseSettings object in a rootDomainNamingContext ([\[MS-ADTS\]](#) section 3.1.1.3.2.16). There SHOULD [<7>](#) always be exactly one mSMQEnterpriseSettings object in a rootDomainNamingContext.

2.2.4 Unused Active Directory Properties

The following directory properties are present in the MSMQ's Active Directory schema ([\[MS-ADA2\]](#) sections 2.574, 2.595, 2.593, 2.545, 2.548, 2.565, and 2.596) but are not used in any Windows version:

- mSMQOwnerID, [MS-ADA2] section 2.574.
- mSMQSiteID, [MS-ADA2] section 2.595.
- mSMQPrevSiteGates, [MS-ADA2] section 2.593.
- MSMQ-SecuredSource, [MS-ADA2] section 2.545.
- mSMQComputerType, [MS-ADA2] section 2.548 (has been replaced with mSMQComputerTypeEx, [MS-ADA2] section 2.549).
- mSMQLabel, [MS-ADA2] section 2.565 (has been replaced with mSMQLabelEx, [MS-ADA2] section 2.566).
- mSMQSiteName, [MS-ADA2] section 2.596 (has been replaced with mSMQSiteNameEx, [MS-ADA2] section 2.597).

2.2.5 Hash String Calculation From Queue Name

The QueueName part of a **path name**, as specified in [\[MS-MQMQ\]](#) section 2.1.1, can exceed 64 **Unicode** characters in length. If this occurs, the QueueName must be split and the overflow stored in a second directory attribute. However, the value of the directory attribute that holds the first 64 characters is still required to be unique among all queues hosted by one **queue manager**, which might not be true if one queue manager hosts multiple queues having long names that differ only by a number at the end, for example. To enforce uniqueness in the first 64 characters, longer names are split at a lower character position, and a hash string computed according to the following algorithm is appended to the first part. See section [3.1.6.1.2.5](#) for more information on when this is used.

1. Let *QName* be a Unicode string initialized to the queue name for which the hash value is calculated.
2. Let *HashValue* be a 32-bit integer, initialized to zero.
3. For each Unicode character in *QName*, these steps MUST be followed:
 1. The character MUST be converted to lowercase.
 2. Let *DownShifted* be a 32-bit integer, initialized to the value of *HashValue* shifted right by eight bits.

3. The most significant eight bits of the current character MUST be bitwise eXclusive-OR (XORed) with the least significant eight bits of *HashValue*, and the result MUST be used as an index into the table at the end of this section.
4. *HashValue* MUST be set to the value of *DownShifted* bitwise eXclusive-OR (XORed) with the table value.
5. *DownShifted* MUST be set to the value of *HashValue* shifted right by eight bits.
6. The least significant eight bits of the current character MUST be bitwise eXclusive-OR (XORed) with the least significant eight bits of *HashValue*, and the result MUST be used as an index into the table at the end of this section.
7. *HashValue* MUST be set to the value of *DownShifted* bitwise eXclusive-OR (XORed) with the table value.
4. The hash string MUST be an eight-character hexadecimal Unicode string representation of the final value of *HashValue*, with the hexadecimal digits 'a' through 'f' represented by lowercase characters.

Index	Table value
0	0x00000000
1	0x82E0FE45
2	0x3302DCCD
3	0xB1E22288
4	0x6605B99A
5	0xE4E547DF
6	0x55076557
7	0xD7E79B12
8	0xCC0B7334
9	0x4EEB8D71
10	0xFF09AFF9
11	0x7DE951BC
12	0xAA0ECAAE
13	0x28EE34EB
14	0x990C1663
15	0x1BECE826
16	0xAED5C62F
17	0x2C35386A
18	0x9DD71AE2
19	0x1F37E4A7
20	0xC8D07FB5

Index	Table value
21	0x4A3081F0
22	0xFBD2A378
23	0x79325D3D
24	0x62DEB51B
25	0xE03E4B5E
26	0x51DC69D6
27	0xD33C9793
28	0x04DB0C81
29	0x863BF2C4
30	0x37D9D04C
31	0xB5392E09
32	0x6B68AC19
33	0xE988525C
34	0x586A70D4
35	0xDA8A8E91
36	0x0D6D1583
37	0x8F8DEBC6
38	0x3E6FC94E
39	0xBC8F370B
40	0xA763DF2D
41	0x25832168
42	0x946103E0
43	0x1681FDA5
44	0xC16666B7
45	0x438698F2
46	0xF264BA7A
47	0x7084443F
48	0xC5BD6A36
49	0x475D9473
50	0xF6BFB6FB
51	0x745F48BE
52	0xA3B8D3AC

Index	Table value
53	0x21582DE9
54	0x90BA0F61
55	0x125AF124
56	0x09B61902
57	0x8B56E747
58	0x3AB4C5CF
59	0xB8543B8A
60	0x6FB3A098
61	0xED535EDD
62	0x5CB17C55
63	0xDE518210
64	0xD6D15832
65	0x5431A677
66	0xE5D384FF
67	0x67337ABA
68	0xB0D4E1A8
69	0x32341FED
70	0x83D63D65
71	0x0136C320
72	0x1ADA2B06
73	0x983AD543
74	0x29D8F7CB
75	0xAB38098E
76	0x7CDF929C
77	0xFE3F6CD9
78	0x4FDD4E51
79	0xCD3DB014
80	0x78049E1D
81	0xFAE46058
82	0x4B0642D0
83	0xC9E6BC95
84	0x1E012787

Index	Table value
85	0x9CE1D9C2
86	0x2D03FB4A
87	0xAF3050F
88	0xB40FED29
89	0x36EF136C
90	0x870D31E4
91	0x05EDCFA1
92	0xD20A54B3
93	0x50EAAAF6
94	0xE108887E
95	0x63E8763B
96	0xBDB9F42B
97	0x3F590A6E
98	0x8EBB28E6
99	0x0C5BD6A3
100	0xDBBC4DB1
101	0x595CB3F4
102	0xE8BE917C
103	0x6A5E6F39
104	0x71B2871F
105	0xF352795A
106	0x42B05BD2
107	0xC050A597
108	0x17B73E85
109	0x9557C0C0
110	0x24B5E248
111	0xA6551C0D
112	0x136C3204
113	0x918CCC41
114	0x206EEEC9
115	0xA28E108C
116	0x75698B9E

Index	Table value
117	0xF78975DB
118	0x466B5753
119	0xC48BA916
120	0xDF674130
121	0x5D87BF75
122	0xEC659DFD
123	0x6E8563B8
124	0xB962F8AA
125	0x3B8206EF
126	0x8A602467
127	0x0880DA22
128	0x9B619023
129	0x19816E66
130	0xA8634CEE
131	0x2A83B2AB
132	0xFD6429B9
133	0x7F84D7FC
134	0xCE66F574
135	0x4C860B31
136	0x576AE317
137	0xD58A1D52
138	0x64683FDA
139	0xE688C19F
140	0x316F5A8D
141	0xB38FA4C8
142	0x026D8640
143	0x808D7805
144	0x35B4560C
145	0xB754A849
146	0x06B68AC1
147	0x84567484
148	0x53B1EF96

Index	Table value
149	0xD15111D3
150	0x60B3335B
151	0xE253CD1E
152	0xF9BF2538
153	0x7B5FDB7D
154	0xCABDF9F5
155	0x485D07B0
156	0x9FBA9CA2
157	0x1D5A62E7
158	0xACB8406F
159	0x2E58BE2A
160	0xF0093C3A
161	0x72E9C27F
162	0xC30BE0F7
163	0x41EB1EB2
164	0x960C85A0
165	0x14EC7BE5
166	0xA50E596D
167	0x27EEA728
168	0x3C024F0E
169	0xBEE2B14B
170	0x0F0093C3
171	0x8DE06D86
172	0x5A07F694
173	0xD8E708D1
174	0x69052A59
175	0xEBE5D41C
176	0x5EDCFA15
177	0xDC3C0450
178	0x6DDE26D8
179	0xEF3ED89D
180	0x38D9438F

Index	Table value
181	0xBA39BDCA
182	0x0BDB9F42
183	0x893B6107
184	0x92D78921
185	0x10377764
186	0xA1D555EC
187	0x2335ABA9
188	0xF4D230BB
189	0x7632CEFE
190	0xC7D0EC76
191	0x45301233
192	0x4DB0C811
193	0xCF503654
194	0x7EB214DC
195	0xFC52EA99
196	0x2BB5718B
197	0xA9558FCE
198	0x18B7AD46
199	0x9A575303
200	0x81BBBB25
201	0x035B4560
202	0xB2B967E8
203	0x305999AD
204	0xE7BE02BF
205	0x655EFCFA
206	0xD4BCDE72
207	0x565C2037
208	0xE3650E3E
209	0x6185F07B
210	0xD067D2F3
211	0x52872CB6
212	0x8560B7A4

Index	Table value
213	0x078049E1
214	0xB6626B69
215	0x3482952C
216	0x2F6E7D0A
217	0xAD8E834F
218	0x1C6CA1C7
219	0x9E8C5F82
220	0x496BC490
221	0xCB8B3AD5
222	0x7A69185D
223	0xF889E618
224	0x26D86408
225	0xA4389A4D
226	0x15DAB8C5
227	0x973A4680
228	0x40DDDD92
229	0xC23D23D7
230	0x73DF015F
231	0xF13FFF1A
232	0xEAD3173C
233	0x6833E979
234	0xD9D1CBF1
235	0x5B3135B4
236	0x8CD6AEA6
237	0x0E3650E3
238	0xBFD4726B
239	0x3D348C2E
240	0x880DA227
241	0x0AED5C62
242	0xBB0F7EEA
243	0x39EF80AF
244	0xEE081BBD

Index	Table value
245	0x6CE8E5F8
246	0xDD0AC770
247	0x5FEA3935
248	0x4406D113
249	0xC6E62F56
250	0x77040DDE
251	0xF5E4F39B
252	0x22036889
253	0xA0E396CC
254	0x1101B444
255	0x93E14A01

2.2.6 LDAP Result Code to DirectoryOperationResult Mapping

The algorithm converts an LDAP result code *TaskReturnStatus* to a **DirectoryOperationResult** according to the following table.

LDAP result code	DirectoryOperationResult enumeration
Success (0)	Success
noSuchObject (32)	ObjectNotFound
noSuchAttribute (16)	AttributeNotFound
entryAlreadyExists (68)	ObjectAlreadyExists
Other codes not listed preceding	GenericError

2.3 Queue Alias

A queue alias is a directory object of the mSMQ-Custom-Recipient class type that associates a directory service path and a user-defined string (alias) with a direct, public, or private format name (see [MS-MQMQ] sections 2.1.2, 2.1.3, and 2.1.4). **<8>** MSMQ instances MUST use the **distinguished name (DN)** for objects of class mSMQ-Custom-Recipient as specified in section 2.2.1 in **LDAP** queries to access the objects and retrieve the format name of a queue from the msMQ-Recipient-FormatName attribute of the queue alias object.

2.4 Directory Service Schema Elements

The algorithm accesses the following Directory Service schema classes and attributes listed in the following table(s).

For the syntactic specifications of the following <Class> or <Class><Attribute> pairs, refer to Active Directory Domain Services (AD DS) ([\[MS-ADA1\]](#), [\[MS-ADA2\]](#), [\[MS-ADA3\]](#)).

Class	Attribute
mSMQQueue	distinguishedName mSMQAuthenticate mSMQBasePriority mSMQJournal mSMQLabelEx MSMQ-MulticastAddress mSMQPrivacyLevel mSMQQueueJournalQuota mSMQQueueNameExt mSMQQueueQuota mSMQQueueType mSMQTransactional objectGUID nTSecurityDescriptor whenChanged whenCreated
mSMQConfiguration	distinguishedName mSMQComputerTypeEx mSMQDependentClientServices mSMQDsServices mSMQEncryptKey mSMQForeign mSMQInRoutingServers mSMQJournalQuota mSMQOSType mSMQOutRoutingServers mSMQQuota mSMQRoutingServices mSMQServiceType mSMQSites nTSecurityDescriptor objectGUID whenChanged whenCreated
computer	dNSHostName mSMQSignCertificates objectSid servicePrincipalName operatingSystemVersion
site	cn distinguishedName mSMQInterval1 mSMQInterval2 mSMQNt4Stub

Class	Attribute
	mSMQSiteForeign nTSecurityDescriptor objectGUID
mSMQEnterpriseSettings	mSMQCSPName mSMQLongLived mSMQNameStyle mSMQVersion nTSecurityDescriptor objectGUID
user	distinguishedName mSMQDigests mSMQSignCertificates objectGUID objectSid
mSMQSiteLink	description distinguishedName mSMQCost mSMQSite1 mSMQSite2 mSMQSiteGates objectGUID
mSMQSettings	mSMQDependentClientService mSMQDsService mSMQQMID mSMQRoutingService mSMQServices

3 Protocol Details

3.1 Algorithm Details

This algorithm is specified in terms of a set of abstract local events that are raised by the processing rules within the common processing rules specified in [\[MS-MQDMPR\]](#). The algorithm converts these events to **LDAP** operations against **Active Directory**. Three operations ([Read Directory Begin \(section 3.1.6.4\)](#), [Read Directory Next \(section 3.1.6.5\)](#), and [Read Directory End \(section 3.1.6.6\)](#)) have shared state; the rest are independent. The processing rules of the abstract events describe the conversion of arguments from abstract to concrete, which can involve some simple computation; the results returned by the transport are returned after conversion of results from concrete to abstract, which again can involve some simple computation.

3.1.1 Abstract Data Model

This section describes a conceptual model of possible data organization that an implementation maintains to participate in this algorithm. The specified organization is provided to facilitate the explanation of how the algorithm behaves. This document does not mandate that implementations adhere to this model as long as their external behavior is consistent with that specified in this document.

The abstract data model for this algorithm comprises elements that are private to this algorithm and consists of elements that are specified in this algorithm and elements that are specified in [\[MS-MQDMPR\]](#) section 3.1.1. Although the abstract data model for this algorithm contains elements that are specified in [\[MS-MQDMPR\]](#), this algorithm does not share instances of any of these elements or state with any protocol. The relationship between this algorithm, a **queue manager**, and MSMQ protocols is described in [\[MS-MQOD\]](#).

Section [3.1.1.1](#) details the elements from the ADM that are specified in [\[MS-MQDMPR\]](#). Sections [3.1.1.2](#) and [3.1.1.3](#) detail the ADM elements that are specific to this algorithm. Section [3.1.1.4](#) details an extension to an ADM element specified in [\[MS-MQDMPR\]](#).

3.1.1.1 Externally Defined Data Elements

This algorithm accesses the following externally defined ADM elements:

- **DirectoryObject**: [\[MS-MQDMPR\]](#) section 3.1.1.
- **QueueManager**: [\[MS-MQDMPR\]](#) section 3.1.1.1.
- **Queue**: [\[MS-MQDMPR\]](#) section 3.1.1.2.
- **Enterprise**: [\[MS-MQDMPR\]](#) section 3.1.1.6.
- **Site**: [\[MS-MQDMPR\]](#) section 3.1.1.7.
- **RoutingLink**: [\[MS-MQDMPR\]](#) section 3.1.1.8.
- **User**: [\[MS-MQDMPR\]](#) section 3.1.1.15.

3.1.1.2 ReadDirectoryIteratorState Data Element

The **ReadDirectoryIteratorState** ADM element represents the state associated with a directory read initiated by a [Read Directory Begin \(section 3.1.6.4\)](#) event. This ADM element MUST contain the following ADM attributes:

- **Handle:** A **HANDLE** ([\[MS-DTYP\]](#) section 2.2.16) that identifies the **ReadDirectoryIteratorState** ADM element instance. This identifier MUST NOT change over the lifetime of the **ReadDirectoryIteratorState** ADM element instance.
- **DataElementType:** A string that is the name of a sub-type of a **DirectoryObject** ADM element.
- **LDAPAttributeList:** The names of the **Active Directory** attributes to be read from the directory.
- **LDAPState:** A list of sublists, where each sublist consists of the values returned by an **LDAP** search for the attributes of one **Active Directory object**, in the same order as the attribute names in the **LDAPAttributeList** ADM attribute.
- **AttributeList:** The names of the attributes to be read from the directory.

3.1.1.3 ReadDirectoryIteratorStateCollection Data Element

The **ReadDirectoryIteratorStateCollection** ADM element is a collection that MUST contain a reference to every existing **ReadDirectoryIteratorState** ADM element instance.

3.1.1.4 User Data Element

The algorithm MUST maintain private state for each **User** ADM element instance in addition to the state specified for the **User** ADM element in [\[MS-MQDMPR\]](#) section 3.1.1.15. The following additional ADM attribute is used to reference this private state:

- **FullPath:** A **distinguished name** that can be used to look up in **Active Directory** the user directory object corresponding to this **User** ADM element instance. This is a directory attribute, as specified in [\[MS-MQDMPR\]](#) section 3.1.1.

3.1.1.5 CachedConfigurationNamingContext

The **CachedConfigurationNamingContext** ADM element is a string that is the **distinguished name** of the root of the configuration tree in **Active Directory**. The value is obtained from Active Directory at algorithm initialization time, as specified in section [3.1.3](#).

3.1.1.6 CachedLDAPConnection

This ADM element represents an LDAP connection that can be reused if multiple directory operations occur simultaneously and defines the following ADM attributes:

Handle: Either an **ADCONNECTION_HANDLE** ([\[MS-DTYP\]](#) section 2.2.2) that is connected to an **Active Directory** server and that is ready to perform LDAP operations or NULL.

RefCount: An integer that represents the count of directory operations using the connection represented by the **Handle** ADM attribute. This count is incremented when an operation starts using the connection, as specified in section [3.1.6.18](#), and is decremented when an operation finishes using the connection, as specified in section [3.1.6.21](#).

3.1.2 Timers

The Directory Service Schema Mapping algorithm does not use timers.

3.1.3 Initialization

The algorithm MUST perform these initialization tasks before any directory operation (sections [3.1.6.1](#) through [3.1.6.9](#)) is called:

- Set the **CachedLDAPConnection.Handle** (section [3.1.1.6](#)) ADM element attribute to NULL and the **CachedLDAPConnection.RefCount** ADM element attribute to zero.
- Raise a [Prepare an LDAP Connection \(section 3.1.6.18\)](#) event. The event takes no arguments. If the *rStatus* returned is not **DirectoryOperationResult.Success**, the **CachedConfigurationNamingContext** ADM element MUST be set to an empty string, and processing MUST end. Otherwise, let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE** ([\[MS-DTYP\]](#) section 2.2.2) initialized to the value returned in *rADConnection*.
- Construct an **LDAPMessage** ([\[RFC2251\]](#) section 4.1):
 - *messageID* = set as described in [\[RFC2251\]](#) section 4.1.1.1.
 - *protocolOp* = *searchRequest*
 - *controls* = none
 - *baseObject* = an empty string
 - *scope* = *baseObject*
 - *derefAliases* = *neverDerefAliases*
 - *sizeLimit* = 0
 - *timeLimit* = 0
 - *typesOnly* = FALSE
 - *filter* = "(objectClass=*)"
 - *attributes* = an empty list
- Perform the Performing an LDAP Operation on an ADConnection task ([\[MS-ADTS\]](#) section 7.6.1.6) with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [\[RFC2251\]](#) section 4.1.10, the **CachedConfigurationNamingContext** ADM element MUST be set to an empty string. If the result message returned in *TaskOutputResultMessages* is not success, as defined in [\[RFC2251\]](#) section 4.1.10, the **CachedConfigurationNamingContext** ADM element MUST be set to an empty string. Otherwise, the **CachedConfigurationNamingContext** ADM element MUST be set to the value of the *configurationNamingContext* attribute extracted from the result message returned in *TaskOutputResultMessages*.
- Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.

3.1.4 Message Processing Events and Sequencing Rules

None.

3.1.5 Timer Events

None.

3.1.6 Other Local Events

3.1.6.1 Create Directory Object

This event MUST be generated with the following arguments:

- *iDirectoryObject*: A **DirectoryObject** ADM element instance to be created in the directory.
- *iAttributeList* (Optional): An array of strings containing the subset of the names of the directory attributes of the *iDirectoryObject* argument to be persisted in the directory. If this argument is not supplied, all directory attributes of the *iDirectoryObject* argument that can be persisted at create time are persisted. The following sections specify the individual types list attributes that can be persisted at create time.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.
- *rObjectGUID*: The **GUID** of the newly created **DirectoryObject** ADM element instance. This value is undefined if *rStatus* does not equal **DirectoryOperationResult.Success**.

The MQDSSM algorithm MUST perform the following actions to process this event:

- Determine the type of the *iDirectoryObject* argument. If the type is not one of **QueueManager**, **Queue**, **Site**, or **RoutingLink**, *rStatus* MUST be set to the **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing MUST end.
- The create operation logic for the type of ADM element specified by the *iDirectoryObject* argument MUST be performed as specified in the sections listed following. The *rStatus* and *rObjectGUID* values specified in each section MUST be returned.
 - [QueueManager \(section 3.1.6.1.1\)](#)
 - [Queue \(section 3.1.6.1.2\)](#)
 - [Site \(section 3.1.6.1.3\)](#)
 - [RoutingLink \(section 3.1.6.1.4\)](#)

3.1.6.1.1 QueueManager

3.1.6.1.1.1 Preconditions

The arguments supplied to the [Create Directory Object \(section 3.1.6.1\)](#) event MUST meet these requirements:

- If *iAttributeList* is provided, the name "SiteIdentifierList" MUST be present in the list.
- The *iDirectoryObject.SiteIdentifierList* and *iDirectoryObject.ComputerName* ADM attributes MUST be populated.
- If *iAttributeList* is provided, it MUST NOT contain the name **ConnectedNetworkIdentifierList**. This attribute is not supported in **Active Directory**-based environments.

If any of these conditions is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.1.1.2 Creation

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **QueueManager** ADM element attributes listed in section [3.1.6.20.1](#), except **ConnectedNetworkIdentifierList**.
2. Let *ConfigurationAttributeList* be a list of mSMQConfiguration attribute names, initialized to be empty. For each mSMQConfiguration attribute listed in the following table, if the corresponding **QueueManager** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the mSMQConfiguration attribute name to *ConfigurationAttributeList* and compute the value for the attribute as shown. **QueueManager** ADM element attribute names that appear in *iAttributeList* but that do not appear in the following table or in subsections referenced by the table MUST be ignored.

mSMQConfiguration attribute	Attribute value computation
mSMQComputerTypeEx ([MS-ADA2] section 2.549)	Directory string transformed from <i>iDirectoryObject.QueueManagerVersion</i> .
mSMQOSType ([MS-ADA2] section 2.572)	Computed from <i>iDirectoryObject.OperatingSystemType</i> . See section 3.1.6.1.1.4 .
mSMQServiceType ([MS-ADA2] section 2.586)	Computed from <i>iDirectoryObject.DirectoryServer</i> , <i>iDirectoryObject.DirectoryServerType</i> , <i>iDirectoryObject.RemoteAccessServer</i> , and <i>iDirectoryObject.RoutingServer</i> . See section 3.1.6.1.1.5 .
mSMQQuota ([MS-ADA2] section 2.582)	Integer copied from <i>iDirectoryObject.QueueManagerQuota</i> .
mSMQJournalQuota ([MS-ADA2] section 2.564)	Integer copied from <i>iDirectoryObject.JournalQuota</i> .
mSMQForeign ([MS-ADA2] section 2.559)	If <i>iDirectoryObject.ForeignSystem</i> is TRUE, mSMQForeign is 0x01 Else mSMQForeign is 0x00.
mSMQSites ([MS-ADA2] section 2.598)	List of GUIDs copied from <i>iDirectoryObject.SiteIdentifierList</i> .
mSMQOutRoutingServers ([MS-ADA2] section 2.573)	Computed from <i>iDirectoryObject.OutRoutingServerIdentifierList</i> . See section 3.1.6.1.1.6 .
mSMQInRoutingServers ([MS-ADA2] section 2.560)	Computed from <i>iDirectoryObject.InRoutingServerIdentifierList</i> . See section 3.1.6.1.1.7 .
mSMQRoutingServices ([MS-ADA2] section 2.584)	If <i>iDirectoryObject.RoutingServer</i> is TRUE, mSMQRoutingServices is 0x01 Else mSMQRoutingServices is 0x00.
mSMQDsServices ([MS-ADA2] section 2.557)	If <i>iDirectoryObject.DirectoryServer</i> is TRUE, mSMQDsServices is 0x01 Else mSMQDsServices is 0x00.
mSMQDependentClientServices ([MS-ADA2] section 2.553)	If <i>iDirectoryObject.SupportingServer</i> is TRUE, mSMQDependentClientServices is 0x01 Else mSMQDependentClientServices is 0x00.

mSMQConfiguration attribute	Attribute value computation
mSMQEncryptKey ([MS-ADA2] section 2.558)	MQDSPUBLICKEYS structure ([MS-MQMQ] section 2.2.2) copied from <i>iDirectoryObject.PublicEncryptionKeyList</i> .
nTSecurityDescriptor ([MS-ADA3] section 2.37)	Computed from <i>iDirectoryObject.Security</i> . See section 3.1.6.1.1.8 .

3. A [Create Object Using LDAP \(section 3.1.6.12\)](#) event MUST be generated with the following arguments:
 - *iParentPath* := a **distinguished name** of the form specified for a computer object in section [2.2.1](#), where "<computer name>" is the value of *iDirectoryObject.ComputerName*
 - *iChildName* := "msmq"
 - *iObjectClass* := "mSMQConfiguration"
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *ConfigurationAttributeList* and the corresponding values, as computed in step 2.
4. If the Create Object Using LDAP event returns an *rStatus* value of **DirectoryOperationResult.ObjectAlreadyExists**, a [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iPath* := a distinguished name of the form specified for mSMQConfiguration in section 2.2.1, where "<computer name>" is the value of *iDirectoryObject.ComputerName*
 - *iAttributes* := a list containing the attribute name "objectGUID"

If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rObjectGUID* is undefined, and processing MUST end. Otherwise, *rStatus* is set to the first element in *rValues* returned by the Get Object Properties Using LDAP event.
5. Else if the Create Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Create Object Using LDAP event, *rObjectGUID* is undefined, and processing MUST end. Otherwise, *rObjectGuid* is set to the *rObjectGuid* returned by the Create Object Using LDAP event.

3.1.6.1.1.3 Postprocessing

1. If the [PublicSigningKeyList \(section 3.1.6.7.1.10\)](#) ADM attribute is present in *iAttributeList*, its value MUST be written to the directory:
 - A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event MUST be generated with the following arguments:
 - *iPath* := **distinguished name** for a computer object MUST be constructed according to the format specified in section [2.2.1](#), where "<computer name>" is the value of *iDirectoryObject.ComputerName*.
 - *iAttributeList* := a list of name-value pairs consisting of the attribute names "mSMQSignCertificates", "mSMQDigests", "mSMQSignCertificatesMig", and "mSMQDigestsMig" and the corresponding values computed as specified in section [3.1.6.1.1.9](#).

- If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, *rObjectGUID* is undefined, and processing MUST end.
2. If one or more of the **RoutingServer**, **DirectoryServer**, or **SupportingServer** ADM attributes on *iDirectoryObject* is TRUE, one or more mSMQSettings objects MUST be created as well. For each GUID in the value of *iDirectoryObject.SiteIdentifierList*, these steps MUST be performed:
- Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from the **SiteIdentifierList**
 - *iAttributeList* := Null
 - If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.ObjectNotFound**:
 - The current **GUID** MUST be skipped.
 - Else if the *rStatus* returned by the Read Directory event is set to anything other than **DirectoryOperationResult.Success**:
 - *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing MUST end.
 - Else if the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**:
 - Generate a [Create Object Using LDAP \(section 3.1.6.12\)](#) event with the following arguments:
 - *iParentPath* := a distinguished name of the form specified for an mSMQSettings object in section 2.2.1 MUST be constructed, where "<computer name>" is the value of **QueueManager.ComputerName**, and "<site name>" is the value of the **Name** ADM attribute of the returned **Site** ADM element instance, then the first comma-separated element removed
 - *iChildName* := "MSMQ Settings"
 - *iObjectClass* := "mSMQSettings"
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in the following table and the corresponding values copied from *ConfigurationAttributeList*

mSMQSettings attribute	mSMQConfiguration attribute
mSMQQMID ([MS-ADA2] section 2.577)	objectGUID
mSMQServices ([MS-ADA2] section 2.585)	mSMQServiceType
mSMQRoutingService ([MS-ADA2] section 2.583)	mSMQRoutingServices
mSMQDsService ([MS-ADA2] section 2.556)	mSMQDsServices
mSMQDependentClientService ([MS-ADA2] section 2.552)	mSMQDependentClientServices

- If the Create Object Using LDAP event returns an *rStatus* value of **DirectoryOperationResult.ObjectAlreadyExists**, a [Delete Object Using LDAP \(section 3.1.6.13\)](#) event MUST be generated with the *iPath* argument set to "CN=MSMQ Settings, <parent path>", where <parent path> is the *iParentPath* argument of the preceding Create Object Using LDAP event. If the *rStatus* returned by the Delete Object Using LDAP event is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Delete Object Using LDAP event, *rObjectGUID* is undefined, and processing MUST end. Otherwise, the Create Object Using LDAP event in the preceding step MUST be invoked again.
- If the Create Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, *rObjectGUID* is undefined, and processing MUST end.

3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.1.1.4 mSMQOSType

The mSMQOSType attribute is an integer that MUST have one of the values listed in the following table. The **OperatingSystemType** ADM attribute is an enumeration. The value MUST be converted according to this table.

OperatingSystemType ADM attribute value	mSMQOSType value
Other	0x00000000
Foreign	0x00000100
Windows 95	0x00000200
WinClient	0x00000300
WinServer	0x00000400
WinEnt	0x00000500

3.1.6.1.1.5 mSMQServiceType

The mSMQServiceType attribute is a bitmap that MUST be computed according to this algorithm:

1. Let *FinalValue* be an integer, initialized to 0x00000000.
2. If *iDirectoryObject.RoutingServer* is populated and TRUE, the 0x00000001 bit of *FinalValue* MUST be set.
3. If *iDirectoryObject.RemoteAccessServer* is populated and TRUE, the 0x00000010 bit of *FinalValue* MUST be set.
4. If *iDirectoryObject.DirectoryServer* is populated and TRUE, and *iDirectoryObject.DirectoryServerType* is populated and does not have the enumeration value **Standalone**, one bit of *FinalValue* MUST be set according to the following table.

DirectoryServerType value	FinalValue bit to be set
BackupSiteController	0x00000002
PrimarySiteController	0x00000004

DirectoryServerType value	FinalValue bit to be set
PrimaryEnterpriseController	0x00000008

5. The value of the mSMQServiceType attribute MUST be the value of *FinalValue*.

3.1.6.1.1.6 mSMQOutRoutingServers

The value of the mSMQOutRoutingServers attribute MUST be computed according to the following algorithm:

- Let *DNList* be a list of **distinguished names**, initialized to be empty.
- For each **GUID** in *iDirectoryObject.OutRoutingServerIdentifierList*, these steps MUST be performed:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - iDirectoryObjectType* := "QueueManager"
 - iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.OutRoutingServerIdentifierList*
 - iAttributeList* := An array of the following **QueueManager** ADM element attribute names:
 - FullPath**
 - If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, this GUID MUST be skipped.
 - If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of the *rDirectoryObject.FullPath* returned by the Read Directory event must be added to *DNList*.
- The value of mSMQOutRoutingServers MUST be the value of *DNList*.

3.1.6.1.1.7 mSMQInRoutingServers

The value of the mSMQInRoutingServers attribute MUST be computed according to the following algorithm:

- Let *DNList* be a list of **distinguished names**, initialized to be empty.
- For each **GUID** in *iDirectoryObject.InRoutingServerIdentifierList*, these steps MUST be performed:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - iDirectoryObjectType* := "QueueManager"
 - iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.InRoutingServerIdentifierList*
 - iAttributeList* := An array of the following **QueueManager** ADM element attribute names:

- **FullPath**

- If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, this GUID MUST be skipped.
- If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of the *rDirectoryObject.FullPath* returned by the Read Directory event must be added to *DNList*.

3. The value of *mSMQInRoutingServers* MUST be the value of *DNList*.

3.1.6.1.1.8 nTSecurityDescriptor

The value of the *nTSecurityDescriptor* attribute MUST be computed according to the following algorithm:

1. Let *OwnerSid* and *MachineSid* be **Security Identifiers (SIDs)** ([MS-DTYP] section 2.4.2), both initialized to zero.
2. A **distinguished name** MUST be formed by removing the first element of the distinguished name computed in step 3 of section 3.1.6.1.1.2. The *objectSid* attribute of the computer object referenced by this distinguished name MUST be read via **LDAP**. If the read operation fails, the values of *MachineSid* and *OwnerSid* MUST be unchanged. Otherwise, the values of *MachineSid* and *OwnerSid* MUST be set to the value read.
3. Let *UserSid* be a SID. If *iDirectoryObject.Security* is populated and the **security descriptor** that is the value of *iDirectoryObject.Security* contains an owner SID, *UserSid* MUST be initialized with the value of that owner SID. Otherwise, *UserSid* MUST be initialized to the SID of the user under whose identity the current thread is running.
4. Let *WorldAccess* be an **MQQMACCESSMASK** ([MS-MQMQ] section 2.2.23) enumerated value, which MUST be initialized to be **MQSEC_MACHINE_WORLD_RIGHTS**.
5. If the ADM attribute name **Foreign** is present in *iAttributeList* and *iDirectoryObject.Foreign* is populated and the value of *iDirectoryObject.Foreign* is TRUE, *WorldAccess* MUST be set to **MQSEC_MACHINE_GENERIC_WRITE**.
6. Let *FinalSecurity* be a **SECURITY_DESCRIPTOR** as specified in [MS-DTYP] section 2.4.6, initialized to be empty.
7. An **ACCESS_ALLOWED_ACE** ([MS-DTYP] section 2.4.4.2) with the **Mask** field set to *WorldAccess* and containing the well-known SID with string representation S-1-1-0 (relative identifier **SECURITY_WORLD_RID** combined with identifier authority **SECURITY_WORLD_SID_AUTHORITY**) MUST be added to the **Dacl** in *FinalSecurity*.
8. If *OwnerSid* is nonzero, an **ACCESS_ALLOWED_ACE** structure with a **Mask** field set to **MQSEC_MACHINE_GENERIC_ALL** and containing the SID in *OwnerSid* MUST be added to the **Dacl** in *FinalSecurity*.
9. If *MachineSid* is nonzero, an **ACCESS_ALLOWED_ACE** structure with a **Mask** set to **MQSEC_MACHINE_SELF_RIGHTS** and containing the SID in *MachineSid* MUST be added to the **Dacl** in *FinalSecurity*.
10. If *UserSid* is nonzero, an **ACCESS_ALLOWED_ACE** structure with a **Mask** field set to **MQSEC_MACHINE_GENERIC_ALL** and containing the SID in *UserSid* MUST be added to the **Dacl** in *FinalSecurity*.
11. The **SECURITY_DESCRIPTOR** structure in *FinalSecurity* MUST be converted to self-relative format (see [MS-DTYP] section 2.4.6).

12. The value of the `nTSecurityDescriptor` attribute MUST be the value of *FinalSecurity*.

3.1.6.1.1.9 mSMQSignCertificates and mSMQDigests

The **PublicSigningKeyList** attribute is stored in two attributes on the computer object. The `mSMQSignCertificates` ([MS-ADA2] section 2.587) attribute contains an **MQUERSIGNCERTS** ([MS-MQMQ] section 2.2.21) structure, which contains **MQUERSIGNCERT** ([MS-MQMQ] section 2.2.22) structures, which in turn contain individual X.509-encoded certificates. The `mSMQDigests` ([MS-ADA2] section 2.554) attribute contains an array of MD5 hashes of the certificates stored in the `mSMQSignCertificates` attribute, mirroring the values of the **Digest** fields of the **MQUERSIGNCERT** structures. Each array element MUST contain the 16-byte output of the MD5 algorithm, as specified in [RFC1321]. The certificates and digests MUST be in the same order, but there is no other sorting requirement.

The values of `mSMQSignCertificates` and `mSMQDigests` MUST be computed according to the following algorithm:

If the **PublicSigningKeyList** ADM attribute name is present in *iAttributeList* and *iDirectoryObject.PublicSigningKeyList* is populated:

- Copy the bytes in *iDirectoryObject.PublicSigningKeyList*, which is an **MQUERSIGNCERTS** structure, to `mSMQSignCertificates`.
- For each **MQUERSIGNCERT** structure in `mSMQSignCertificates`:
 - Append the bytes of the **Digest** field in the structure to `mSMQDigests`.

The attributes `mSMQSignCertificatesMig` ([MS-ADA2] section 2.588) and `mSMQDigestsMig` ([MS-ADA2] section 2.555) MAY <9> be set to the values of `mSMQSignCertificates` and `mSMQDigests`, respectively.

3.1.6.1.2 Queue

3.1.6.1.2.1 Preconditions

The arguments supplied to the [Create Directory Object \(section 3.1.6.1\)](#) event MUST meet these requirements:

- If *iAttributeList* is provided, the name "Pathname" MUST be present in the list.
- If *iAttributeList* is provided, it MUST NOT contain the name "Scope". This name is not supported in **Active Directory**-based environments.
- *iDirectoryObject.Pathname* MUST be populated.

If any of these conditions is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.1.2.2 Creation

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **Queue** ADM element attributes listed in section [3.1.6.20.2](#).
2. Let *QueueAttributeList* be a list of `mSMQQueue` attribute names, initialized to be empty. For each `mSMQQueue` attribute listed in the following table, if the corresponding **Queue** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the `mSMQQueue` attribute name to *QueueAttributeList* and compute the value for the attribute as

shown. **Queue** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table **MUST** be ignored.

mSMQQueue attribute	Attribute value computation
mSMQLabelEx ([MS-ADA2] section 2.566)	Directory string transformed from <i>iDirectoryObject.Label</i> .
mSMQQueueType ([MS-ADA2] section 2.581)	GUID copied from <i>iDirectoryObject.Type</i> .
mSMQJournal ([MS-ADA2] section 2.563)	If <i>iDirectoryObject.Journaling</i> is TRUE, mSMQJournal is 1 Else mSMQJournal is 0.
mSMQQueueQuota ([MS-ADA2] section 2.580)	Integer number of kilobytes equal to <i>iDirectoryObject.Quota</i> .
mSMQQueueJournalQuota ([MS-ADA2] section 2.578)	Integer number of kilobytes equal to <i>iDirectoryObject.JournalQuota</i> .
mSMQAuthenticate ([MS-ADA2] section 2.546)	If <i>iDirectoryObject.Authentication</i> is TRUE, mSMQAuthenticate is 1 Else mSMQAuthenticate is 0.
mSMQPrivacyLevel ([MS-ADA2] section 2.576)	Computed from <i>iDirectoryObject.PrivacyLevel</i> . See section 3.1.6.1.2.4 .
mSMQTransactional ([MS-ADA2] section 2.599)	If <i>iDirectoryObject.Transactional</i> is TRUE, mSMQTransactional is 1 Else mSMQTransactional is 0.
MSMQ-MulticastAddress ([MS-ADA2] section 2.543)	Directory string transformed from <i>iDirectoryObject.MulticastAddress</i> .
nTSecurityDescriptor ([MS-ADA3] section 2.37)	SECURITY_DESCRIPTOR ([MS-DTYP] section 2.4.6) structure copied from <i>iDirectoryObject.Security</i> .
mSMQBasePriority ([MS-ADA2] section 2.547)	Integer copied from <i>iDirectoryObject.BasePriority</i> .

3. The queue name **MUST** be modified as specified in section [3.1.6.1.2.5](#).
4. A [Create Object Using LDAP \(section 3.1.6.12\)](#) event **MUST** be generated with the following arguments:
 - *iParentPath* := a **distinguished name** of the form specified for an mSMQConfiguration object in section [2.2.1](#), where "<computer name>" is the computer name portion of the value of *iDirectoryObject.Pathname*, as specified in [\[MS-MQMQ\]](#) section 2.1.1.
 - *iChildName* := *FinalChildName*, as computed in step 3.
 - *iObjectClass* := "mSMQQueue"
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *QueueAttributeList* and the corresponding values, as computed in step 2.

5. If the Create Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Create Object Using LDAP event, *rObjectGUID* is undefined, and processing MUST end. Otherwise, *rObjectGuid* is set to the *rObjectGuid* returned by the Create Object Using LDAP event.
6. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.1.2.3 Postprocessing

None.

3.1.6.1.2.4 mSMQPrivacyLevel

The mSMQPrivacyLevel attribute is an integer that MUST have one of the values listed in the following table. The **PrivacyLevel** ADM attribute of the **Queue** ADM element is an enumeration. The value MUST be converted according to this table.

mSMQPrivacyLevel value	PrivacyLevel ADM attribute value
0	None
1	Optional
2	Body

3.1.6.1.2.5 Name

The name of the queue is modified before the mSMQQueue object is created. The modified name is used to create the object. The modification is performed according to the following algorithm:

1. Let *FinalChildName* and *FinalExtension* be **Unicode** strings, initialized to be empty.
2. Let *QName* be a Unicode string that MUST be the "<queue name>" portion of *iDirectoryObject.Pathname*, as specified in section [2.2.1](#).
3. A Unicode backslash ('\') MUST be inserted in *QName* before every instance of a character belonging to this set of special characters: '\n', '/', '#', '>', '<', '=', and '\x0A'.
4. If the length of *QName* does not exceed 63 characters, the value of *QName* MUST be copied to *FinalChildName*. Otherwise, the first 55 characters of *QName* MUST be copied to *FinalChildName*; the remaining characters of *QName* MUST be copied to *FinalExtension*; and a nine-character extension MUST be added to the end of *FinalChildName*, which MUST be exactly 64 characters long after adding the extension. The extension consists of a Unicode dash ('-') followed by an eight-character hash string that is calculated from the values of the Unicode characters in *QName* as specified in section [2.2.5](#).
5. If *FinalExtension* is not empty, the mSMQQueueNameExt attribute ([\[MS-ADA2\]](#) section 2.579) MUST be added to *QueueAttributeList*, with the value of *FinalExtension* transformed to a **directory string** as the associated value.

3.1.6.1.3 Site

3.1.6.1.3.1 Preconditions

The arguments supplied to the [Create Directory Object \(section 3.1.6.1\)](#) event MUST meet these requirements:

- If *iAttributeList* is provided, the name "Name" MUST be present in the list.
- The *iDirectoryObject*.**Name** ADM attribute MUST be populated.
- *iAttributeList* MUST NOT contain the names **PrimarySiteController** or **PublicSigningKey**. These two ADM element attributes are not supported in **Active Directory**-based environments.

If any of these conditions is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.1.3.2 Creation

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **Site** ADM element attributes listed in section [3.1.6.20.4](#), except **PrimarySiteController** and **PublicSigningKey**.
2. Let *SiteAttributeList* be a list of site attribute names, initialized to be empty. For each site attribute listed in the following table, if the corresponding **Site** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the site attribute name to *SiteAttributeList* and compute the value of the attribute as shown. **Site** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table MUST be ignored.

site attribute	Attribute value computation
mSMQInterval1 ([MS-ADA2] section 2.561)	Integer copied from <i>iDirectoryObject</i> . IntraSiteReplicationInterval .
mSMQInterval2 ([MS-ADA2] section 2.562)	Integer copied from <i>iDirectoryObject</i> . InterSiteReplicationInterval .
mSMQSiteForeign ([MS-ADA2] section 2.592)	If <i>iDirectoryObject</i> . ForeignSite is TRUE, mSMQSiteForeign is 0x01 Else mSMQSiteForeign is 0x00.
mSMQnt4Stub ([MS-ADA2] section 2.571)	If <i>iDirectoryObject</i> . MigratedFromMsmq10 is TRUE, mSMQnt4Stub is 0x01 Else mSMQnt4Stub is 0x00.
nTSecurityDescriptor ([MS-ADA3] section 2.37)	See section 3.1.6.1.3.4 .

3. A [Create Object Using LDAP \(section 3.1.6.12\)](#) event MUST be generated with the following arguments:
 - *iParentPath* := "CN=Services, CN=Configuration, <root>", where <root> is as specified in section [2.2.1](#).
 - *iChildName* := *iDirectoryObject*.**Name**
 - *iObjectClass* := "site"
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *SiteAttributeList* and the corresponding values, as computed in step 2.

4. If the Create Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Create Object Using LDAP event, *rObjectGUID* is undefined, and processing MUST end. Otherwise, *rObjectGuid* is set to the *rObjectGuid* returned by the Create Object Using LDAP event.
5. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.1.3.3 Postprocessing

None.

3.1.6.1.3.4 nTSecurityDescriptor

If *iAttributeList* (either supplied or constructed) includes the ADM attribute name **Security**, the value of the *iDirectoryObject.Security* ADM attribute MUST be modified before it is written to the directory. If *iAttributeList* does not include the ADM attribute name **Security**, a default value MUST be computed and written. The algorithm for doing both is as follows:

1. Let *FinalSecurity* and *SuppliedSecurity* be **SECURITY_DESCRIPTOR** structures, as specified in [\[MS-DTYP\]](#) section 2.4.6, initialized to be empty.
2. If *iAttributeList* includes the ADM attribute name **Security**, the value of *iDirectoryObject.Security* MUST be copied to the *SuppliedSecurity* structure.
3. Let *OwnerSid* be a **SID** ([MS-DTYP] section 2.4.2) structure initialized to zero.
4. If the *SuppliedSecurity* structure is not empty:
 1. *OwnerSid* MUST be set to the owner **SID** from the *SuppliedSecurity* structure.
 2. If the **Dacl** field is populated in the *SuppliedSecurity* structure, the value MUST be copied to the *FinalSecurity* structure.
 3. If the **Sacl** field is populated in the *SuppliedSecurity* structure, the value MUST be copied to the *FinalSecurity* structure.
5. If *OwnerSid* is zero, it MUST be set to the SID of the user under whose identity the current thread is running.
6. If the user referenced by the SID in the *OwnerSid* structure is not a domain user, the *OwnerSid* structure MUST be set to the well-known SID with string representation S-1-5-7 (relative identifier SECURITY_ANONYMOUS_LOGON_RID combined with identifier authority SECURITY_NT_AUTHORITY).
7. If the **Dacl** field was not copied to the *FinalSecurity* structure in step 4:
 1. Let *WorldAccess* and *OwnerAccess* be **MQSITEACCESSMASK** ([\[MS-MQMQ\]](#) section 2.2.23) enumerated values, initialized to zero.
 2. If *OwnerSid* is a guest SID (equal to the SID designated by DOMAIN_USER_RID_GUEST, as specified in [\[MS-SAMR\]](#) section 2.2.1.14) or the user referenced by the SID in *OwnerSid* is not a domain user, *WorldAccess* MUST be set to MQSEC_SITE_GENERIC_ALL. Otherwise, *WorldAccess* MUST be set to MQSEC_SITE_GENERIC_READ, and *OwnerAccess* MUST be set to MQSEC_SITE_GENERIC_ALL.
 3. An **ACCESS_ALLOWED_ACE** ([MS-DTYP] section 2.4.4.2) structure with a **Mask** field set to *WorldAccess* and containing the well-known SID with string representation S-1-1-0 (relative identifier SECURITY_WORLD_RID combined with identifier authority SECURITY_WORLD_SID_AUTHORITY) MUST be added to the **Dacl** field in the *FinalSecurity* structure.

4. If *OwnerAccess* is nonzero, an **ACCESS_ALLOWED_ACE** structure with a **Mask** field set to *OwnerAccess* and containing the SID in *OwnerSid* MUST be added to the **Dacl** field in the *FinalSecurity* structure.
8. The **SECURITY_DESCRIPTOR** structure in *FinalSecurity* MUST be converted to self-relative format (see [MS-DTYP] section 2.4.6).
9. The value of the *nTSecurityDescriptor* attribute MUST be the value of the *FinalSecurity* structure.

3.1.6.1.4 RoutingLink

3.1.6.1.4.1 Preconditions

The arguments supplied to the [Create Directory Object \(section 3.1.6.1\)](#) event MUST meet these requirements:

- If *iAttributeList* is provided, the names "Site1Identifier", "Site2Identifier", and "ActualCost" MUST be present in the list.
- The *iDirectoryObject.Site1Identifier*, *iDirectoryObject.Site2Identifier*, and *iDirectoryObject.ActualCost* ADM attributes MUST be populated.

If any of these conditions is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.1.4.2 Creation

1. If *iAttributeList* was not provided as an argument, then an *iAttributeList* MUST be constructed that MUST contain the names of all **RoutingLink** ADM element attributes listed in section [3.1.6.20.5](#).
2. Let *SiteLinkAttributeList* be a list of mSMQSiteLink attribute names, initialized to be empty. For each mSMQSiteLink attribute listed in the following table, if the corresponding **RoutingLink** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the mSMQSiteLink attribute name to *SiteLinkAttributeList* and compute the value for the attribute as shown. **RoutingLink** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table MUST be ignored.

mSMQSiteLink attribute	Attribute value computation
description ([MS-ADA1] section 2.153)	Directory string transformed from <i>iDirectoryObject.Description</i>
mSMQCost ([MS-ADA2] section 2.550)	Integer copied from <i>iDirectoryObject.ActualCost</i>
mSMQSite1 ([MS-ADA2] section 2.590)	Computed from <i>iDirectoryObject.Site1Identifier</i> . See section 3.1.6.1.4.4 .
mSMQSite2 ([MS-ADA2] section 2.591)	Computed from <i>iDirectoryObject.Site2Identifier</i> . See section 3.1.6.1.4.5 .
mSMQSiteGates ([MS-ADA2] section 2.593)	Computed from <i>iDirectoryObject.SiteGateIdentifierList</i> . See section 3.1.6.1.4.6 .

3. A [Create Object Using LDAP \(section 3.1.6.12\)](#) event MUST be generated with the following arguments:

- *iParentPath* := "CN=MsmqServices, CN=Services, CN=Configuration, <root>", where <root> is as specified in section [2.2.1](#).
 - *iChildName* := a string composed from the values of *iDirectoryObject.Site1Identifier* and *iDirectoryObject.Site2Identifier*. The two GUIDs MUST be converted to curly braced GUID string form ([\[MS-DTYP\]](#) section 2.3.4.3); next, they MUST be compacted by removing all characters that are not hexadecimal digits; finally, the string for Site2 MUST be appended to the string for Site1.
 - *iObjectClass* := "mSMQSiteLink"
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in SiteLinkAttributeList and the corresponding values, as computed in step 2.
4. If the Create Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Create Object Using LDAP event, *rObjectGUID* is undefined, and processing MUST end. Otherwise, *rObjectGuid* is set to the *rObjectGuid* returned by the Create Object Using LDAP event.
 5. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.1.4.3 Postprocessing

None.

3.1.6.1.4.4 mSMQSite1

The value of the mSMQSite1 attribute MUST be computed according to the following algorithm:

1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS *iDirectoryObject.Site1Identifier*
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **FullPath**
2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing MUST end.
3. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of mSMQSite1 MUST be the value of the *rDirectoryObject.FullPath* returned by the Read Directory event.

3.1.6.1.4.5 mSMQSite2

The value of the mSMQSite2 attribute MUST be computed according to the following algorithm:

1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):

- "Identifier" EQUALS *iDirectoryObject.Site2Identifier*
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **FullPath**
2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing MUST end.
 3. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of *mSMQSite2* MUST be the value of the *rDirectoryObject.FullPath* returned by the Read Directory event.

3.1.6.1.4.6 mSMQSiteGates

The value of the *mSMQSiteGates* attribute MUST be computed according to the following algorithm:

1. Let *DNList* be a list of **distinguished names**, initialized to be empty.
2. For each GUID in *iDirectoryObject.SiteGateIdentifierList*, these steps MUST be performed:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.SiteGateIdentifierList*
 - *iAttributeList* := An array of the following **QueueManager** ADM element attribute names:
 - **FullPath**
 - If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, this GUID MUST be skipped.
 - If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of the *rDirectoryObject.FullPath* returned by the Read Directory event MUST be added to *DNList*.
3. The value of *mSMQSiteGates* MUST be the value of *DNList*.

3.1.6.2 Delete Directory Object

This event MUST be generated with the following arguments:

- *iDirectoryObject*: A **DirectoryObject** ADM element instance to be deleted from the directory.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.

The algorithm MUST perform the following actions to process this event:

- Determine the type of *iDirectoryObject*. If the type is not one of **QueueManager**, **Queue**, **Site**, or **RoutingLink**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.

- Perform the steps of the delete operation for that type of ADM element, as specified in the sections listed following, and return the *rStatus* value specified in that section.
 - [QueueManager \(section 3.1.6.2.1\)](#)
 - [Queue \(section 3.1.6.2.2\)](#)
 - [Site \(section 3.1.6.2.3\)](#)
 - [RoutingLink \(section 3.1.6.2.4\)](#)

3.1.6.2.1 QueueManager

3.1.6.2.1.1 Preconditions

The arguments supplied to the [Delete Directory Object \(section 3.1.6.2\)](#) event MUST meet the following condition:

- At least one of *iDirectoryObject.FullPath*, *iDirectoryObject.ComputerName*, or *iDirectoryObject.Identifier* MUST be populated.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing of the event MUST end.

3.1.6.2.1.2 Delete

1. Let *DeleteDN* be a **distinguished name**, initialized to be empty.
2. If *iDirectoryObject.FullPath* is populated, the value of *iDirectoryObject.FullPath* MUST be copied to *DeleteDN*.
3. If *DeleteDN* is empty and *iDirectoryObject.ComputerName* is populated, a distinguished name for the mSMQConfiguration object to be deleted MUST be constructed according to the format specified in section [2.2.1](#), where "<computer name>" is the value of *iDirectoryObject.ComputerName*.
4. A [Delete Object Using LDAP \(section 3.1.6.13\)](#) event MUST be generated with the following arguments:
 - *iPath* := *DeleteDN*
 - *iGuid* := *iDirectoryObject.Identifier*
5. If the Delete Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Delete Object Using LDAP event, and processing MUST end.

3.1.6.2.1.3 Postprocessing

1. If one or more of *iDirectoryObject.RoutingServer*, *iDirectoryObject.DirectoryServer*, or *iDirectoryObject.SupportingServer* is TRUE, for each **GUID** in *iDirectoryObject.SiteIdentifierList*, these steps MUST be performed:
 1. Search the Directory for a **Site** ADM element instance that has an **Identifier** ADM attribute that matches the GUID:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"

- *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from the **SiteIdentifierList**
 - *iAttributeList* := An array of the following **Site** ADM attribute names:
 - **Name**
 - If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.ObjectNotFound**, the current GUID MUST be skipped.
 - If *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
2. A **distinguished name** MUST be constructed, of the form specified for an mSMQSettings object in section [2.2.1](#), where "<computer name>" is the value of *iDirectoryObject.ComputerName*, and "<site name>" is the value of the **Name** ADM attribute of the **Site** ADM element instance found in the preceding step.
 3. Generate a [Delete Object Using LDAP \(section 3.1.6.13\)](#) event with the following arguments:
 - *iPath* := the distinguished name constructed in the preceding step.
 4. If the Delete Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Delete Object Using LDAP event, and processing MUST end.
2. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.2.2 Queue

3.1.6.2.2.1 Preconditions

The arguments supplied to the [Delete Directory Object \(section 3.1.6.2\)](#) event MUST meet the following condition:

- At least one of *iDirectoryObject.FullPath*, *iDirectoryObject.Pathname*, or *iDirectoryObject.Identifier* MUST be populated.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing of the event MUST end.

3.1.6.2.2.2 Delete

The delete operation is performed in these steps:

1. Let *DeleteDN* be a **distinguished name** initialized to be empty.
2. If *iDirectoryObject.FullPath* is populated, the value of *iDirectoryObject.FullPath* MUST be copied to *DeleteDN*.
3. If *DeleteDN* is empty and *iDirectoryObject.Pathname* is populated, a distinguished name for the mSMQQueue object to be deleted MUST be constructed according to the format specified in section [2.2.1](#), where "<computer name>" is the ComputerName portion of the value of *iDirectoryObject.Pathname* and "<queue name>" is the QueueName portion of the value of *iDirectoryObject.Pathname*, as specified in [\[MS-MQMQ\]](#) section 2.1.1.

4. A [Delete Object Using LDAP \(section 3.1.6.13\)](#) event MUST be generated with the following arguments:
 - *iPath* := *DeleteDN*
 - *iGuid* := *iDirectoryObject.Identifier*
5. If the Delete Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Delete Object Using LDAP event, and processing MUST end.
6. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.2.2.3 Postprocessing

None.

3.1.6.2.3 Site

3.1.6.2.3.1 Preconditions

The arguments supplied to the [Delete Directory Object \(section 3.1.6.2\)](#) event MUST meet the following condition:

- At least one of *iDirectoryObject.FullPath*, *iDirectoryObject.Name*, or *iDirectoryObject.Identifier* MUST be populated.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing of the event MUST end.

3.1.6.2.3.2 Delete

1. Let *DeleteDN* be a **distinguished name** initialized to be empty.
2. If *iDirectoryObject.FullPath* is populated, the value of *iDirectoryObject.FullPath* MUST be copied to *DeleteDN*.
3. If *DeleteDN* is empty and *iDirectoryObject.Name* is populated, a distinguished name for the site object to be deleted MUST be constructed according to the format specified in section [2.2.1](#), where "<site name>" is the value of *iDirectoryObject.Name*.
4. A [Delete Object Using LDAP \(section 3.1.6.13\)](#) event MUST be generated with the following arguments:
 - *iPath* := *DeleteDN*
 - *iGuid* := *iDirectoryObject.Identifier*
5. If the Delete Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Delete Object Using LDAP event, and processing MUST end.
6. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.2.3.3 Postprocessing

None.

3.1.6.2.4 RoutingLink

3.1.6.2.4.1 Preconditions

The arguments supplied to the [Delete Directory Object \(section 3.1.6.2\)](#) event MUST meet the following condition:

- At least one of *iDirectoryObject.FullPath* or *iDirectoryObject.Identifier* MUST be populated.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing of the event MUST end.

3.1.6.2.4.2 Delete

1. A [Delete Object Using LDAP \(section 3.1.6.13\)](#) event MUST be generated with the following arguments:
 - *iPath* := *iDirectoryObject.FullPath*
 - *iGuid* := *iDirectoryObject.Identifier*
2. If the Delete Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Delete Object Using LDAP event, and processing MUST end.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.2.4.3 Postprocessing

None.

3.1.6.3 Read Directory

This event MUST be generated with the following arguments:

- *iDirectoryObjectType*: A string that specifies the name of the sub-type of **DirectoryObject** ADM element to be read from the directory.
- *iFilter*: An array of **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20) where the valid attributes are the set of directory ADM attributes associated with a **DirectoryObject** ADM element instance of type *iDirectoryObjectType*. Each **DirectoryObject** ADM element instance that is returned by the read operation MUST satisfy all attribute-filter expressions in this array.
- *iAttributeList (Optional)*: An array of strings containing the names of the directory attributes associated with a **DirectoryObject** ADM element instance of type *iDirectoryObjectType* to be read from the directory and included in the *rDirectoryObject* return value. If this argument is not supplied, all directory ADM attributes are read.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.
- *rDirectoryObject*: A **DirectoryObject** ADM element instance of type *iDirectoryObjectType* populated with the attributes specified in *iAttributeList*, the attributes of which match the expression specified by *iFilter*. This value is undefined if *rStatus* does not equal **DirectoryOperationResult.Success**.

The algorithm MUST perform the following operations to process this event:

1. If the *iFilter* specified contains more than one element, or if it contains one element and that element is not of any of the forms listed in the following table, perform the steps specified in

section [3.1.6.3.1](#) and return the *rStatus* value specified in that section. The directory contents MUST NOT be modified by this operation.

iFilter form
"Identifier" EQUALS <i>aGuid</i>
"FullPath" EQUALS <i>aDistinguishedName</i>

2. Let *UseDN* be a string variable initialized to be empty. If the single element in *iFilter* is of the form "FullPath" EQUALS *aDistinguishedName*, set the value of *UseDN* to *aDistinguishedName*.
3. Let *UseGuid* be a GUID variable initialized to be empty. If the single element in *iFilter* is of the form "Identifier" EQUALS *aGuid*, set the value of *UseGuid* to *aGuid*.
4. If *iDirectoryObjectType* has the value **ConnectedNetwork**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.
5. Perform the steps of the read operation for the type of ADM element indicated by *iDirectoryObjectType*, as specified in the following sections, and return the *rStatus* and *rDirectoryObject* values specified in that section. The directory contents must NOT be modified by this operation.
 - [QueueManager \(section 3.1.6.3.2\)](#)
 - [Queue \(section 3.1.6.3.3\)](#)
 - [Enterprise \(section 3.1.6.3.4\)](#)
 - [Site \(section 3.1.6.3.5\)](#)
 - [RoutingLink \(section 3.1.6.3.6\)](#)
 - [User \(section 3.1.6.3.7\)](#)

3.1.6.3.1 Search For One Object

1. Generate a [Read Directory Begin \(section 3.1.6.4\)](#) event with the following arguments:
 - *iDirectoryObjectType* := *iDirectoryObjectType*
 - *iFilter* := *iFilter*
 - *iAttributeList* := *iAttributeList*
 - *iAttributeSortOrder* := Null
2. If the *rStatus* value returned by the Read Directory Begin event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Read Directory Begin event, and processing MUST end.
3. Let *IteratorHandle* be a **HANDLE** ([\[MS-DTYP\]](#) section 2.2.16) that MUST be initialized to the value returned by the Read Directory Begin (section 3.1.6.4) event in *rQueryHandle*.
4. Generate a [Read Directory Next \(section 3.1.6.5\)](#) event with the following arguments:
 - *iQueryHandle* := *IteratorHandle*

5. If the *rStatus* value returned by the Read Directory Next event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Read Directory Next event, and processing MUST end.
6. Let *DirObj* be a **DirectoryObject** ADM element instance that MUST be initialized to the value returned by the Read Directory Next event in *rDirectoryObject*.
7. Generate a [Read Directory End \(section 3.1.6.6\)](#) event with the following arguments:
 - *iQueryHandle* := *IteratorHandle*
8. If the *rStatus* value returned by the Read Directory End event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Read Directory End event, and processing MUST end.
9. *rStatus* MUST be set to **DirectoryOperationResult.Success**, *rDirectoryObject* MUST be set to the value of *DirObj*, and processing MUST end.

3.1.6.3.2 QueueManager

3.1.6.3.2.1 Preconditions

The arguments supplied to the [Read Directory \(section 3.1.6.3\)](#) event MUST meet the following requirement:

- If *iAttributeList* is provided, it MUST NOT contain the name **ConnectedNetworkIdentifierList**. This attribute is not supported in **Active Directory**-based environments.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.3.2.2 Read

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **QueueManager** ADM element attributes listed in section [3.1.6.20.1](#), except **ConnectedNetworkIdentifierList**.
2. If any of the ADM attribute names **QualifiedComputerName**, **PublicSigningKeyList**, **OperatingSystemVersion**, or **Clustered** appears in *iAttributeList*, the ADM attribute name **FullPath** MUST be added to *iAttributeList* if it is not already present.
3. If the ADM attribute name **DirectoryServerType** appears in *iAttributeList*, the ADM attribute name **DirectoryServer** MUST be added to *iAttributeList* if it is not already present.
4. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iADMAAttributeList* := *iAttributeList*
5. Let *ConfigurationAttributeList* be a list of mSMQConfiguration attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event.
6. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iGuid* := *UseGuid*
 - *iPath* := *UseDN*

- *iAttributes* := *ConfigurationAttributeList*
7. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rDirectoryObject* is undefined, and processing MUST end.
 8. A [Create ADM Element From LDAP Values \(section 3.1.6.11\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iADMAAttributes* := *iAttributeList*
 - *iLDAPAttributeList* := *ConfigurationAttributeList*
 - *iValues* := *rValues* returned by the Get Object Properties Using LDAP event
 9. *rStatus* MUST be set to the *rStatus* returned by the Create ADM Element From LDAP Values event. *rDirectoryObject* MUST be set to the *rDirectoryObject* returned by the Create ADM Element From LDAP Values event. Processing MUST end.

3.1.6.3.3 Queue

3.1.6.3.3.1 Preconditions

None.

3.1.6.3.3.2 Read

1. If *iAttributeList* is not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **Queue** ADM element attributes listed in section [3.1.6.20.2](#).
2. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Queue"
 - *iADMAAttributeList* := *iAttributeList*
3. Let *QueueAttributeList* be a list of mSMQQueue attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event.
4. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iGuid* := *UseGuid*
 - *iPath* := *UseDN*
 - *iAttributes* := *QueueAttributeList*
5. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rDirectoryObject* is undefined, and processing MUST end.
6. A [Create ADM Element From LDAP Values \(section 3.1.6.11\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Queue"
 - *iADMAAttributes* := *iAttributeList*

- *iLDAPAttributeList* := *QueueAttributeList*
 - *iValues* := *rValues* returned by the Get Object Properties Using LDAP event
7. *rStatus* MUST be set to the *rStatus* returned by the Create ADM Element From LDAP Values event. *rDirectoryObject* MUST be set to the *rDirectoryObject* returned by the Create ADM Element From LDAP Values event. Processing MUST end.

3.1.6.3.4 Enterprise

3.1.6.3.4.1 Preconditions

None.

3.1.6.3.4.2 Read

1. If *iAttributeList* is not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **Enterprise** ADM element attributes listed in section [3.1.6.20.3](#).
2. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Enterprise"
 - *iADMAAttributeList* := *iAttributeList*
3. Let *EnterpriseAttributeList* be a list of mSMQEnterpriseSettings attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event.
4. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iGuid* := *UseGuid*
 - *iPath* := *UseDN*
 - *iAttributes* := *EnterpriseAttributeList*
5. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rDirectoryObject* is undefined, and processing MUST end.
6. A [Create ADM Element From LDAP Values \(section 3.1.6.11\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Enterprise"
 - *iADMAAttributes* := *iAttributeList*
 - *iLDAPAttributeList* := *EnterpriseAttributeList*
 - *iValues* := *rValues* returned by the Get Object Properties Using LDAP event
7. *rStatus* MUST be set to the *rStatus* returned by the Create ADM Element From LDAP Values event. *rDirectoryObject* MUST be set to the *rDirectoryObject* returned by the Create ADM Element From LDAP Values event. Processing MUST end.

3.1.6.3.5 Site

3.1.6.3.5.1 Preconditions

The arguments supplied to the [Read Directory \(section 3.1.6.3\)](#) event MUST meet the following requirement:

- If *iAttributeList* is provided, it MUST NOT contain the names **PrimarySiteController** or **PublicSigningKey**. These **Site** ADM element attributes are not supported in **Active Directory**-based environments.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.3.5.2 Read

1. If *iAttributeList* is not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **Site** ADM element attributes listed in section [3.1.6.20.4](#), except **PrimarySiteController** and **PublicSigningKey**.
2. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iADMAAttributeList* := *iAttributeList*
3. Let *SiteAttributeList* be a list of site attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event.
4. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iGuid* := *UseGuid*
 - *iPath* := *UseDN*
 - *iAttributes* := *SiteAttributeList*
5. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rDirectoryObject* is undefined, and processing MUST end.
6. A [Create ADM Element From LDAP Values \(section 3.1.6.11\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iADMAAttributes* := *iAttributeList*
 - *iLDAPAttributeList* := *SiteAttributeList*
 - *iValues* := *rValues* returned by the Get Object Properties Using LDAP event
7. *rStatus* MUST be set to the *rStatus* returned by the Create ADM Element From LDAP Values event. *rDirectoryObject* MUST be set to the *rDirectoryObject* returned by the Create ADM Element From LDAP Values event. Processing MUST end.

3.1.6.3.6 RoutingLink

3.1.6.3.6.1 Preconditions

None.

3.1.6.3.6.2 Read

1. If *iAttributeList* is not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **RoutingLink** ADM element attributes listed in section [3.1.6.20.5](#).
2. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "RoutingLink"
 - *iADMAAttributeList* := *iAttributeList*
3. Let *SiteLinkAttributeList* be a list of mSMQSiteLink attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event.
4. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iGuid* := *UseGuid*
 - *iPath* := *UseDN*
 - *iAttributes* := *SiteLinkAttributeList*
5. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rDirectoryObject* is undefined, and processing MUST end.
6. A [Create ADM Element From LDAP Values \(section 3.1.6.11\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "RoutingLink"
 - *iADMAAttributes* := *iAttributeList*
 - *iLDAPAttributeList* := *SiteLinkAttributeList*
 - *iValues* := *rValues* returned by the Get Object Properties Using LDAP event
7. *rStatus* MUST be set to the *rStatus* returned by the Create ADM Element From LDAP Values event. *rDirectoryObject* MUST be set to the *rDirectoryObject* returned by the Create ADM Element From LDAP Values event. Processing MUST end.

3.1.6.3.7 User

3.1.6.3.7.1 Preconditions

None.

3.1.6.3.7.2 Read

1. If *iAttributeList* is not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **User** ADM element attributes listed in section [3.1.6.20.6](#).
2. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "User"
 - *iADMAAttributeList* := *iAttributeList*

3. Let *UserAttributeList* be a list of user attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event.
4. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iGuid* := *UseGuid*
 - *iPath* := *UseDN*
 - *iAttributes* := *UserAttributeList*
5. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rDirectoryObject* is undefined, and processing MUST end.
6. A [Create ADM Element From LDAP Values \(section 3.1.6.11\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "User"
 - *iADMAAttributes* := *iAttributeList*
 - *iLDAPAttributeList* := *UserAttributeList*
 - *iValues* := *rValues* returned by the Get Object Properties Using LDAP event
7. *rStatus* MUST be set to the *rStatus* returned by the Create ADM Element From LDAP Values event. *rDirectoryObject* MUST be set to the *rDirectoryObject* returned by the Create ADM Element From LDAP Values event. Processing MUST end.

3.1.6.4 Read Directory Begin

This event MUST be generated with the following arguments:

- *iDirectoryObjectType*: A string that specifies the name of the sub-type of **DirectoryObject** ADM element to be read from the directory.
- *iFilter*: An array of **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20), where the valid attributes comprise the set of directory attributes associated with a **DirectoryObject** ADM element instance of type *iDirectoryObjectType*. Each object that is returned by the read operation MUST satisfy all attribute-filter expressions in this array.
- *iAttributeList* (Optional): An array of strings containing the names of the directory attributes associated with a **DirectoryObject** ADM element instance of type *iDirectoryObjectType* to be read from the directory and included in the *rDirectoryObject* return value. If this argument is not supplied, all directory attributes are read.
- *iAttributeSortOrder* (Optional): An array of **attribute-sort-order values** (as specified following). The elements of this array have a one-to-one correspondence with the elements of *iAttributeList* array. This array represents the precedence that the corresponding attributes in the *iAttributeList* take when ordering the returned results.

An **attribute-sort-order** value consists of the following:

- **SortPriority**: A numeric value that indicates the order in which results would be sorted with respect to the corresponding attribute. Attributes that have a higher sort priority are given higher precedence when sorting the result set. A value of 0 indicates that the corresponding attribute is not used when sorting the results.

- **SortAscending**: A Boolean value that indicates whether the corresponding attribute will be sorted in ascending order.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.
- *rQueryHandle*: A **HANDLE** ([\[MS-DTYP\]](#) section 2.2.16) that the invoker can use to iterate through the result set. This value is undefined if *rStatus* does not equal **DirectoryOperationResult.Success**.

The algorithm MUST perform the following actions to process this event:

- If *iDirectoryObjectType* has the value "ConnectedNetwork", *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rQueryHandle* is undefined, and processing MUST end.
- Let *ReadIterator* be a [ReadDirectoryIteratorState \(section 3.1.1.2\)](#) ADM element instance, which MUST be created. The **ReadDirectoryIteratorState** ADM element instance referenced by *ReadIterator* MUST be added to the [ReadDirectoryIteratorStateCollection \(section 3.1.1.3\)](#) ADM element. *ReadIterator.Handle* MUST be set to a value that is unique within the **ReadDirectoryIteratorStateCollection** ADM element. The value of *iDirectoryObjectType* MUST be copied to *ReadIterator.DataElementType*.
- Based on *iDirectoryObjectType*, perform the steps of the read-begin operation for that type of ADM element, as specified in the following sections, and return the *rStatus* value specified in that section. The directory contents must NOT be modified by this operation.
 - [QueueManager \(section 3.1.6.4.1\)](#)
 - [Queue \(section 3.1.6.4.2\)](#)
 - [Enterprise \(section 3.1.6.4.3\)](#)
 - [Site \(section 3.1.6.4.4\)](#)
 - [RoutingLink \(section 3.1.6.4.5\)](#)
 - [User \(section 3.1.6.4.6\)](#)
- If the value of *rStatus* is not **DirectoryOperationResult.Success**, the **ReadDirectoryIteratorState** ADM element instance referenced by *ReadIterator* MUST be removed from the **ReadDirectoryIteratorStateCollection** ADM element and discarded, *rQueryHandle* is undefined, and processing MUST end.
- If the value of *rStatus* is **DirectoryOperationResult.Success**, *rQueryHandle* MUST be set to the value of *ReadIterator.Handle*, and processing MUST end.

3.1.6.4.1 QueueManager

3.1.6.4.1.1 Preconditions

The arguments supplied to the [Read Directory Begin \(section 3.1.6.4\)](#) event MUST meet the following requirement:

- If *iAttributeList* is provided, it MUST NOT contain the name **ConnectedNetworkIdentifierList**. This attribute is not supported in **Active Directory**-based environments.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.4.1.2 Read Begin

1. If *iAttributeList* is not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **QueueManager** ADM element attributes listed in section [3.1.6.20.1](#), except **ConnectedNetworkIdentifierList**.
2. If any of the ADM attribute names **QualifiedComputerName**, **PublicSigningKeyList**, **OperatingSystemVersion**, or **Clustered** appears in *iAttributeList*, the ADM attribute name **FullPath** MUST be added to *iAttributeList* if it is not already present.
3. If the ADM attribute name **DirectoryServerType** appears in *iAttributeList*, the attribute name **DirectoryServer** MUST be added to *iAttributeList* if it is not already present.
4. The value of *iAttributeList* MUST be copied to *ReadIterator.AttributeList*.
5. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iADMAAttributeList* := *iAttributeList*
6. Let *ConfigurationAttributeList* be a list of mSMQConfiguration attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event. The populated *ConfigurationAttributeList* MUST be copied to *ReadIterator.LDAPAttributeList*. If any ADM attribute present in *iAttributeList* does not appear in the *rLDAPAttributesList* returned by the Create LDAP Attribute List event, the entry in *iAttributeSortOrder* corresponding to that *iAttributeList* entry, if any, MUST be removed.
7. Let *ConfigurationFilterList* be a list of attribute-filter expressions modified to refer to mSMQConfiguration attributes and values, initialized to be empty. For each expression in *iFilter*, add an expression to *ConfigurationFilterList* that consists of:
 - The mSMQConfiguration attribute that corresponds to the **QueueManager** ADM element attribute in the original expression, as shown in the following table.
 - The same operator as the original expression.
 - A value transformed from the value in the original expression as shown in the following table.

If the **QueueManager** ADM element attribute in the original expression does not appear in the following table, that expression MUST be ignored.

QueueManager ADM element attribute	mSMQConfiguration attribute	Value transformation
Identifier	objectGUID ([MS-ADA3] section 2.44)	Copy
QueueManagerVersion	mSMQComputerTypeEx ([MS-ADA2] section 2.549)	Copy
OperatingSystemType	mSMQOSType ([MS-ADA2] section 2.572)	See the table in section 3.1.6.1.1.4 .
CreateTime	whenCreated ([MS-ADA3] section 2.371)	Converts to midnight (00:00:00), January 1, 1970 UTC + CreateTime seconds.
ModifyTime	whenChanged ([MS-ADA3] section 2.370)	Converts to midnight (00:00:00), January 1, 1970 UTC + ModifyTime

QueueManager ADM element attribute	mSMQConfiguration attribute	Value transformation
		seconds.
QueueManagerQuota	mSMQQuota ([MS-ADA2] section 2.582)	Copy
JournalQuota	mSMQJournalQuota ([MS-ADA2] section 2.564)	Copy
ForeignSystem	mSMQForeign ([MS-ADA2] section 2.559)	TRUE converts to 0x01; FALSE converts to 0x00.
FullPath	distinguishedName ([MS-ADA1] section 2.177)	Copy
SiteIdentifierList	mSMQSites ([MS-ADA2] section 2.598)	Copy
RoutingServer	mSMQRoutingServices ([MS-ADA2] section 2.584))	TRUE converts to 0x01; FALSE converts to 0x00.
DirectoryServer	mSMQDsServices ([MS-ADA2] section 2.557)	TRUE converts to 0x01; FALSE converts to 0x00.
SupportingServer	mSMQDependentClientServices ([MS-ADA2] section 2.553)	TRUE converts to 0x01; FALSE converts to 0x00.
PublicEncryptionKeyList	mSMQEncryptKey ([MS-ADA2] section 2.558)	Copy
Security	nTSecurityDescriptor ([MS-ADA3]section 2.37)	Copy

8. A [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be generated with the following arguments:
 - *iObjectClass* := mSMQConfiguration
 - *iFilter* := ConfigurationFilterList
 - *iAttributes* := ConfigurationAttributeList
9. If the Search Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success** and *rStatus* is set to **DirectoryOperationResult.ObjectNotFound**, *ReadIterator.LDAPState* MUST be set to an empty list, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.
10. If the search fails for any other reason, *rStatus* MUST be set to the *rStatus* returned by the Search Using LDAP event, and processing MUST end.

3.1.6.4.1.3 Postprocessing

1. If *iAttributeSortOrder* is supplied, the sublists in the *rValues* list returned by the [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be sorted according to the priorities and orders specified in *iAttributeSortOrder*, as specified at the start of section [3.1.6.4](#), where the priorities and orders apply to the corresponding attributes named in *QueueAttributeList*.
2. The sorted *rValues* list MUST be stored in *ReadIterator.LDAPState*.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.4.2 Queue

3.1.6.4.2.1 Preconditions

None.

3.1.6.4.2.2 Read Begin

1. If *iAttributeList* was not provided as an argument, then an *iAttributeList* MUST be constructed that MUST contain the names of all **Queue** ADM element attributes listed in section [3.1.6.20.2](#).
2. The value of *iAttributeList* MUST be copied to *ReadIterator.AttributeList*.
3. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Queue"
 - *iADMAAttributeList* := *iAttributeList*
4. Let *QueueAttributeList* be a list of mSMQQueue attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event. The populated *QueueAttributeList* MUST be copied to *ReadIterator.LDAPAttributeList*. If any ADM attribute present in *iAttributeList* does not appear in the *rLDAPAttributesList* returned by the Create LDAP Attribute List event, the entry in *iAttributeSortOrder* corresponding to that *iAttributeList* entry, if any, MUST be removed.
5. Let *QueueFilterList* be a list of attribute-filter expressions modified to refer to mSMQQueue attributes and values, initialized to be empty. For each expression in *iFilter*, add an expression to *QueueFilterList* that consists of:
 - The mSMQQueue attribute that corresponds to the **Queue** ADM element attribute in the original expression, as shown in the following table.
 - The same operator as the original expression.
 - A value transformed from the value in the original expression as shown in the following table.

If the **Queue** ADM element attribute in the original expression does not appear in the following table, the expression MUST be ignored.

Queue ADM element attribute	mSMQQueue attribute	Value transformation
Identifier	objectGUID ([MS-ADA3] section 2.44)	Copy
Label	mSMQLabelEx ([MS-ADA2] section 2.566)	Copy
CreateTime	whenCreated ([MS-ADA3] section 2.371)	Converts to midnight (00:00:00), January 1, 1970 UTC + CreateTime seconds.
ModifyTime	whenChanged ([MS-ADA3] section 2.370)	Converts to midnight (00:00:00), January 1, 1970 UTC + ModifyTime seconds.
Type	mSMQQueueType ([MS-ADA2] section 2.581)	Copy
Journaling	mSMQJournal ([MS-ADA2] section 2.582)	TRUE converts to 1; FALSE converts to 0

Queue ADM element attribute	mSMQQueue attribute	Value transformation
	2.563)	converts to 0.
Quota	mSMQQueueQuota ([MS-ADA2] section 2.580)	Copy
JournalQuota	mSMQQueueJournalQuota ([MS-ADA2] section 2.564)	Copy
Authentication	mSMQAuthenticate ([MS-ADA2] section 2.546)	TRUE converts to 1; FALSE converts to 0.
PrivacyLevel	mSMQPrivacyLevel ([MS-ADA2] section 2.576)	See the table in section 3.1.6.1.2.4 .
Transactional	mSMQTransactional ([MS-ADA2] section 2.599)	TRUE converts to 1; FALSE converts to 0.
MulticastAddress	MSMQ-MulticastAddress ([MS-ADA2] section 2.543)	Copy
Security	nTSecurityDescriptor ([MS-ADA3] section 2.37)	Copy
BasePriority	mSMQBasePriority ([MS-ADA2] section 2.547)	Copy
FullPath	distinguishedName ([MS-ADA1] section 2.177)	Copy

6. A [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be generated with the following arguments:
 - *iObjectClass* := mSMQQueue
 - *iFilter* := QueueFilterList
 - *iAttributes* := QueueAttributeList
7. If the Search Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success** and *rStatus* is set to **DirectoryOperationResult.ObjectNotFound**, *ReadIterator.LDAPState* MUST be set to an empty list, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.
8. If the search fails for any other reason, *rStatus* MUST be set to the *rStatus* returned by the Search Using LDAP event, and processing MUST end.

3.1.6.4.2.3 Postprocessing

1. If *iAttributeSortOrder* is supplied, the sublists in the *rValues* list returned by the [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be sorted according to the priorities and orders specified in *iAttributeSortOrder*, as specified at the start of section [3.1.6.4](#), where the priorities and orders apply to the corresponding attributes named in *QueueAttributeList*.
2. The sorted *rValues* list MUST be stored in *ReadIterator.LDAPState*.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.4.3 Enterprise

3.1.6.4.3.1 Preconditions

None.

3.1.6.4.3.2 Read Begin

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **Enterprise** ADM element attributes listed in section [3.1.6.20.3](#).
2. The value of *iAttributeList* MUST be copied to *ReadIterator.AttributeList*.
3. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Enterprise"
 - *iADMAAttributeList* := *iAttributeList*
4. Let *EnterpriseAttributeList* be a list of mSMQEnterpriseSettings attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event. The populated *EnterpriseAttributeList* MUST be copied to *ReadIterator.LDAPAttributeList*. If any ADM attribute present in *iAttributeList* does not appear in the *rLDAPAttributesList* returned by the Create LDAP Attribute List event, the entry in *iAttributeSortOrder* corresponding to that *iAttributeList* entry, if any, MUST be removed.
5. Let *EnterpriseFilterList* be a list of attribute-filter expressions modified to refer to mSMQEnterpriseSettings attributes and values, initialized to be empty. For each expression in *iFilter*, add an expression to *EnterpriseFilterList* that consists of:
 - The mSMQEnterpriseSettings attribute that corresponds to the **Enterprise** ADM element attribute in the original expression, as shown in the following table.
 - The same operator as the original expression.
 - A value transformed from the value in the original expression as shown in the following table.

If the **Enterprise** ADM element attribute in the original expression does not appear in the following table, the expression MUST be ignored.

Enterprise ADM element attribute	mSMQEnterpriseSettings attribute	Value transformation
Identifier	objectGUID ([MS-ADA3] section 2.44)	Copy
WeakenedSecurity	mSMQNameStyle ([MS-ADA2] section 2.569)	See section 3.1.6.4.3.4 .
NonLDAPCapableQueueManagerNotification	mSMQCSPName ([MS-ADA2] section 2.551)	See section 3.1.6.4.3.5 .
DefaultTimeToLive	mSMQLongLived ([MS-ADA2] section 2.567)	Copy
OldDirectory	mSMQVersion ([MS-ADA2] section 2.601)	TRUE converts to 3; FALSE converts to 200.
Security	nTSecurityDescriptor ([MS-ADA3] section 2.37)	Copy

6. A [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be generated with the following arguments:

- *iObjectClass* := mSMQEnterpriseSettings
 - *iFilter* := EnterpriseFilterList
 - *iAttributes* := EnterpriseAttributeList
7. If the Search Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success** and *rStatus* is set to **DirectoryOperationResult.ObjectNotFound**, *ReadIterator.LDAPState* MUST be set to an empty list, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.
 8. If the search fails for any other reason, *rStatus* MUST be set to the *rStatus* returned by the Search Using LDAP event, and processing MUST end.

3.1.6.4.3.3 Postprocessing

1. If *iAttributeSortOrder* is supplied, the sublists in the *rValues* list returned by the [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be sorted according to the priorities and orders specified in *iAttributeSortOrder*, as specified at the start of section [3.1.6.4](#), where the priorities and orders apply to the corresponding attributes named in *EnterpriseAttributeList*.
2. The sorted *rValues* list MUST be stored in *ReadIterator.LDAPState*.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.4.3.4 WeakenedSecurity

The mSMQNameStyle attribute is an integer that MUST have one of the values listed in the following table. The WeakenedSecurity attribute is a Boolean. The values MUST be converted according to this table.

mSMQNameStyle value	WeakenedSecurity value
0x00	FALSE
0x01	TRUE

3.1.6.4.3.5 NonLDAPCapableQueueManagerNotification

The mSMQCSPName attribute is a string that MUST have one of the values listed in the following table. The attribute is a Boolean. The values MUST be converted according to this table.

mSMQCSPName value	NonLDAPCapableQueueManagerNotification ADM attribute value
"Y"	true
"N"	false

3.1.6.4.4 Site

3.1.6.4.4.1 Preconditions

The arguments supplied to the [Read Directory Begin \(section 3.1.6.4\)](#) event MUST meet the following requirement:

- If *iAttributeList* is provided, it MUST NOT contain the names **PrimarySiteController** or **PublicSigningKey**. These **Site** ADM element attributes are not supported in **Active Directory**-based environments.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.4.4.2 Read Begin

1. If *iAttributeList* was not provided as an argument, then an *iAttributeList* MUST be constructed that MUST contain the names of all **Site** ADM element attributes listed in section [3.1.6.20.4](#), except **PrimarySiteController** and **PublicSigningKey**.
2. The value of *iAttributeList* MUST be copied to *ReadIterator.AttributeList*.
3. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iADMAAttributeList* := *iAttributeList*
4. Let *SiteAttributeList* be a list of site attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event. The populated *SiteAttributeList* MUST be copied to *ReadIterator.LDAPAttributeList*. If any ADM attribute present in *iAttributeList* does not appear in the *rLDAPAttributesList* returned by the Create LDAP Attribute List event, the entry in *iAttributeSortOrder* corresponding to that *iAttributeList* entry, if any, MUST be removed.
5. Let *SiteFilterList* be a list of attribute-filter expressions modified to refer to site attributes and values, initialized to be empty. For each expression in *iFilter*, add an expression to *SiteFilterList* that consists of:
 - The site attribute that corresponds to the **Site** ADM element attribute in the original expression, as shown in the following table.
 - The same operator as the original expression.
 - A value transformed from the value in the original expression as shown in the following table.

If the **Site** ADM element attribute in the original expression does not appear in the following table, the expression MUST be ignored.

Site ADM element attribute	site attribute	Value transformation
Identifier	objectGUID ([MS-ADA3] section 2.44)	Copy
Name	cn ([MS-ADA1] section 2.110)	Copy
IntraSiteReplicationInterval	mSMQInterval1 ([MS-ADA2] section 2.561)	Copy
InterSiteReplicationInterval	mSMQInterval2 ([MS-ADA2] section 2.562)	Copy
FullPath	distinguishedName ([MS-ADA1] section 2.177)	Copy
Foreign	mSMQSiteForeign ([MS-ADA2] section 2.592)	TRUE converts to 0x01; FALSE converts to 0x00.

Site ADM element attribute	site attribute	Value transformation
MigratedFromMsmq10	mSMQnt4Stub ([MS-ADA2] section 2.571)	TRUE converts to 0x01; FALSE converts to 0x00.
Security	nTSecurityDescriptor ([MS-ADA3] section 2.37)	Copy

6. A [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be generated with the following arguments:
 - *iObjectClass* := site
 - *iFilter* := *SiteFilterList*
 - *iAttributes* := *SiteAttributeList*
7. If the Search Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success** and *rStatus* is set to **DirectoryOperationResult.ObjectNotFound**, *ReadIterator.LDAPState* MUST be set to an empty list, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.
8. If the search fails for any other reason, *rStatus* MUST be set to the *rStatus* returned by the Search Using LDAP event, and processing MUST end.

3.1.6.4.4.3 Postprocessing

1. If *iAttributeSortOrder* is supplied, the sublists in the *rValues* list returned by the [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be sorted according to the priorities and orders specified in *iAttributeSortOrder*, as specified at the start of section [3.1.6.4](#), where the priorities and orders apply to the corresponding attributes named in *SiteAttributeList*.
2. The sorted *rValues* list MUST be stored in *ReadIterator.LDAPState*.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.4.5 RoutingLink

3.1.6.4.5.1 Preconditions

None.

3.1.6.4.5.2 Read Begin

1. If *iAttributeList* was not provided as an argument, then an *iAttributeList* MUST be constructed that MUST contain the names of all **RoutingLink** ADM element attributes listed in section [3.1.6.20.5](#).
2. The value of *iAttributeList* MUST be copied to *ReadIterator.AttributeList*.
3. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "RoutingLink"
 - *iADMAAttributeList* := *iAttributeList*
4. Let *SiteLinkAttributeList* be a list of mSMQSiteLink attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event. The populated *SiteLinkAttributeList* MUST be copied to *ReadIterator.LDAPAttributeList*. If any ADM attribute present in *iAttributeList* does not appear in the *rLDAPAttributesList* returned by the

Create LDAP Attribute List event, the entry in *iAttributeSortOrder* corresponding to that *iAttributeList* entry, if any, MUST be removed.

5. Let *SiteLinkFilterList* be a list of attribute-filter expressions modified to refer to mSMQSiteLink attributes and values, initialized to be empty. For each expression in *iFilter*, add an expression to *SiteLinkFilterList* that consists of:
 - The mSMQSiteLink attribute that corresponds to the **RoutingLink** ADM element attribute in the original expression, as shown in the following table.
 - The same operator as the original expression.
 - A value transformed from the value in the original expression as shown in the following table.

If the **RoutingLink** ADM element attribute in the original expression does not appear in the following table, the expression MUST be ignored.

RoutingLink ADM element attribute	mSMQSiteLink attribute	Value transformation
Identifier	objectGUID ([MS-ADA3] section 2.44)	Copy
Description	description ([MS-ADA1] section 2.153)	Copy
FullPath	distinguishedName ([MS-ADA1] section 2.177)	Copy
ActualCost	mSMQCost ([MS-ADA2] section 2.550)	Copy
Site1Identifier	mSMQSite1 ([MS-ADA2] section 2.590)	See section 3.1.6.4.5.4.
Site2Identifier	mSMQSite2 ([MS-ADA2] section 2.591)	See section 3.1.6.4.5.5.

6. A [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be generated with the following arguments:
 - *iObjectClass* := mSMQSiteLink
 - *iFilter* := *SiteLinkFilterList*
 - *iAttributes* := *SiteLinkAttributeList*
7. If the Search Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success** and *rStatus* is set to **DirectoryOperationResult.ObjectNotFound**, *ReadIterator.LDAPState* MUST be set to an empty list, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.
8. If the search fails for any other reason, *rStatus* MUST be set to the *rStatus* returned by the Search Using LDAP event, and processing MUST end.

3.1.6.4.5.3 Postprocessing

1. If *iAttributeSortOrder* is supplied, the sublists in the *rValues* list returned by the [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be sorted according to the priorities and orders specified in *iAttributeSortOrder*, as specified at the start of section 3.1.6.4, where the priorities and orders apply to the corresponding attributes named in *SiteLinkAttributeList*.
2. The sorted *rValues* list MUST be stored in *ReadIterator.LDAPState*.

3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.4.5.4 Site1Identifier Filtering

The filter value to be used when filtering on the **Site1Identifier** ADM attribute MUST be computed according to the following algorithm:

1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the value of the original filter value for **Site1Identifier**
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **FullPath**
2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**:
 - *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
3. The filter value for the **Site1Identifier** ADM attribute MUST be the value of the *rDirectoryObject.FullPath* returned by the Read Directory event.

3.1.6.4.5.5 Site2Identifier Filtering

The filter value to be used when filtering on the **Site2Identifier** ADM attribute MUST be computed according to the following algorithm:

1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the value of the original filter value for **Site2Identifier**
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **FullPath**
2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**:
 - *rStatus* must be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
3. The filter value for **Site2Identifier** ADM attribute MUST be the value of the *rDirectoryObject.FullPath* returned by the Read Directory event.

3.1.6.4.6 User

3.1.6.4.6.1 Preconditions

None.

3.1.6.4.6.2 Read Begin

1. If *iAttributeList* was not provided as an argument, then an *iAttributeList* MUST be constructed that MUST contain the names of all **User** ADM element attributes listed in section [3.1.6.20.6](#).
2. The value of *iAttributeList* MUST be copied to *ReadIterator.AttributeList*.
3. A [Create LDAP Attribute List \(section 3.1.6.10\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := "User"
 - *iADMAAttributeList* := *iAttributeList*
4. Let *UserAttributeList* be a list of user attribute names, initialized to be the LDAP attribute names returned in *rLDAPAttributeList* by the Create LDAP Attribute List event. If any ADM attribute present in *iAttributeList* does not appear in the *rLDAPAttributesList* returned by the Create LDAP Attribute List event, the entry in *iAttributeSortOrder* corresponding to that *iAttributeList* entry, if any, MUST be removed.
5. Let *UserFilterList* be a list of attribute-filter expressions modified to refer to user attributes and values, initialized to be empty. For each expression in *iFilter*, add an expression to *UserFilterList* that consists of:
 - The user attribute that corresponds to the **User** ADM element attribute in the original expression, as shown in the following table.
 - The same operator as the original expression.
 - A value transformed from the value in the original expression as shown in the following table.

If the **User** ADM element attribute in the original expression does not appear in the following table, the expression MUST be ignored.

User ADM element attribute	user attribute	Value transformation
Identifier	objectGUID ([MS-ADA3] section 2.44)	Copy
SecurityIdentifier	objectSid ([MS-ADA3] section 2.45)	Copy
CertificateDigestList	mSMQDigests ([MS-ADA2] section 2.554)	Copy
Certificates	mSMQSignCertificates ([MS-ADA2] section 2.587)	Copy
FullPath	distinguishedName ([MS-ADA1] section 2.177)	Copy

6. A [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be generated with the following arguments:
 - *iObjectClass* := user
 - *iFilter* := *UserFilterList*
 - *iAttributes* := *UserAttributeList*
7. If the Search Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, and *rStatus* is set to **DirectoryOperationResult.ObjectNotFound**, *ReadIterator.LDAPState* MUST be set to an empty list, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

8. If the search fails for any other reason, *rStatus* MUST be set to the *rStatus* returned by the Search Using LDAP event, and processing MUST end.

3.1.6.4.6.3 Postprocessing

1. If *iAttributeSortOrder* is supplied, the sublists in the *rValues* list returned by the [Search Using LDAP \(section 3.1.6.15\)](#) event MUST be sorted according to the priorities and orders specified in *iAttributeSortOrder*, as specified at the start of section [3.1.6.4](#), where the priorities and orders apply to the corresponding attributes named in *UserAttributeList*.
2. The sorted *rValues* list MUST be stored in *ReadIterator.LDAPState*.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.5 Read Directory Next

This event MUST be generated with the following arguments:

- *iQueryHandle*: The **HANDLE** that was generated by the [Read Directory Begin \(section 3.1.6.4\)](#) event.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.
- *rDirectoryObject*: The next **DirectoryObject** ADM element instance from the collection of **DirectoryObject** ADM element instances that match the *iFilter* expressions supplied to the Read Directory Begin event that created the supplied *iQueryHandle*. The collection is ordered based on *iAttributeSortOrder* supplied to the Read Directory Begin event that created the supplied *iQueryHandle*. This value is populated with the attributes specified in *iAttributeList* supplied to the Read Directory Begin event that created the supplied *iQueryHandle*. This value is undefined if *rStatus* does not equal **DirectoryOperationResult.Success**.

The algorithm MUST perform the following actions to process this event:

- Let *ReadIterator* be the **ReadDirectoryIteratorState** ADM element instance found in the **ReadDirectoryIteratorStateCollection** ADM element for which the value of the **Handle** ADM attribute equals the value of *iQueryHandle*. If no such **ReadDirectoryIteratorState** ADM element is found, processing MUST end. *rStatus* MUST be set to **DirectoryOperationResult.GenericError**.
- Let *NextResult* be a list of attribute values initialized by removing the first sublist from *ReadIterator.LDAPState* and placing it in *NextResult*. If *ReadIterator.LDAPState* is empty, *rStatus* MUST be set to **DirectoryOperationResult.EndOfData**, and processing MUST end.
- A [Create ADM Element From LDAP Values \(section 3.1.6.11\)](#) event MUST be generated with the following arguments:
 - *iDirectoryObjectType* := *ReadIterator.DataElementType*
 - *iADMAAttributes* := *ReadIterator.AttributeList*
 - *iLDAPAttributeList* := *ReadIterator.LDAPAttributeList*
 - *iValues* := *NextResult*
- *rStatus* MUST be set to the *rStatus* returned by the Create ADM Element From LDAP Values event. *rDirectoryObject* MUST be set to the *rDirectoryObject* returned by the Create ADM Element From LDAP Values event. Processing MUST end.

3.1.6.6 Read Directory End

This event MUST be generated with the following arguments:

- *iQueryHandle*: The **HANDLE** that was generated by the [Read Directory Begin \(section 3.1.6.4\)](#) event.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.

The algorithm MUST perform the following actions to process this event:

- Let *ReadIterator* be the **ReadDirectoryIteratorState** ADM element instance in the **ReadDirectoryIteratorStateCollection** ADM element for which the value of the **Handle** ADM attribute equals the value of *iQueryHandle*. If there is no such **ReadDirectoryIteratorState** ADM element instance in the collection, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
- The **ReadDirectoryIteratorState** ADM element instance referenced by *ReadIterator* MUST be removed from the **ReadDirectoryIteratorStateCollection** ADM element.
- The search results stored in *ReadIterator.LDAPState* MUST be discarded.
- The **ReadDirectoryIteratorState** ADM element instance referenced by *ReadIterator* MUST be discarded.
- *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.7 Write Directory

This event MUST be generated with the following arguments:

- *iDirectoryObject*: The **DirectoryObject** ADM element instance to be written to the directory.
- *iAttributeList (Optional)*: An array of strings containing the names of the directory attributes associated with *iDirectoryObject* to be updated to the directory. If this argument is not supplied, all directory attributes are updated.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.

The algorithm MUST perform the following actions to process this event:

- Determine the type of *iDirectoryObject*. If the type is **ConnectedNetwork** ([\[MS-MQDMPR\]](#) section 3.1.1.9), *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
- Perform the steps of the write operation for that type of ADM element, as specified in the sections listed following, and return the *rStatus* value specified in that section.
 - [QueueManager \(section 3.1.6.7.1\)](#)
 - [Queue \(section 3.1.6.7.2\)](#)
 - [Enterprise \(section 3.1.6.7.3\)](#)
 - [Site \(section 3.1.6.7.4\)](#)
 - [RoutingLink \(section 3.1.6.7.5\)](#)

- [User \(section 3.1.6.7.6\)](#)

3.1.6.7.1 QueueManager

3.1.6.7.1.1 Preconditions

The arguments supplied to the [Write Directory \(section 3.1.6.7\)](#) event MUST meet these requirements:

- If *iAttributeList* is provided, it MUST NOT contain the name **ConnectedNetworkIdentifierList**. This attribute is not supported in **Active Directory**-based environments.
- At least one of *iDirectoryObject.FullPath*, *iDirectoryObject.ComputerName*, or *iDirectoryObject.Identifier* MUST be populated.

If any of these conditions is violated, *rStatus* MUST be set to

DirectoryOperationResult.GenericError, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.7.1.2 Write

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **QueueManager** ADM element attributes listed in section [3.1.6.20.1](#), except **ConnectedNetworkIdentifierList**.
2. Let *ConfigurationAttributeList* be a list of mSMQConfiguration attribute names, initialized to be empty. For each mSMQConfiguration attribute listed in the following table, if the corresponding **QueueManager** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the mSMQConfiguration attribute name to *ConfigurationAttributeList* and compute the value for the attribute as shown. **QueueManager** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table MUST be ignored in this step.

mSMQConfiguration attribute	Attribute value computation
mSMQComputerTypeEx ([MS- ADA2] section 2.549)	Directory string transformed from <i>iDirectoryObject.QueueManagerVersion</i> .
mSMQOSType ([MS- ADA2] section 2.572)	Computed from <i>iDirectoryObject.OperatingSystemType</i> . See section 3.1.6.7.1.4 .
mSMQServiceType ([MS- ADA2] section 2.586)	Computed from <i>iDirectoryObject.DirectoryServer</i> , <i>iDirectoryObject.DirectoryServerType</i> , <i>iDirectoryObject.RemoteAccessServer</i> , and <i>iDirectoryObject.RoutingServer</i> . See section 3.1.6.7.1.5 .
mSMQQuota ([MS- ADA2] section 2.582)	Integer copied from <i>iDirectoryObject.QueueManagerQuota</i> .
mSMQJournalQuota ([MS- ADA2] section 2.564)	Integer copied from <i>iDirectoryObject.JournalQuota</i> .
mSMQForeign ([MS- ADA2] section 2.559)	If <i>iDirectoryObject.ForeignSystem</i> is TRUE, mSMQForeign is 0x01 Else mSMQForeign is 0x00.
mSMQSites ([MS- ADA2] section 2.598)	List of GUIDs copied from <i>iDirectoryObject.SiteIdentifierList</i> .

mSMQConfiguration attribute	Attribute value computation
mSMQOutRoutingServers ([MS-ADA2] section 2.573)	Computed from <i>iDirectoryObject</i> . OutRoutingServerIdentifierList . See section 3.1.6.7.1.6 .
mSMQInRoutingServers ([MS-ADA2] section 2.560)	Computed from <i>iDirectoryObject</i> . InRoutingServerIdentifierList . See section 3.1.6.7.1.7 .
mSMQRoutingServices ([MS-ADA2] section 2.584)	If <i>iDirectoryObject</i> . RoutingServer is TRUE, mSMQRoutingServices is 0x01 Else mSMQRoutingServices is 0x00.
mSMQDsServices ([MS-ADA2] section 2.557)	If <i>iDirectoryObject</i> . DirectoryServer is TRUE, mSMQDsServices is 0x01 Else mSMQDsServices is 0x00.
mSMQDependentClientServices ([MS-ADA2] section 2.553)	If <i>iDirectoryObject</i> . SupportingServer is TRUE, mSMQDependentClientServices is 0x01 Else mSMQDependentClientServices is 0x00.
mSMQEncryptKey ([MS-ADA2] section 2.558)	MQDSPUBLICKEYS ([MS-MQMQ] section 2.2.2) copied from <i>iDirectoryObject</i> . PublicEncryptionKeyList .

3. Let *WriteGUID* be a distinguished name, initialized to be empty. If *iDirectoryObject*.**Identifier** is populated, the value of *iDirectoryObject*.**Identifier** MUST be copied to *WriteGUID*.
4. Let *WriteDN* be a **distinguished name**, initialized to be empty.
5. If *iDirectoryObject*.**FullPath** is populated, the value of *iDirectoryObject*.**FullPath** MUST be copied to *WriteDN*.
6. If *WriteDN* is empty and *iDirectoryObject*.**ComputerName** is populated, the value of *WriteDN* MUST be a distinguished name for a mSMQConfiguration object, constructed according to the format specified in section [2.2.1](#), where "<computer name>" is the value of *iDirectoryObject*.**ComputerName**.
7. A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event MUST be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *ConfigurationAttributeList* and the corresponding values, as computed in step 2.
8. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.

3.1.6.7.1.3 Postprocessing

1. If *iAttributeList* contains the name "Security" and the *iDirectoryObject*.**Security** ADM attribute is populated, a [Set Object Security Using LDAP \(section 3.1.6.17\)](#) event MUST be generated with the following arguments:

- *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iSecurity* := *iDirectoryObject.Security*
2. If the Set Object Security Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Security Using LDAP event, and processing MUST end.
 3. If the **PublicSigningKeyList** ADM attribute is present in *iAttributeList*, a [Set Object Properties Using LDAP](#) event MUST be generated with the following arguments:
 - *iPath* := a **distinguished name** for a computer object of the format specified in section [2.2.1](#), where "<computer name>" is the value of *iDirectoryObject.ComputerName*.
 - *iAttributes* := a list of attribute name-value pairs, where the names are "mSMQSignCertificates" and "mSMQDigests", and the values are computed as specified in section [3.1.6.7.1.8](#).
 4. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.
 5. The steps in section [3.1.6.7.1.9](#) MUST be performed to determine whether any mSMQSettings objects are affected and any necessary updates MUST be performed.
 6. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.7.1.4 mSMQOSType

The mSMQOSType attribute is an integer that MUST have one of the values listed in the following table. The **OperatingSystemType** ADM attribute is an enumeration. The values MUST be converted according to this table.

OperatingSystemType ADM attribute value	mSMQOSType value
Unknown	0x00000000
Foreign	0x00000100
Windows 95	0x00000200
WinClient	0x00000300
WinServer	0x00000400
WinEnt	0x00000500

3.1.6.7.1.5 mSMQServiceType

The mSMQServiceType attribute is a bitmap that MUST be computed according to this algorithm:

1. Let *FinalValue* be a 32-bit unsigned integer, initialized to 0x00000000.
2. If *iDirectoryObject.RoutingServer* is populated and TRUE, the 0x00000001 bit of *FinalValue* MUST be set.

3. If *iDirectoryObject.RemoteAccessServer* is populated and TRUE, the 0x00000010 bit of *FinalValue* MUST be set.
4. If *iDirectoryObject.DirectoryServer* is populated and TRUE, and *iDirectoryObject.DirectoryServerType* is populated and is not set to the enumeration value **Standalone**, one bit of *FinalValue* MUST be set according to the following table.

DirectoryServerType value	FinalValue bit to be set
BackupSiteController	0x00000002
PrimarySiteController	0x00000004
PrimaryEnterpriseController	0x00000008

5. The value of the mSMQServiceType attribute MUST be the value of *FinalValue*.

3.1.6.7.1.6 mSMQOutRoutingServers

The value of the mSMQOutRoutingServers attribute MUST be computed according to the following algorithm:

1. Let *DNList* be a list of **distinguished names**, initialized to be empty.
2. For each **GUID** in *iDirectoryObject.OutRoutingServerIdentifierList*, these steps MUST be performed:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.OutRoutingServerIdentifierList*
 - *iAttributeList* := An array of the following **QueueManager** ADM element attribute names:
 - **FullPath**
 - If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, this GUID MUST be skipped.
 - If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of the *rDirectoryObject.FullPath* returned by the Read Directory event must be added to *DNList*.
3. The value of mSMQOutRoutingServers MUST be the value of *DNList*.

3.1.6.7.1.7 mSMQInRoutingServers

The value of the mSMQInRoutingServers attribute MUST be computed according to the following algorithm:

1. Let *DNList* be a list of **distinguished names**, initialized to be empty.
2. For each **GUID** in *iDirectoryObject.InRoutingServerIdentifierList*, these steps MUST be performed:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:

- *iDirectoryObjectType* := "QueueManager"
- *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.InRoutingServerIdentifierList*
- *iAttributeList* := An array of the following **QueueManager** ADM element attribute names:
 - **FullPath**
- If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, this GUID MUST be skipped.
- If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of the *rDirectoryObject.FullPath* returned by the Read Directory event must be added to *DNList*.

3. The value of *mSMQInRoutingServers* MUST be the value of *DNList*.

3.1.6.7.1.8 mSMQSignCertificates and mSMQDigests

The **PublicSigningKeyList** ADM attribute is stored in two attributes on the computer object. The *mSMQSignCertificates* ([\[MS-ADA2\]](#) section 2.587) attribute contains an **MQUERSIGNCERTS** ([\[MS-MQMQ\]](#) section 2.2.21) structure, which contains **MQUERSIGNCERT** ([\[MS-MQMQ\]](#) section 2.2.22) structures, which in turn contain individual X.509-encoded certificates. The *mSMQDigests* ([\[MS-ADA2\]](#) section 2.554) attribute contains an array of MD5 hashes of the certificates stored in the *mSMQSignCertificates* attribute, mirroring the values of the **Digest** fields of the **MQUERSIGNCERT** structures. Each array element MUST contain the 16-byte output of the MD5 algorithm, as specified in [\[RFC1321\]](#). The certificates and digests MUST be in the same order, but there is no other sorting requirement.

The values of *mSMQSignCertificates* and *mSMQDigests* MUST be computed according to the following algorithm:

If the **PublicSigningKeyList** ADM attribute name is present in *iAttributeList* and *iDirectoryObject.PublicSigningKeyList* is populated:

- Copy the bytes in *iDirectoryObject.PublicSigningKeyList*, which is an **MQUERSIGNCERTS** structure, to *mSMQSignCertificates*.
- For each **MQUERSIGNCERT** structure in *mSMQSignCertificates*:
 - Append the bytes of the **Digest** field in the structure to *mSMQDigests*.

3.1.6.7.1.9 mSMQSettings Objects

If, as a result of this write operation, one or more of the **RoutingServer**, **DirectoryServer**, or **SupportingServer** ADM attributes is TRUE where all were previously FALSE, one or more *mSMQSettings* objects MUST be created. For each **GUID** in *iDirectoryObject.SiteIdentifierList*, do the following:

1. Search the Directory for a **Site** ADM element instance that has an **Identifier** ADM attribute that matches the GUID:
 1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"

- *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.SiteIdentifierList*
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **Name**
2. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.ObjectNotFound**, the current GUID MUST be skipped.
 3. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
 2. If any of the mSMQConfiguration attributes listed in the following table does not appear in *ConfigurationAttributeList*, a [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iAttributes* := a list of attribute names consisting of the mSMQConfiguration attributes listed in the following table that do not also appear in *ConfigurationAttributeList*

mSMQSettings attribute	mSMQConfiguration attribute
mSMQQMID ([MS-ADA2] section 2.577)	objectGUID
mSMQServices ([MS-ADA2] section 2.585)	mSMQServiceType
mSMQRoutingService ([MS-ADA2] section 2.583)	mSMQRoutingServices
mSMQDsService ([MS-ADA2] section 2.556)	mSMQDsServices
mSMQDependentClientService ([MS-ADA2] section 2.552)	mSMQDependentClientServices

3. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, and processing MUST end.
4. Generate a [Create Object Using LDAP \(section 3.1.6.12\)](#) event with the following arguments:
 - *iParentPath* := a **distinguished name** that MUST be constructed of the form specified for an mSMQSettings object in section [2.2.1](#), where "<computer name>" is the value of *iDirectoryObject.ComputerName*, and "<site name>" is the value of the *rDirectoryObject.Name* returned by the Read Directory event, with the first comma-separated element removed.
 - *iChildName* := "MSMQ Settings"
 - *iObjectClass* := "mSMQSettings"
 - *iAttributes* := a list of name-value pairs consisting of the mSMQSettings attribute names listed in the table in step 2 and the corresponding mSMQConfiguration values copied from *ConfigurationAttributeList* or the values returned by the Get Object Properties Using LDAP event in step 3.

5. If the Create Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Create Object Using LDAP event, and processing MUST end.

If, as a result of this write operation, all of the **RoutingServer**, **DirectoryServer**, or **SupportingServer** ADM attributes are FALSE where previously one or more were TRUE, one or more mSMQSettings objects MUST be deleted. For each GUID in the **SiteIdentifierList** ADM attribute, do the following:

1. Search the Directory for a **Site** ADM element instance that has an **Identifier** ADM attribute that matches the GUID:
 1. Generate a Read Directory (section 3.1.6.3) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions**:
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.SiteIdentifierList*
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **Name**
 2. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.ObjectNotFound**, the current GUID MUST be skipped.
 3. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
2. A [Delete Object Using LDAP \(section 3.1.6.13\)](#) event MUST be generated with the following arguments:
 - *iPath* := a distinguished name MUST be constructed of the form specified for an mSMQSettings object in section 2.2.1, where "<computer name>" is the value of *iDirectoryObject.ComputerName*, and "<site name>" is the value of the *rDirectoryObject.Name* returned by the Read Directory event.
3. If the Delete Object Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Delete Object Using LDAP event, and processing MUST end.

If, as a result of this write operation, one or more of the **RoutingServer**, **DirectoryServer**, or **SupportingServer** ADM element attributes is TRUE, where this condition was also previously true, and one or more of those ADM attributes was written, or the **DirectoryServerType** or **RemoteAccessServer** ADM attributes were written, one or more mSMQSettings objects MUST be updated. For each GUID in the **SiteIdentifierList** ADM attribute, do the following:

1. Search the Directory for a **Site** ADM element instance that has an **Identifier** ADM attribute that matches the GUID:
 1. Generate a Read Directory event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions**:
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.SiteIdentifierList*
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:

- **Name**
2. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.ObjectNotFound**, the current GUID MUST be skipped.
 3. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
2. A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event MUST be generated with the following arguments:
 - *iPath* := a distinguished name MUST be constructed of the form specified for an mSMQSettings object in section 2.2.1, where "<computer name>" is the value of **QueueManager.ComputerName**, and "<site name>" is the value of the *rDirectoryObject.Name* returned by the Read Directory event.
 - *iAttributes* := a list of attribute name-value pairs consisting of the mSMQSettings attributes listed in the preceding table in this section for which the corresponding mSMQConfiguration attributes appear in *ConfigurationAttributeList* and the values of those mSMQConfiguration attributes.
 3. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.

3.1.6.7.1.10 PublicSigningKeyList

The **PublicSigningKeyList** ADM attribute is stored in two attributes on the computer object, as specified in section [3.1.6.11.1](#). When writing the **Certificates** ADM attribute, both mSMQSignCertificates and mSMQDigests MUST be written. In addition, the **MQUERSIGNCERTS** ([MS-MQMQ] section 2.2.21) structure contained in the mSMQSignCertificates attribute can be written only atomically; so to add, delete, or write an individual certificate, the mSMQSignCertificates attribute MUST be read first; the **MQUERSIGNCERTS** structure modified (by adding, deleting, or writing a **MQUERSIGNCERT** structure within it); then the mSMQSignCertificates attribute written with the modified structure. The appropriate operation (add, delete, or write) MUST be performed on the corresponding MD5 digest in the mSMQDigests attribute. The certificates and the digests MUST be in the same order.

3.1.6.7.2 Queue

3.1.6.7.2.1 Preconditions

The arguments supplied to the [Write Directory \(section 3.1.6.7\)](#) event MUST meet the following requirement:

- At least one of the *iDirectoryObject.FullPath*, *iDirectoryObject.DirectoryPath*, *iDirectoryObject.Identifier*, or *iDirectoryObject.Pathname* ADM attributes MUST be populated.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing of the event MUST end.

3.1.6.7.2.2 Write

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **Queue** ADM element attributes listed in section [3.1.6.20.2](#).
2. Let *QueueAttributeList* be a list of mSMQQueue attribute names, initialized to be empty. For each mSMQQueue attribute listed in the following table, if the corresponding **Queue** ADM element

attribute appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the mSMQQueue attribute name to *QueueAttributeList* and compute the value for the attribute as shown. **Queue** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table **MUST** be ignored in this step.

mSMQQueue attribute	Attribute value computation
mSMQLabelEx ([MS-ADA2] section 2.566)	Directory string transformed from <i>iDirectoryObject.Label</i> .
mSMQQueueType ([MS-ADA2] section 2.581)	GUID copied from <i>iDirectoryObject.Type</i> .
mSMQJournal ([MS-ADA2] section 2.563)	If <i>iDirectoryObject.Journaling</i> is TRUE, mSMQJournal is 1 Else mSMQJournal is 0.
mSMQQueueQuota ([MS-ADA2] section 2.580)	Integer copied from <i>iDirectoryObject.Quota</i> .
mSMQQueueJournalQuota ([MS-ADA2] section 2.564)	Integer copied from <i>iDirectoryObject.JournalQuota</i> .
mSMQAuthenticate ([MS-ADA2] section 2.546)	If <i>iDirectoryObject.Authentication</i> is TRUE, mSMQAuthenticate is 1 Else mSMQAuthenticate is 0.
mSMQPrivacyLevel ([MS-ADA2] section 2.576)	Computed from <i>iDirectoryObject.PrivacyLevel</i> . See section 3.1.6.7.2.4.
MSMQ-MulticastAddress ([MS-ADA2] section 2.543)	Directory string transformed from <i>iDirectoryObject.MulticastAddress</i> .
mSMQBasePriority ([MS-ADA2] section 2.547)	Integer copied from <i>iDirectoryObject.BasePriority</i> .

3. Let *WriteGUID* be a distinguished name, initialized to be empty. If *iDirectoryObject.Identifier* is populated, the value of *iDirectoryObject.Identifier* **MUST** be copied to *WriteGUID*.
4. Let *WriteDN* be a **distinguished name**, initialized to be empty.
5. If *iDirectoryObject.FullPath* is populated, the value of *iDirectoryObject.FullPath* **MUST** be copied to *WriteDN*.
6. If *WriteDN* is empty and *iDirectoryObject.DirectoryPath* is populated, the value of *WriteDN* **MUST** be the value of *iDirectoryObject.DirectoryPath* with the prefix "LDAP://" removed.
7. If *WriteDN* is empty and *iDirectoryObject.Pathname* is populated, the value of *WriteDN* **MUST** be a distinguished name for an mSMQQueue object, constructed according to the format specified in section 2.2.1, where "<computer name>" is the ComputerName portion of the value of *iDirectoryObject.Pathname* and "<queue name>" is the QueueName portion of the value of *iDirectoryObject.Pathname*, as specified in [MS-MQMQ] section 2.1.1, and "<queue name>" is further modified as specified in section 3.1.6.7.2.5.
8. A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event **MUST** be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*

- *iAttributes* := a list of name-value pairs consisting of the attribute names in *QueueAttributeList* and the corresponding values, as computed in step 2.
9. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.

3.1.6.7.2.3 Postprocessing

1. If *iAttributeList* contains the name "Security" and the *iDirectoryObject.Security* ADM attribute is populated, a [Set Object Security Using LDAP \(section 3.1.6.17\)](#) event MUST be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iSecurity* := *iDirectoryObject.Security*
2. If the Set Object Security Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Security Using LDAP event, and processing MUST end.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.7.2.4 PrivacyLevel

The *mSMQPrivacyLevel* attribute is an integer that MUST have one of the values listed in the following table. The **PrivacyLevel** ADM attribute is an enumeration. The values MUST be converted according to the following table.

PrivacyLevel ADM attribute value	mSMQPrivacyLevel value
None	0
Optional	1
Body	2

3.1.6.7.2.5 <queue name>

If the length of the "<queue name>" does not exceed 63 characters, it MUST be used unmodified. Otherwise, it is modified according to the following algorithm:

1. Let *QName* be a **Unicode** string, initialized to be empty.
2. The first 55 characters of the "<queue name>" MUST be copied to *QName*.
3. A nine-character extension MUST be added to the end of *QName*. *QName* MUST be exactly 64 characters long after adding the extension. The extension consists of a Unicode dash ('-') followed by an eight-character hash string calculated from the values of the Unicode characters in the "<queue name>" as specified in section [2.2.5](#).
4. The original "<queue name>" MUST be replaced with the value of *QName*.

3.1.6.7.3 Enterprise

3.1.6.7.3.1 Preconditions

There are no preconditions for the [Write Directory \(section 3.1.6.7\)](#) event for **Enterprise** ADM element objects.

3.1.6.7.3.2 Write

1. If *iAttributeList* was not provided as an argument, then an *iAttributeList* MUST be constructed that MUST contain the names of all **Enterprise** ADM element attributes listed in section [3.1.6.20.3](#).
2. Let *EnterpriseAttributeList* be a list of mSMQEnterpriseSettings attribute names, initialized to be empty. For each mSMQEnterpriseSettings attribute listed in the following table, if the corresponding **Enterprise** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the mSMQEnterpriseSettings attribute name to *EnterpriseAttributeList* and compute the value for the attribute as shown. **Enterprise** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table MUST be ignored in this step.

mSMQEnterpriseSettings attribute	Attribute value computation
mSMQNameStyle ([MS-ADA2] section 2.569)	If <i>iDirectoryObject</i> . WeakenedSecurity equals disabled, mSMQNameStyle is 0x00 Else if <i>iDirectoryObject</i> . WeakenedSecurity equals enabled, mSMQNameStyle is 0x01.
mSMQCSPName ([MS-ADA2] section 2.551)	Computed from <i>iDirectoryObject</i> . NonLDAPCapableQueueManagerNotification . See section 3.1.6.7.3.4 .
mSMQLongLived ([MS-ADA2] section 2.567)	Integer copied from <i>iDirectoryObject</i> . DefaultTimeToLive .
mSMQVersion ([MS-ADA2] section 2.601)	If <i>iDirectoryObject</i> . OldDirectory is TRUE, mSMQVersion is 3 Else mSMQVersion is 200.

3. A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event MUST be generated with the following arguments:
 - *iPath* := the **distinguished name** specified in section [2.2.1](#) for an mSMQEnterpriseSettings object.
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *EnterpriseAttributeList* and the corresponding values, as computed in step 2.
4. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.

3.1.6.7.3.3 Postprocessing

1. If *iAttributeList* contains the name "Security" and the *iDirectoryObject*.**Security** ADM attribute is populated, a [Set Object Security Using LDAP \(section 3.1.6.17\)](#) event MUST be generated with the following arguments:
 - *iPath* := the **distinguished name** specified in section [2.2.1](#) for an mSMQEnterpriseSettings object.

- *iSecurity* := *iDirectoryObject.Security*
2. If the Set Object Security Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Security Using LDAP event, and processing MUST end.
 3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.7.3.4 mSMQCSPName

The mSMQCSPName attribute is a string that MUST have one of the values listed in the following table. The **NonLDAPCapableQueueManagerNotification** ADM attribute is a Boolean. The values MUST be converted according to the following table.

NonLDAPCapableQueueManagerNotification ADM attribute value	mSMQCSPName value
true	"Y"
false	"N"

3.1.6.7.4 Site

3.1.6.7.4.1 Preconditions

The arguments supplied to the [Write Directory \(section 3.1.6.7\)](#) event MUST meet these requirements:

- If *iAttributeList* is provided, it MUST NOT contain the names **PrimarySiteController** or **PublicSigningKeyList**. These attributes are not supported in **Active Directory**-based environments.
- At least one of the *iDirectoryObject.Identifier*, *iDirectoryObject.Name*, or *iDirectoryObject.FullPath* ADM attributes MUST be populated.

If any of these conditions is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.7.4.2 Write

1. If *iAttributeList* was not provided as an argument, then an *iAttributeList* MUST be constructed that MUST contain the names of all **Site** ADM element attributes listed in section [3.1.6.20.4](#), except **PrimarySiteController** and **PublicSigningKeyList**.
2. Let *SiteAttributeList* be a list of site attribute names, initialized to be empty. For each site attribute listed in the following table, if the corresponding **Site** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the site attribute name to *SiteAttributeList* and compute the value for the attribute as shown. **Site** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or subsections referenced by the table MUST be ignored in this step.

site attribute	Attribute value computation
mSMQInterval1 ([MS-ADA2] section 2.561)	Integer copied from <i>iDirectoryObject.IntraSiteReplicationInterval</i> .
mSMQInterval2 ([MS-ADA2] section 2.562)	Integer copied from

site attribute	Attribute value computation
	<i>iDirectoryObject</i> . InterSiteReplicationInterval .
mSMQSiteForeign ([MS-ADA2] section 2.592)	If <i>iDirectoryObject</i> . ForeignSite is TRUE, mSMQSiteForeign is 0x01 Else mSMQSiteForeign is 0x00.
mSMQNT4Stub ([MS-ADA2] section 2.571)	If <i>iDirectoryObject</i> . MigratedFromMsmq10 is TRUE, mSMQNT4Stub is 0x01 Else mSMQNT4Stub is 0x00.

3. Let *WriteGUID* be a **distinguished name**, initialized to be empty. If *iDirectoryObject*.**Identifier** is populated, the value of *iDirectoryObject*.**Identifier** MUST be copied to *WriteGUID*.
4. Let *WriteDN* be a distinguished name, initialized to be empty.
5. If *iDirectoryObject*.**FullPath** is populated, the value of *iDirectoryObject*.**FullPath** MUST be copied to *WriteDN*.
6. If *WriteDN* is empty and *iDirectoryObject*.**Name** is populated, the value of *WriteDN* MUST be a distinguished name for a site object, constructed according to the format specified in section [2.2.1](#), where "<site name>" is the value of *iDirectoryObject*.**Name**.
7. A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event MUST be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *SiteAttributeList* and the corresponding values, as computed in step 2.
8. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.

3.1.6.7.4.3 Postprocessing

1. If *iAttributeList* contains the name "Security" and the *iDirectoryObject*.**Security** attribute is populated, a [Set Object Security Using LDAP \(section 3.1.6.17\)](#) event MUST be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iSecurity* := *iDirectoryObject*.**Security**
2. If the Set Object Security Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Security Using LDAP event, and processing MUST end.
3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.7.5 RoutingLink

3.1.6.7.5.1 Preconditions

The arguments supplied to the [Write Directory \(section 3.1.6.7\)](#) event MUST meet the following requirement:

- At least one of the *iDirectoryObject.FullPath* or *iDirectoryObject.Identifier* ADM attributes MUST be populated.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.7.5.2 Write

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **RoutingLink** ADM element attributes listed in section [3.1.6.20.5](#).
2. Let *SiteLinkAttributeList* be a list of mSMQSiteLink attribute names, initialized to be empty. For each mSMQSiteLink attribute listed in the following table, if the corresponding **RoutingLink** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the mSMQSiteLink attribute name to *SiteLinkAttributeList* and compute the value for the attribute as shown. **RoutingLink** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table MUST be ignored.

mSMQSiteLink attribute	Attribute value computation
description ([MS-ADA1] section 2.153)	Directory string transformed from <i>iDirectoryObject.Description</i> .
mSMQCost ([MS-ADA2] section 2.550)	Integer copied from <i>iDirectoryObject.ActualCost</i> .
mSMQSite1 ([MS-ADA2] section 2.590)	Computed from <i>iDirectoryObject.Site1Identifier</i> . See section 3.1.6.7.5.4 .
mSMQSite2 ([MS-ADA2] section 2.591)	Computed from <i>iDirectoryObject.Site2Identifier</i> . See section 3.1.6.7.5.5 .
mSMQSiteGates ([MS-ADA2] section 2.593)	Computed from <i>iDirectoryObject.SiteGateIdentifierList</i> . See section 3.1.6.7.5.6 .

3. Let *WriteGUID* be a **distinguished name**, initialized to be empty. If *iDirectoryObject.Identifier* is populated, the value of *iDirectoryObject.Identifier* MUST be copied to *WriteGUID*.
4. Let *WriteDN* be a distinguished name, initialized to be empty.
5. If *iDirectoryObject.FullPath* is populated, the value of *iDirectoryObject.FullPath* MUST be copied to *WriteDN*.
6. A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event MUST be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *SiteLinkAttributeList* and the corresponding values, as computed in step 2.

7. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.
8. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.7.5.3 Postprocessing

None.

3.1.6.7.5.4 mSMQSite1

The value of the mSMQSite1 attribute MUST be computed according to the following algorithm:

1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS *iDirectoryObject.Site1Identifier*
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **FullPath**
2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing MUST end.
3. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of mSMQSite1 MUST be the value of the *rDirectoryObject.FullPath* returned by the Read Directory event.

3.1.6.7.5.5 mSMQSite2

The value of the mSMQSite2 attribute MUST be computed according to the following algorithm:

1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "Site"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS *iDirectoryObject.Site2Identifier*
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **FullPath**
2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing MUST end.
3. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of mSMQSite2 MUST be the value of the *rDirectoryObject.FullPath* returned by the Read Directory event.

3.1.6.7.5.6 mSMQSiteGates

The value of the mSMQSiteGates attribute MUST be computed according to the following algorithm:

1. Let *DNList* be a list of **distinguished names**, initialized to be empty.
2. For each **GUID** in *iDirectoryObject.SiteGateIdentifierList*, these steps MUST be performed:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "Identifier" EQUALS the current GUID from *iDirectoryObject.SiteGateIdentifierList*
 - *iAttributeList* := An array of the following **QueueManager** ADM element attribute names:
 - **FullPath**
 - If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**, this GUID MUST be skipped.
 - If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of the *rDirectoryObject.FullPath* returned by the Read Directory event must be added to *DNList*.
3. The value of mSMQSiteGates MUST be the value of *DNList*.

3.1.6.7.6 User

3.1.6.7.6.1 Preconditions

The arguments supplied to the [Write Directory \(section 3.1.6.7\)](#) event MUST meet the following requirement:

- At least one of the *iDirectoryObject.Identifier* or *iDirectoryObject.SecurityIdentifier* ADM attributes MUST be populated.

If this condition is violated, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rObjectGUID* is undefined, and processing of the event MUST end.

3.1.6.7.6.2 Write

1. If *iAttributeList* was not provided as an argument, an *iAttributeList* MUST be constructed that MUST contain the names of all **User** ADM element attributes listed in [section 3.1.6.20.6](#).
2. Let *UserAttributeList* be a list of user attribute names, initialized to be empty. For each user attribute listed in the following table, if the corresponding **User** ADM element attribute name appears in *iAttributeList* and that attribute is populated in *iDirectoryObject*, add the user attribute to *UserAttributeList* and compute the value for the attribute as shown. **User** ADM element attribute names that appear in *iAttributeList* but do not appear in the following table or in subsections referenced by the table MUST be ignored.

User ADM element attribute	Attribute value computation
mSMQSignCertificates ([MS-ADA2] section 2.587)	MQUSERSIGNCERTS structure ([MS-MQMQ] section 2.2.21) copied from <i>iDirectoryObject.Certificates</i> .

User ADM element attribute	Attribute value computation
mSMQDdigests ([MS-ADA2] section 2.554)	List of GUIDs copied from <i>iDirectoryObject</i> . CertificateDigestList .

3. Let *WriteGUID* be a **distinguished name**, initialized to be empty. If *iDirectoryObject*.**Identifier** is populated, the value of *iDirectoryObject*.**Identifier** MUST be copied to *WriteGUID*.
4. Let *WriteDN* be a distinguished name, initialized to be empty.
5. If *WriteGUID* is empty, search the directory for the user object with that **SecurityIdentifier**:
 - Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "User"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\] section 3.1.7.1.20](#)):
 - "SecurityIdentifier" EQUALS *iDirectoryObject*.**SecurityIdentifier**
 - *iAttributeList* := An array of the following **User ADM element attribute names**:
 - **FullPath**
 - If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**, the value of *rDirectoryObject*.**FullPath** MUST be copied to *WriteDN*.
 - If the *rStatus* returned by the Read Directory event is set to any other value, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
6. A [Set Object Properties Using LDAP \(section 3.1.6.16\)](#) event MUST be generated with the following arguments:
 - *iPath* := *WriteDN*
 - *iGuid* := *WriteGUID*
 - *iAttributes* := a list of name-value pairs consisting of the attribute names in *UserAttributeList* and the corresponding values, as computed in step 2.
7. If the Set Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Set Object Properties Using LDAP event, and processing MUST end.
8. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.7.6.3 Postprocessing

None.

3.1.6.7.6.4 Note on mSMQSignCertificates and mSMQDdigests

The mSMQDdigests attribute holds a list of MD5 digests that correspond to the **Digest** fields of the **MQUERSIGNCERT** ([\[MS-MQMQ\] section 2.2.22](#)) structures within the **MQUERSIGNCERTS** ([MS-MQMQ] section 2.2.21) structure found in the mSMQSignCertificates attribute. The digests in mSMQDdigests MUST be in the same order as the certificates in mSMQSignCertificates. Therefore, when modifying one of these attributes, the other MUST be modified at the same time. This algorithm does not enforce that requirement; it is up to implementers of this algorithm to ensure that it is met.

3.1.6.8 Resolve Queue Alias

This event MUST be generated with the following arguments:

- *iFullPath*: A **Distinguished Name (DN)** that can be used to lookup a directory object of type msMQ-Custom-Recipient ([MS-ADSC] section 2.161).

Return Values

- *rStatus*: A status code that indicates success or failure.
- *rFormatName*: The msMQ-Recipient-FormatName ([MS-ADA2] section 2.544) attribute of the msMQ-Custom-Recipient object. This value is defined only if *rStatus* has a value of **DirectoryOperationResult.Success**.

The algorithm MUST perform the following processing steps to resolve a queue alias:

- A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iPath* := *iFullPath*
 - *iAttributes* := a list of attribute names consisting of one element, "msMQ-Recipient-FormatName"
- If the *rStatus* returned by the Get Object Properties Using LDAP event is **DirectoryOperationResult.Success**, set *rFormatName* to the value returned in *rValues* for the msMQ-Recipient-FormatName attribute.
- Set *rStatus* to the *rStatus* returned by the Get Object Properties Using LDAP event. Processing MUST end.

3.1.6.9 Resolve Distribution List

This event MUST be generated with the following arguments:

- *iDLFormatName*: A distribution list format name as specified in [MS-MQMQ] section 2.1.5.

Return Values

- *rStatus*: A status code that indicates success or failure.
- *rFormatNameCollection*: A list of queue format names. This value is defined only if *rStatus* has a value of **DirectoryOperationResult.Success**.

The algorithm MUST perform the following processing steps to resolve a queue alias:

1. Instantiate a new list referred to as *newDistinguishedNameCollection*.
2. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iGuid* := the DistributionListGuid portion of *iDLFormatName*, as specified in [MS-MQMQ] section 2.1.5
 - *iAttributes* := a list of attribute names consisting of one element, "member"
3. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rFormatNameCollection* is undefined, and processing MUST end.

4. Copy the values returned by the Get Object Properties Using LDAP event for the member attribute into *newDistinguishedNameCollection*.
5. For each **distinguished name** *dName* in *newDistinguishedNameCollection*:
 1. Generate a Get Object Properties Using LDAP (section 3.1.6.14) event with the following arguments:
 - *iPath* := *dName*
 - *iAttributes* := a list of attribute names consisting of two elements, "objectClass" and "objectGuid"
 2. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rFormatNameCollection* is undefined, and processing MUST end.
 3. Let *LoopGuid* be a GUID variable, initialized to the value returned by the Get Object Properties Using LDAP event in *rValues* for the objectGuid attribute.
 4. If the value returned by the Get Object Properties Using LDAP event in *rValues* for the objectClass attribute is "mSMQQueue", construct a public format name, as specified in [MS-MQMQ] section 2.1.3, where *QueueGuid* is the value of *LoopGuid*, and add the format name to *rFormatNameCollection*.
 5. Else if the value returned by the Get Object Properties Using LDAP event in *rValues* for the objectClass attribute is "mSMQ-Custom-Recipient", perform the following steps:
 1. Generate a [Resolve Queue Alias \(section 3.1.6.8\)](#) event with the following arguments:
 - *iFullPath* := *dName*
 2. If the *rStatus* returned by the Resolve Queue Alias event is **DirectoryOperationResult.Success**, add the returned *rFormatName* to *rFormatNameCollection*.
 3. Else *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
 6. Else if the value returned by the Get Object Properties Using LDAP event in *rValues* for the objectClass attribute is "group", perform the following steps:
 1. Recursively generate a Resolve Distribution List event with the following arguments:
 - *iDLFormatName* := a distribution list format name constructed as specified in [MS-MQMQ] section 2.1.5, where *DistributionListGuid* is the value of *LoopGuid*
 2. If the *rStatus* returned by the Resolve DistributionList event is **DirectoryOperationResult.Success**, add the *rFormatNameCollection* returned by the recursive Resolve Distribution List event to *rFormatNameCollection*.
 3. Else *rStatus* MUST be set to the *rStatus* returned by the recursive Resolve Distribution List event, and processing MUST end.
 7. Else *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
6. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.10 Create LDAP Attribute List

This event MUST be generated with the following arguments:

- *iDirectoryObjectType*: A string that specifies the name of the sub-type of **DirectoryObject** ADM element.
- *iADMAAttributeList*: An array of strings containing the subset of the names of the directory attributes of the *iDirectoryObjectType* to be converted to the names of the LDAP attributes necessary to populate those directory attributes on an ADM element of type *iDirectoryObjectType*.

Return Values:

- *rLDAPAttributeList*: A list of name pairs, consisting of the directory attribute name and the corresponding LDAP attribute name.

The algorithm MUST perform the following actions to process this event:

- Based on *iDirectoryObjectType*, perform the conversion steps for that type of ADM element, as specified in the following sections.
 - [QueueManager \(section 3.1.6.10.1\)](#)
 - [Queue \(section 3.1.6.10.2\)](#)
 - [Enterprise \(section 3.1.6.10.3\)](#)
 - [Site \(section 3.1.6.10.4\)](#)
 - [RoutingLink \(section 3.1.6.10.5\)](#)
 - [User \(section 3.1.6.10.6\)](#)

3.1.6.10.1 QueueManager

For each **QueueManager** ADM element attribute listed in the following table that appears in the *iADMAAttributeList* argument, add the ADM element attribute name paired with the corresponding mSMQConfiguration attribute name to *rLDAPAttributeList*. The **QueueManager** ADM element attributes that appear in the *iADMAAttributeList* argument but not in the following table MUST be ignored.

QueueManager ADM element attribute	mSMQConfiguration attribute
Identifier	objectGUID ([MS-ADA3] section 2.44)
ComputerName	distinguishedName ([MS-ADA1] section 2.177)
QueueManagerVersion	mSMQComputerTypeEx ([MS-ADA2] section 2.549)
OperatingSystemType	mSMQOSType ([MS-ADA2] section 2.572)
CreateTime	whenCreated ([MS-ADA3] section 2.371)
ModifyTime	whenChanged ([MS-ADA3] section 2.370)
QueueManagerQuota	mSMQQuota ([MS-ADA2] section 2.582)
JournalQuota	mSMQJournalQuota ([MS-ADA2] section 2.564)
ForeignSystem	mSMQForeign ([MS-ADA2] section 2.559)

QueueManager ADM element attribute	mSMQConfiguration attribute
FullPath	distinguishedName ([MS-ADA1] section 2.177)
SiteIdentifierList	mSMQSites ([MS-ADA2] section 2.598)
OutRoutingServerIdentifierList	mSMQOutRoutingServers ([MS-ADA2] section 2.573)
InRoutingServerIdentifierList	mSMQInRoutingServers ([MS-ADA2] section 2.560)
RoutingServer	mSMQRoutingServices ([MS-ADA2] section 2.584))
DirectoryServer	mSMQDsServices ([MS-ADA2] section 2.557)
DirectoryServerType	mSMQDsServices ([MS-ADA2] section 2.557) and mSMQServiceType ([MS-ADA2] section 2.586)
RemoteAccessServer	mSMQServiceType ([MS-ADA2] section 2.586)
SupportingServer	mSMQDependentClientServices ([MS-ADA2] section 2.553)
PublicEncryptionKeyList	mSMQEncryptKey ([MS-ADA2] section 2.558)
Security	nTSecurityDescriptor ([MS-ADA3] section 2.37)

3.1.6.10.2 Queue

For each **Queue** ADM element attribute listed in the following table that appears in the *iADMAtributeList* argument, add the ADM element attribute name paired with the corresponding mSMQConfiguration attribute name to *rLDAPAttributeList*. The **Queue** ADM element attributes that appear in the *iADMAtributeList* argument but not in the following table MUST be ignored.

Queue ADM element attribute	mSMQQueue attribute
Identifier	objectGUID ([MS-ADA3] section 2.44)
Label	mSMQLabelEx ([MS-ADA2] section 2.566)
CreateTime	whenCreated ([MS-ADA3] section 2.371)
ModifyTime	whenChanged ([MS-ADA3] section 2.370)
Type	mSMQQueueType ([MS-ADA2] section 2.581)
Pathname	distinguishedName ([MS-ADA1] section 2.177), mSMQQueueNameExt ([MS-ADA2] section 2.579)
QualifiedPathname	distinguishedName ([MS-ADA1] section 2.177), mSMQQueueNameExt ([MS-ADA2] section 2.579)
Journaling	mSMQJournal ([MS-ADA2] section 2.563)
Quota	mSMQQueueQuota ([MS-ADA2] section 2.580)
JournalQuota	mSMQQueueJournalQuota ([MS-ADA2] section 2.564)
Authentication	mSMQAuthenticate ([MS-ADA2] section 2.546)
PrivacyLevel	mSMQPrivacyLevel ([MS-ADA2] section 2.576)

Queue ADM element attribute	mSMQQueue attribute
Transactional	mSMQTransactional ([MS-ADA2] section 2.599)
MulticastAddress	MSMQ-MulticastAddress ([MS-ADA2] section 2.543)
Security	nTSecurityDescriptor ([MS-ADA3] section 2.37)
BasePriority	mSMQBasePriority ([MS-ADA2] section 2.547)
FullPath	distinguishedName ([MS-ADA1] section 2.177)
DirectoryPath	distinguishedName ([MS-ADA1] section 2.177)

3.1.6.10.3 Enterprise

For each **Enterprise** ADM element attribute listed in the following table that appears in the *iADMAtributeList* argument, add the ADM element attribute name paired with the corresponding mSMQConfiguration attribute name to *rLDAPAttributeList*. The **Enterprise** ADM element attributes that appear in the *iADMAtributeList* argument but not in the following table **MUST** be ignored.

Enterprise ADM element attribute	mSMQEnterpriseSettings attribute
Identifier	objectGUID ([MS-ADA3] section 2.44)
Name	None
WeakenedSecurity	mSMQNameStyle ([MS-ADA2] section 2.569)
NonLDAPCapableQueueManagerNotification	mSMQCSPName ([MS-ADA2] section 2.551)
DefaultTimeToLive	mSMQLongLived ([MS-ADA2] section 2.567)
OldDirectory	mSMQVersion ([MS-ADA2] section 2.601)
Security	nTSecurityDescriptor ([MS-ADA3] section 2.37)

3.1.6.10.4 Site

For each **Site** ADM element attribute listed in the following table that appears in the *iADMAtributeList* argument, add the ADM element attribute name paired with the corresponding mSMQConfiguration attribute name to *rLDAPAttributeList*. The **Site** ADM element attributes that appear in the *iADMAtributeList* argument but not in the following table **MUST** be ignored.

Site ADM element attribute	site attribute
Identifier	objectGUID ([MS-ADA3] section 2.44)
Name	cn ([MS-ADA1] section 2.110)
IntraSiteReplicationInterval	mSMQInterval1 ([MS-ADA2] section 2.561)
InterSiteReplicationInterval	mSMQInterval2 ([MS-ADA2] section 2.562)
FullPath	distinguishedName ([MS-ADA1] section 2.177)

Site ADM element attribute	site attribute
Foreign	mSMQSiteForeign ([MS-ADA2] section 2.592)
MigratedFromMsmq10	mSMQNT4Stub ([MS-ADA2] section 2.571)
Security	nTSecurityDescriptor ([MS-ADA3] section 2.37)

3.1.6.10.5 RoutingLink

For each **RoutingLink** ADM element attribute listed in the following table that appears in the *iADMAAttributeList* argument, add the ADM element attribute name paired with the corresponding mSMQConfiguration attribute name to *rLDAPAttributeList*. The **RoutingLink** ADM element attributes that appear in the *iADMAAttributeList* argument but not in the following table MUST be ignored.

RoutingLink ADM element attribute	mSMQSiteLink attribute
Identifier	objectGUID ([MS-ADA3] section 2.44)
Description	description ([MS-ADA1] section 2.153)
FullPath	distinguishedName ([MS-ADA1] section 2.177)
ActualCost	mSMQCost ([MS-ADA2] section 2.550)
Site1Identifier	mSMQSite1 ([MS-ADA2] section 2.590)
Site2Identifier	mSMQSite2 ([MS-ADA2] section 2.591)
SiteGateIdentifierList	mSMQSiteGates ([MS-ADA2] section 2.593)

3.1.6.10.6 User

For each **User** ADM element attribute listed in the following table that appears in the *iADMAAttributeList* argument, add the ADM element attribute name paired with the corresponding mSMQConfiguration attribute name to *rLDAPAttributeList*. The **User** ADM element attributes that appear in the *iADMAAttributeList* argument but not in the following table MUST be ignored.

User ADM element attribute	user attribute
Identifier	objectGUID ([MS-ADA3] section 2.44)
SecurityIdentifier	objectSid ([MS-ADA3] section 2.45)
CertificateDigestList	mSMQDigests ([MS-ADA2] section 2.554)
Certificates	mSMQSignCertificates ([MS-ADA2] section 2.587)
FullPath	distinguishedName ([MS-ADA1] section 2.177)

3.1.6.11 Create ADM Element From LDAP Values

This event MUST be generated with the following arguments:

- *iDirectoryObjectType*: A string that specifies the name of the sub-type of the **DirectoryObject** ADM element to be created.
- *iADMAAttributeList*: An array of ADM element attribute names indicating which attributes to set on the created ADM element.
- *iLDAPAttributeList*: An array of LDAP attribute names.
- *iValues*: An array of LDAP attribute values, in the same order as the names in *iLDAPAttributeList*.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this operation.
- *rDirectoryObject*: The created ADM element instance. This value is undefined if *rStatus* is not **DirectoryOperationResult.Success**.

3.1.6.11.1 QueueManager

1. A **QueueManager** ADM element instance MUST be created, and *rDirectoryObject* MUST be set to this new **QueueManager** ADM element instance.
2. The attributes listed in the *iADMAAttributeList* argument MUST be set on *rDirectoryObject*, using the attribute names in the *iLDAPAttributeList* argument and the corresponding values in the *iValues* argument, according to the following table. If the value of the LDAP attribute required to compute the **QueueManager** ADM element attribute is empty in the *iValues* argument, depending on the attribute, either this condition is an error, or a default implementation-specific value SHOULD [<10>](#) be supplied. If an error condition occurs, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.

QueueManager ADM element attribute	Attribute value computation	If not set, default value or error?
Identifier	GUID copied from objectGUID ([MS-ADA3] section 2.44).	Error
ComputerName	Computed from distinguishedName ([MS-ADA1] section 2.177); see section 3.1.6.11.1.1 .	Error
QueueManagerVersion	Unicode string transformed from mSMQComputerTypeEx ([MS-ADA2] section 2.549).	Empty string
OperatingSystemType	Computed from mSMQOSType ([MS-ADA2] section 2.572); see section 3.1.6.11.1.2 .	The enumeration value Unknown .
QualifiedComputerName	Set in a following step.	N/A
CreateTime	Integer time value expressed as the number of seconds elapsed from midnight (00:00:00), January 1, 1970 UTC to whenCreated ([MS-ADA3] section 2.371).	Error
ModifyTime	Integer time value expressed as the number of seconds elapsed from midnight (00:00:00), January 1, 1970 UTC to whenChanged ([MS-ADA3] section 2.370).	Error
QueueManagerQuota	Integer copied from mSMQQuota ([MS-ADA2] section 2.582).	0x00100000

QueueManager ADM element attribute	Attribute value computation	If not set, default value or error?
JournalQuota	Integer copied from mSMQJournalQuota ([MS-ADA2] section 2.564).	0xFFFFFFFF
ForeignSystem	If mSMQForeign ([MS-ADA2] section 2.559) equals 0x01, ForeignSystem is TRUE Else if mSMQForeign equals 0x00, ForeignSystem is FALSE.	FALSE
FullPath	Distinguished name copied from distinguishedName ([MS-ADA1] section 2.177).	Error
SiteIdentifierList	List of GUIDs copied from mSMQSites ([MS-ADA2] section 2.598).	Empty list
OutRoutingServerIdentifierList	Computed from mSMQOutRoutingServers ([MS-ADA2] section 2.573); see section 3.1.6.11.1.3 .	Empty list
InRoutingServerIdentifierList	Computed from mSMQInRoutingServers ([MS-ADA2] section 2.560); see section 3.1.6.11.1.4 .	Empty list
RoutingServer	If mSMQRoutingServices ([MS-ADA2] section 2.584) equals 0x01, RoutingServer is TRUE Else if mSMQRoutingServices equals 0x00, RoutingServer is FALSE.	FALSE
DirectoryServer	If mSMQDsServices ([MS-ADA2] section 2.557) equals 0x01, DirectoryServer is TRUE Else if mSMQDsServices equals 0x00, DirectoryServer is FALSE.	FALSE
DirectoryServerType	Computed from mSMQDsServices ([MS-ADA2] section 2.557) and mSMQServiceType ([MS-ADA2] section 2.586); see section 3.1.6.11.1.5 .	The enumeration value Standalone if DirectoryServer equals TRUE; undefined if DirectoryServer equals FALSE or is unpopulated.
RemoteAccessServer	If bit 0x00000010 of mSMQServiceType ([MS-ADA2] section 2.586) is set, RemoteAccessServer is TRUE Else if bit 0x00000010 of mSMQServiceType is not set, RemoteAccessServer is FALSE.	FALSE
SupportingServer	If mSMQDependentClientServices ([MS-ADA2] section 2.553) equals 0x01, SupportingServer is TRUE Else if mSMQDependentClientServices equals 0x00, SupportingServer is FALSE.	FALSE

QueueManager ADM element attribute	Attribute value computation	If not set, default value or error?
PublicEncryptionKeyList	MQDSPUBLICKEYS structure ([MS-MQMQ] section 2.2.2) copied from mSMQEncryptKey ([MS-ADA2] section 2.558).	Empty list
PublicSigningKeyList	Set in a following step.	N/A
Security	Security descriptor copied from nTSecurityDescriptor ([MS-ADA3] section 2.37).	Error
Clustered	Set in a following step.	N/A

3. If none of the **QueueManager** ADM element attribute names **QualifiedComputerName**, **PublicSigningKeyList**, **OperatingSystemVersion**, or **Clustered** appears in the *iADMAAttributeList* argument, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.
4. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iPath* := a distinguished name of the form for a computer object as specified in section [2.2.1](#), created by removing the first comma-separated element of the value of *rDirectoryObject.FullPath*.
 - *iAttributes* := a list of attribute names. The following table lists the computer attributes required to compute the values of the **QueueManager** ADM element attributes **QualifiedComputerName**, **PublicSigningKeyList**, **OperatingSystemVersion**, and **Clustered**. For each **QueueManager** ADM element attribute that appears in *ConfigurationAttributeList*, the required computer attribute name MUST appear in the *iAttributes* argument.

QueueManager ADM element attribute	Required computer attribute	Attribute value computation
QualifiedComputerName	dNSHostName ([MS-ADA1] section 2.185)	Unicode string transformed from dNSHostName.
PublicSigningKeyList	mSMQSignCertificates ([MS-ADA2] section 2.587)	MQUSERSIGNCERTS ([MS-MQMQ] section 2.2.21) structure copied from mSMQSignCertificates.
OperatingSystemVersion	operatingSystemVersion ([MS-ADA3] section 2.56)	Unicode string transformed from operatingSystemVersion.
Clustered	servicePrincipalName ([MS-ADA3] section 2.253)	See section 3.1.6.11.1.6 .

5. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, *rDirectoryObject* is undefined, and processing MUST end. If dNSHostName or servicePrincipalName is required, but the value returned in *rValues* for that attribute is empty, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end. If operatingSystemVersion is required but not set on the computer object, a default value MUST be supplied. [<11>](#)
6. For each of the **QueueManager** ADM element attributes **QualifiedComputerName**, **PublicSigningKeyList**, and **Clustered** that appears in the *iADMAAttributeList* argument, that

attribute must be set on *rDirectoryObject* according to the computation rule specified in the table in step 4. If **PublicSigningKeyList** appears in **ReadIterator.AttributeList**, but **mSMQSignCertificates** is not set on the computer object, a default value MUST be supplied for the **PublicSigningKeyList** ADM attribute. <12>

7. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.11.1.1 ComputerName

The value of *rDirectoryObject.ComputerName* MUST be computed from the value of the distinguishedName attribute. *rDirectoryObject.ComputerName* must be set to the "<computer name>" portion of the value of the distinguishedName attribute, as specified in section 2.2.1 for the distinguished name of an mSMQConfiguration object.

3.1.6.11.1.2 OperatingSystemType

The mSMQOSType attribute is an integer that MUST have one of the values listed in the following table. The **OperatingSystemType** ADM attribute is an enumeration. The values MUST be converted according to this table.

mSMQOSType value	OperatingSystemType ADM attribute value
0x00000000	Other
0x00000100	Foreign
0x00000200	Win95
0x00000300	WinClient
0x00000400	WinServer
0x00000500	WinEnt

3.1.6.11.1.3 OutRoutingServerIdentifierList

The value of *rDirectoryObject.OutRoutingServerIdentifierList* MUST be computed from the value of the mSMQOutRoutingServers attribute. For each distinguished name in the mSMQOutRoutingServers attribute, these steps MUST be followed:

- Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - iDirectoryObjectType* := "QueueManager"
 - iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "FullPath" EQUALS the current distinguished name
 - iAttributeList* := An array of the following **QueueManager** ADM element attribute names:
 - Identifier**
- If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.ObjectNotFound**:
 - The current distinguished name MUST be skipped.

3. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**:
 - *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and *rDirectoryObject* is undefined.
 - Processing MUST end.
4. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**:
 - The **GUID** returned in *rDirectoryObject.Identifier* by the Read Directory event MUST be added to the value of the **OutRoutingServerIdentifierList** ADM attribute.

3.1.6.11.1.4 InRoutingServerIdentifierList

The value of *rDirectoryObject.InRoutingServerIdentifierList* MUST be computed from the value of the mSMQInRoutingServers attribute. For each **distinguished name** in the mSMQInRoutingServers attribute, these steps MUST be followed:

1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "FullPath" EQUALS the current distinguished name
 - *iAttributeList* := An array of the following **QueueManager** ADM attribute names:
 - **Identifier**
2. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.ObjectNotFound**:
 - The current distinguished name MUST be skipped.
3. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**:
 - *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and *rDirectoryObject* is undefined.
 - Processing MUST end.
4. If the *rStatus* returned by the Read Directory event is set to **DirectoryOperationResult.Success**:
 - The **GUID** returned in *rDirectoryObject.Identifier* by the Read Directory event MUST be added to the value of the **InRoutingServerIdentifierList** ADM attribute.

3.1.6.11.1.5 DirectoryServerType

The *rDirectoryObject.DirectoryServerType* attribute MUST NOT be set if *rDirectoryObject.DirectoryServer* is returned as FALSE or is not populated.

The mSMQServiceType attribute is an integer containing a bitmap. If one of the bits listed in the following table is set, *rDirectoryObject.DirectoryServerType* MUST be set to the corresponding enumeration value. If none of the bits listed in the following table is set, *rDirectoryObject.DirectoryServerType* MUST be set to the enumeration value **Standalone**.

mSMQServiceType value	DirectoryServerType value
0x00000002	BackupSiteController
0x00000004	PrimarySiteController
0x00000008	PrimaryEnterpriseController

3.1.6.11.1.6 Clustered

The value of *rDirectoryObject.Clustered* MUST be computed from the servicePrincipalName attribute. If the value of the servicePrincipalName attribute contains the substring "MSClusterVirtualServer", the value of the **Clustered** ADM attribute MUST be TRUE; otherwise, it MUST be FALSE.

3.1.6.11.2 Queue

1. A **Queue** ADM element instance MUST be created, and *rDirectoryObject* MUST be set to this new **Queue** ADM element instance.
2. The attributes listed in the *iADMAAttributeList* argument MUST be set on *rDirectoryObject*, using the attribute names in the *iLDAPAttributeList* argument and the corresponding values in the *iValues* argument, according to the following table. If the value of the LDAP attribute required to compute the **Queue** ADM element attribute is empty in the *iValues* argument, depending on the attribute, either that is an error condition, or a default value MUST be used. If an error condition occurs, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.

Queue attribute	Attribute value computation	If not set, default value or error?
Identifier	GUID copied from objectGUID ([MS-ADA3] section 2.44).	Error.
Label	Unicode string transformed from mSMQLabelEx ([MS-ADA2] section 2.566).	Default value: empty Unicode string.
CreateTime	Integer time value expressed as the number of seconds elapsed from midnight (00:00:00), January 1, 1970 UTC to whenCreated ([MS-ADA3] section 2.371).	Error.
ModifyTime	Integer time value expressed as the number of seconds elapsed from midnight (00:00:00), January 1, 1970 UTC to whenChanged ([MS-ADA3] section 2.370).	Error.
Type	GUID copied from mSMQQueueType ([MS-ADA2] section 2.581).	Default value: a GUID with all fields set to zero.
Pathname	Computed from distinguishedName ([MS-ADA1] section 2.177) and mSMQQueueNameExt ([MS-ADA2] section 2.579); see section 3.1.6.11.2.1.	See section 3.1.6.11.2.1.
QualifiedPathName	Set in a following step. distinguishedName ([MS-ADA1] section 2.177) and mSMQQueueNameExt ([MS-ADA2] section 2.579) are required.	N/A
Journaling	If mSMQJournal ([MS-ADA2] section 2.563) equals 1,	Default value:

Queue attribute	Attribute value computation	If not set, default value or error?
	Journaling is TRUE Else if mSMQJournal equals 0, Journaling is FALSE.	FALSE.
Quota	Integer copied from mSMQQueueQuota ([MS-ADA2] section 2.580).	Default value: 0xFFFFFFFF.
JournalQuota	Integer copied from mSMQQueueJournalQuota ([MS-ADA2] section 2.578).	Default value: 0xFFFFFFFF.
Authentication	If mSMQAuthenticate ([MS-ADA2] section 2.546) equals 1, Authentication is TRUE Else if mSMQAuthenticate equals 0, Authentication is FALSE.	Default value: FALSE.
PrivacyLevel	Computed from mSMQPrivacyLevel ([MS-ADA2] section 2.576); see section 3.1.6.11.2.3 .	Default value: enumeration value Optional .
Transactional	If mSMQTransactional ([MS-ADA2] section 2.599) equals 1, Transactional is TRUE Else if mSMQTransactional equals 0, Transactional is FALSE.	Default value: FALSE.
MulticastAddress	Unicode string transformed from MSMQ-MulticastAddress ([MS-ADA2] section 2.543).	Default value: empty Unicode string.
Security	Security descriptor copied from nTSecurityDescriptor ([MS-ADA3] section 2.37).	Error
BasePriority	Integer copied from mSMQBasePriority ([MS-ADA2] section 2.547).	Default value: zero.
FullPath	Distinguished name copied from distinguishedName ([MS-ADA1] section 2.177).	Error
DirectoryPath	Unicode string computed from distinguishedName ([MS-ADA1] section 2.177) by transforming to Unicode and prepending the Unicode string "LDAP://".	Error

3. If the **Queue** ADM element attribute name **QualifiedPathname** does not appear in *iADMAtributeList*, *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.
4. A [Get Object Properties Using LDAP \(section 3.1.6.14\)](#) event MUST be generated with the following arguments:
 - *iPath* := a distinguished name of the form for a computer object as specified in section [2.2.1](#), created by removing the first two comma-separated elements of the value in the *iValues* argument for the mSMQQueue attribute distinguishedName.
 - *iAttributes* := a list of attribute names consisting of one element, "dNSHostName"
5. If the Get Object Properties Using LDAP event returns an *rStatus* that is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the *rStatus* returned by the Get Object Properties Using LDAP event, and processing MUST end. If the value returned in *rValues* for

the `dNSHostName` attribute is empty, `rDirectoryObject.QualifiedPathname` MUST be set to an empty string. Otherwise, let `DNSname` be a string initialized to the value returned in `rValues` for the computer attribute `dNSHostName`.

6. `rDirectoryObject.QualifiedPathname` MUST be computed from the values in the `iValues` argument of the `mSMQueue` attributes `distinguishedName` and `mSMQueueNameExt` and the value of `DNSname`, as specified in section [3.1.6.11.2.2](#).
7. `rStatus` MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.11.2.1 Pathname

The **Pathname** ADM attribute MUST be computed from the values in the `iValues` argument of the `distinguishedName` and `mSMQueueNameExt` attributes according to the following algorithm:

1. Let `CompName`, `QName`, `ExtName`, and `OutputName` be **Unicode** strings, initialized to be empty.
2. `CompName` and `QName` MUST be the "<computer name>" and "<queue name>", respectively, extracted from the value of the `distinguishedName` attribute in the `iValues` argument and transformed to Unicode strings. The "<computer name>" and "<queue name>" as elements of the **distinguished name** for an `mSMQueue` object are specified in section [2.2.1](#). If the value in the `iValues` argument of the `distinguishedName` attribute is empty, `rStatus` MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
3. If the value of `QName` is 64 Unicode characters long, `ExtName` MUST be set to the value of the `mSMQueueNameExt` attribute transformed to a Unicode string; then `QName` MUST be truncated by removing the last nine Unicode characters. If the value in the `iValues` argument of the `mSMQueueNameExt` attribute is empty, `ExtName` MUST remain empty.
4. The value of `CompName` MUST be copied to `OutputName`.
5. A Unicode backslash character ('\\') MUST be appended to `OutputName`.
6. All Unicode backslash characters in the value of `QName` MUST be removed.
7. The value of `QName` MUST be appended to the value of `OutputName`.
8. If `ExtName` is not an empty string, the value of `ExtName` MUST be appended to the value of `OutputName`.
9. All alphabetic characters in the value of `OutputName` MUST be converted to lowercase.
10. The value of `rDirectoryObject.Pathname` MUST be the value of `OutputName`.

3.1.6.11.2.2 QualifiedPathname

The **QualifiedPathname** ADM attribute MUST be computed from the values in the `iValues` argument of the `mSMQueue` attributes `distinguishedName` and `mSMQueueNameExt` and the value of `DNSname`:

1. Let `QName`, `ExtName`, and `OutputName` be **Unicode** strings, initially set to empty.
2. `QName` MUST be the "<queue name>" extracted from the `distinguishedName` attribute in the `iValues` argument and transformed to a Unicode string. The "<queue name>" as an element of the **distinguished name** of an `mSMQueue` object is specified in section [2.2.1](#). If the value in the `iValues` argument of the `distinguishedName` attribute is empty, `rStatus` MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
3. If the `QName` is 64 Unicode characters long, `ExtName` MUST be set to the value of the `mSMQueueNameExt` attribute and transformed to a Unicode string; then `QName` MUST be

truncated by removing the last nine Unicode characters. If the value in the *iValues* argument of the *mSMQQueueNameExt* attribute is empty, *ExtName* MUST remain empty.

4. The value of the *dNSHostName* attribute MUST be copied to *OutputName*.
5. A Unicode backslash character ('\') MUST be appended to *OutputName*.
6. The value of *QName* MUST be appended to *OutputName*.
7. If *ExtName* is not an empty string, the value of *ExtName* MUST be appended to *OutputName*.
8. All alphabetic characters in the value *OutputName* MUST be converted to lowercase.
9. The value of *rDirectoryObject.QualifiedPathname* MUST be set to the value of *OutputName*.

3.1.6.11.2.3 PrivacyLevel

The *mSMQPrivacyLevel* attribute is an integer that MUST have one of the values listed in the following table. The **PrivacyLevel** ADM attribute is an enumeration. The values MUST be converted according to this table.

mSMQPrivacyLevel value	PrivacyLevel ADM attribute value
0	None
1	Optional
2	Body

3.1.6.11.3 Enterprise

1. An **Enterprise** ADM element instance MUST be created, and *rDirectoryObject* MUST be set to this new **Enterprise** ADM element instance.
2. The attributes listed in the *iADMAAttributeList* argument MUST be set on *rDirectoryObject*, using the attribute names in the *iLDAPAttributeList* argument and the corresponding values in the *iValues* argument, according to the following table. If the value of the LDAP attribute required to compute the **Enterprise** ADM element attribute is empty in the *iValues* argument, depending on the attribute, either a default value MUST be supplied, or that is an error condition. [<13>](#) If an error condition occurs, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.

Enterprise ADM element attribute	Attribute value computation	If not set, default value or error?
Identifier	GUID copied from objectGUID ([MS-ADA3] section 2.44).	Error
Name	See section 3.1.6.11.3.1 .	See section 3.1.6.11.3.1
WeakenedSecurity	Computed from <i>mSMQNameStyle</i> ([MS-ADA2] section 2.569); see section 3.1.6.11.3.2 .	Default value
NonLDAPCapableQueueManagerNotification	Computed from <i>mSMQCSPName</i> ([MS-ADA2] section 2.551); see section 3.1.6.11.3.3 .	Default value

Enterprise ADM element attribute	Attribute value computation	If not set, default value or error?
DefaultTimeToLive	Integer copied from mSMQLongLived ([MS-ADA2] section 2.567).	Default value
OldDirectory	If mSMQVersion ([MS-ADA2] section 2.601) equals 3, OldDirectory is TRUE Else if mSMQVersion equals 200, OldDirectory is FALSE.	Default value
Security	Security descriptor copied from nTSecurityDescriptor ([MS-ADA3] section 2.37).	Error

3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.11.3.1 Name

The **Name** ADM attribute MUST be computed by retrieving the rootDomainNamingContext as specified in [MS-ADTS] section 3.1.1.3.2.16. If the rootDomainNamingContext cannot be retrieved, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.

3.1.6.11.3.2 WeakenedSecurity

The mSMQNameStyle attribute is an integer that MUST have one of the values listed in the following table. The **WeakenedSecurity** ADM attribute is a Boolean. The values MUST be converted according to this table.

mSMQNameStyle value	WeakenedSecurity ADM attribute value
0x00	False
0x01	True
0x02	Use internal default. <14>

3.1.6.11.3.3 NonLDAPCapableQueueManagerNotification

The mSMQCSPName attribute is a string that MUST have one of the values listed in the following table. The **NonLDAPCapableQueueManagerNotification** ADM attribute is a Boolean. The values MUST be converted according to this table.

mSMQCSPName value	NonLDAPCapableQueueManagerNotification ADM attribute value
"Y"	True
"N"	False

3.1.6.11.4 Site

1. A **Site** ADM element instance MUST be created, and *rDirectoryObject* MUST be set to this new **Site** ADM element instance.
2. The attributes listed in the *iADMAAttributeList* argument MUST be set on *rDirectoryObject*, using the attribute names in the *iLDAPAttributeList* argument and the corresponding values in the *iValues* argument, according to the following table. If the value of the LDAP attribute required to compute the **Site** ADM element attribute is empty in the *iValues* argument, depending on the attribute, either a default value MUST be supplied, or that is an error condition. <15> If an error condition occurs, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.

Site ADM element attribute	Attribute value computation	If not set, default value or error?
Identifier	GUID copied from objectGUID ([MS-ADA3] section 2.44).	Error
Name	Unicode string transformed from cn ([MS-ADA1] section 2.110).	Error
IntraSiteReplicationInterval	Integer copied from mSMQInterval1 ([MS-ADA2] section 2.561).	Default value
InterSiteReplicationInterval	Integer copied from mSMQInterval2 ([MS-ADA2] section 2.562).	Default value
FullPath	Distinguished name copied from distinguishedName ([MS-ADA1] section 2.177).	Error
Foreign	If mSMQSiteForeign ([MS-ADA2] section 2.592) equals 0x01, Foreign is TRUE Else if mSMQSiteForeign equals 0x00, Foreign is FALSE.	Default value
MigratedFromMsmq10	If mSMQNT4Stub ([MS-ADA2] section 2.571) equals 1, MigratedFromMsmq10 is TRUE Else if mSMQNT4Stub equals 0 MigratedFromMsmq10 is FALSE.	Error
Security	Security descriptor copied from nTSecurityDescriptor ([MS-ADA3] section 2.37).	Error

3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.11.5 RoutingLink

1. A **RoutingLink** ADM element instance MUST be created, and *rDirectoryObject* MUST be set to this new **RoutingLink** ADM element instance.
2. The attributes listed in *iADMAAttributeList* MUST be set on *rDirectoryObject*, using the attribute names in *iLDAPAttributeList* and the corresponding values in *iValues*, according to the following table. If the value of the LDAP attribute required to compute the **QueueManager** ADM element attribute is empty in *iValues*, depending on the attribute, either a default value MUST be supplied, or that is an error condition. <16> If an error condition occurs, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.

RoutingLink ADM element attribute	Attribute value computation	If not set, default value or error?
Identifier	GUID copied from objectGUID ([MS-ADA3] section 2.44).	Error
Description	Unicode string transformed from description ([MS-ADA1] section 2.153).	Default value
FullPath	Distinguished name copied from distinguishedName ([MS-ADA1] section 2.177).	Error
ActualCost	Integer copied from mSMQCost ([MS-ADA2] section 2.550).	Error
Site1Identifier	Computed from mSMQSite1 ([MS-ADA2] section 2.590); see section 3.1.6.11.5.1.	Error
Site2Identifier	Computed from mSMQSite2 ([MS-ADA2] section 2.591); see section 3.1.6.11.5.2.	Error
SiteGateIdentifierList	Computed from mSMQSiteGates ([MS-ADA2] section 2.593); see section 3.1.6.11.5.3.	Default value

3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.11.5.1 Site1Identifier

The value of the *rDirectoryObject.Site1Identifier* ADM attribute MUST be computed from the value of mSMQSite1:

- Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - iDirectoryObjectType* := "Site"
 - iFilter* := An array of the following **attribute-filter expressions** ([MS-MQDMPR] section 3.1.7.1.20):
 - "FullPath" EQUALS the value of mSMQSite1
 - iAttributeList* := An array of the following **Site** ADM element attribute names:
 - Identifier**
- If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**:
 - rStatus* must be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.
- The value of the *rDirectoryObject.Site1Identifier* ADM attribute MUST be set to the value of the **Identifier** ADM attribute of the *rDirectoryObject* returned by the Read Directory event.

3.1.6.11.5.2 Site2Identifier

The value of the *rDirectoryObject.Site2Identifier* ADM attribute MUST be computed from the value of mSMQSite2:

- Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - iDirectoryObjectType* := "Site"

- *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "FullPath" EQUALS the value of mSMQSite2
 - *iAttributeList* := An array of the following **Site** ADM element attribute names:
 - **Identifier**
2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**:
 - *rStatus* must be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.
 3. The value of the *rDirectoryObject.Site2Identifier* ADM attribute MUST be set to the value of the **Identifier** ADM attribute of the *rDirectoryObject* returned by the Read Directory event.

3.1.6.11.5.3 SiteGateIdentifierList

The value of *rDirectoryObject.SiteGateIdentifierList* MUST be computed from the value of mSMQSiteGates according to the following algorithm:

1. Let *TempList* be a list of **distinguished names**, initialized to the value of *rDirectoryObject.mSMQSiteGates*.
2. Let *FinalList* be a list of GUIDs, initialized to be empty.
3. For each distinguished name in *TempList*:
 1. Generate a [Read Directory \(section 3.1.6.3\)](#) event with the following arguments:
 - *iDirectoryObjectType* := "QueueManager"
 - *iFilter* := An array of the following **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20):
 - "FullPath" EQUALS the current distinguished name from *TempList*
 - *iAttributeList* := An array of the following **QueueManager** ADM element attribute names:
 - **Identifier**
 2. If the *rStatus* returned by the Read Directory event is not set to **DirectoryOperationResult.Success**:
 - *rStatus* must be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.
 3. The GUID in the **Identifier** ADM attribute of the *rDirectoryObject* returned by the Read Directory event MUST be placed in *FinalList*.
4. The value of the *rDirectoryObject.SiteGateIdentifierList* attribute is the value of *FinalList*.

3.1.6.11.6 User

1. A **User** ADM element instance MUST be created, and *rDirectoryObject* MUST be set to this new **User** ADM element instance.
2. The attributes listed in *iADMAAttributeList* MUST be set on *rDirectoryObject*, using the attribute names in *iLDAPAttributeList* and the corresponding values in *iValues*, according to the following

table. If the value of the LDAP attribute required to compute the **User** ADM element attribute is empty in *iValues*, depending on the attribute, either a default value MUST be supplied, or that is an error condition. <17> If an error condition occurs, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, *rDirectoryObject* is undefined, and processing MUST end.

User ADM element attribute	Attribute value computation	If not set, default value or error?
Identifier	GUID copied from objectGUID ([MS-ADA3] section 2.44).	Error
SecurityIdentifier	SID copied from objectSid ([MS-ADA3] section 2.45).	Error
CertificateDigestList	List of GUIDs copied from mSMQDigests ([MS-ADA2] section 2.554).	Default value
Certificates	MQUSERSIGNCERTS ([MS-MQMQ] section 2.2.21) structure copied from mSMQSignCertificates ([MS-ADA2] section 2.587).	Default value
FullPath	Distinguished name copied from distinguishedName ([MS-ADA1] section 2.177).	Error

3. *rStatus* MUST be set to **DirectoryOperationResult.Success**, and processing MUST end.

3.1.6.12 Create Object Using LDAP

This event MUST be generated with the following arguments:

- *iParentPath*: the **distinguished name** of the parent of the object to be created.
- *iChildName*: the name of the object to be created.
- *iObjectClass*: the **Active Directory schema** class of the new object.
- *iAttributes*: a list of name-value pairs.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.
- *rObjectGuid*: if *rStatus* is **DirectoryOperationResult.Success**, the value of the objectGUID attribute of the created object. Otherwise, this value is undefined and MUST NOT be used.

The algorithm MUST perform the following actions to process this event:

- If the value of the **CachedConfigurationNamingContext** ADM element is an empty string, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end.
- Raise a [Prepare an LDAP Connection \(section 3.1.6.18\)](#) event. The event takes no arguments. If the *rStatus* returned is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end. Otherwise, let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE** ([MS-DTYP] section 2.2.2), which is initialized to the value returned in *rADConnection*.
- Construct an **LDAPMessage** ([RFC2251] section 4.1):
 - messageID = set as described in [RFC2251] section 4.1.1.1.
 - protocolOp = searchRequest

- controls = none
- baseObject = *iParentPath*
- scope = baseObject
- derefAliases = neverDerefAliases
- sizeLimit = 0
- timeLimit = 0
- typesOnly = FALSE
- filter = "(objectClass=*)"
- attributes = a one-element list consisting of the string "objectClass"
- Perform the Performing an LDAP Operation on an ADConnection ([\[MS-ADTS\]](#) section 7.6.1.6) task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section [2.2.6](#).
 - Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.
 - Processing MUST end.
- Let *ChildDistinguishedName* be a string variable, initialized by concatenating the strings "CN=", the value of *iChildName*, a comma ",", and the value of *iParentPath*.
- Construct a set of attribute name-value pairs:
 - The attribute objectClass MUST be paired with the value of *iObjectClass*.
 - All pairs in *iAttributes*.
- Construct an **LDAPMessage**:
 - messageID = set as described in [RFC2251] section 4.1.1.1.
 - protocolOp = addRequest
 - controls = none
 - entry = *ChildDistinguishedName*
 - attributes = the set of attribute name-value pairs constructed in the previous step
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step

- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
 - Raise a Shut Down an LDAP Connection event.
 - Processing MUST end.
- Construct an **LDAPMessage**:
 - *messageID* = set as described in [RFC2251] section 4.1.1.1.
 - *protocolOp* = *searchRequest*
 - *controls* = none
 - *baseObject* = *ChildDistinguishedName*
 - *scope* = *baseObject*
 - *derefAliases* = *neverDerefAliases*
 - *sizeLimit* = 0
 - *timeLimit* = 0
 - *typesOnly* = FALSE
 - *filter* = "(objectClass=*)"
 - *attributes* = an empty list
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
 - Raise a Shut Down an LDAP Connection event.
 - Processing MUST end.
- Extract the value of the *objectGuid* attribute from the result message returned in *TaskOutputResultMessages*, and set *rObjectGuid* to that value.
- Raise a Shut Down an LDAP Connection event.
- Processing MUST end.

3.1.6.13 Delete Object Using LDAP

This event MUST be generated with the following arguments:

- *iPath*: the **distinguished name** of the object to be deleted; optional if *iGuid* is supplied.
- *iGuid*: the value of the objectGuid attribute of the object to be deleted; optional if *iPath* is supplied.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.

The algorithm MUST perform the following actions to process this event:

- If the value of the **CachedConfigurationNamingContext** ADM element is an empty string, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end.
- Raise a [Prepare an LDAP Connection \(section 3.1.6.18\)](#) event. The event takes no arguments. If the *rStatus* returned is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end. Otherwise, let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE** ([\[MS-DTYP\]](#) section 2.2.2), which is initialized to the value returned in *rADConnection*.
- If *iGuid* is supplied, perform the following steps:
 - Raise a [Find Object By GUID Using LDAP \(section 3.1.6.19\)](#) event with the following arguments:
 - *iGuid* = *iGuid*
 - *iADConnection* = *DirectoryServerConnection*
 - If the value of *rStatus* returned by the event is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the value of *rStatus* returned by the event, and processing MUST end.
 - Set *iPath* to the value of *rDN*.
- Let *ParentPath* be a string variable that contains the distinguished name of the parent object of the object identified by *iPath* and that is initialized by copying the value of *iPath* and removing the leftmost comma-separated element.
- Construct an **LDAPMessage** ([\[RFC2251\]](#) section 4.1):
 - messageID = set as described in [\[RFC2251\]](#) section 4.1.1.1.
 - protocolOp = searchRequest
 - controls = none
 - baseObject = *ParentPath*
 - scope = baseObject
 - derefAliases = neverDerefAliases
 - sizeLimit = 0
 - timeLimit = 0
 - typesOnly = FALSE
 - filter = "(objectClass=*)"
 - attributes = a one-element list consisting of the string "objectClass"

- Perform the Performing an LDAP Operation on an ADConnection ([\[MS-ADTS\]](#) section 7.6.1.6) task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section [2.2.6](#).
 - Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.
 - Processing MUST end.
- Construct an **LDAPMessage**:
 - *messageID* = set as described in [RFC2251] section 4.1.1.1.
 - *protocolOp* = *searchRequest*
 - *controls* = none
 - *baseObject* = *iPath*
 - *scope* = *baseObject*
 - *derefAliases* = *neverDerefAliases*
 - *sizeLimit* = 0
 - *timeLimit* = 0
 - *typesOnly* = FALSE
 - *filter* = "(objectClass=*)"
 - *attributes* = a one-element list consisting of the string "objectClass"
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section [2.2.6](#).
 - Raise a Shut Down an LDAP Connection event.
 - Processing MUST end.
- Construct an **LDAPMessage**:
 - *messageID* = set as described in [RFC2251] section 4.1.1.1.
 - *protocolOp* = *delRequest*

- controls = none
- baseObject = *iPath*
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
 - Raise a Shut Down an LDAP Connection event.
 - Processing MUST end.
- Raise a Shut Down an LDAP Connection event.
- Processing MUST end.

3.1.6.14 Get Object Properties Using LDAP

This event MUST be generated with the following arguments:

- *iPath*: the **distinguished name** of the object; optional if *iGuid* is supplied.
- *iGuid*: the value of the objectGuid attribute of the object; optional if *iPath* is supplied.
- *iAttributes*: a list of attribute names.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.
- *rValues*: if *rStatus* is **Success**, the values of the attributes listed in *iAttributes*, in the same order. Otherwise, this value is undefined and MUST NOT be used.

The algorithm MUST perform the following actions to process this event:

- If the value of the **CachedConfigurationNamingContext** ADM element is an empty string, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end.
- If the *iAttributes* list does not contain the attribute name "objectGuid", add it to the list. If the *iAttributes* list does not contain the attribute name "distinguishedName", add it to the list.
- Raise a [Prepare an LDAP Connection \(section 3.1.6.18\)](#) event. The event takes no arguments. If the *rStatus* returned is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end. Otherwise, let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE** ([\[MS-DTYP\]](#) section 2.2.2), which is initialized to the value returned in *rADConnection*.
- If *iGuid* is supplied, perform the following steps:
 - Raise a [Find Object By GUID Using LDAP \(section 3.1.6.19\)](#) event with the following arguments:

- *iGuid* = *iGuid*
- *iADConnection* = *DirectoryServerConnection*
- If the value of *rStatus* returned by the event is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the value of *rStatus* returned by the event, and processing MUST end.
- Set *iPath* to the value of *rDN*.
- Construct an **LDAPMessage** ([\[RFC2251\]](#) section 4.1):
 - *messageID* = set as described in [\[RFC2251\]](#) section 4.1.1.1.
 - *protocolOp* = *searchRequest*
 - *controls* = *none*
 - *baseObject* = *iPath*
 - *scope* = *baseObject*
 - *derefAliases* = *neverDerefAliases*
 - *sizeLimit* = 0
 - *timeLimit* = 0
 - *typesOnly* = *FALSE*
 - *filter* = "(objectClass=*)"
 - *attributes* = an empty list
- Perform the Performing an LDAP Operation on an ADConnection ([\[MS-ADTS\]](#) section 7.6.1.6) task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [\[RFC2251\]](#) section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in [section 2.2.6](#).
 - Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.
 - Processing MUST end.
- For each attribute name in *iAttributes*, extract the value for that attribute from the result message returned in *TaskOutputResultMessages* and add it to the *rValues* list. If there is no matching value in the results, add an empty entry to the *rValues* list.
- Raise a Shut Down an LDAP Connection event.
- Processing MUST end.

3.1.6.15 Search Using LDAP

This event MUST be generated with the following arguments:

- *iObjectClass*: the **Active Directory schema** class of the objects being searched for.
- *iFilter*: An array of **attribute-filter expressions** ([\[MS-MQDMPR\]](#) section 3.1.7.1.20), but modified so that the valid attributes comprise the set of **Active Directory** attributes associated with an Active Directory schema class instance of type *iObjectClass* instead of ADM element attributes. Each sublist that is returned in *rValues* MUST satisfy all attribute-filter expressions in this array.
- *iAttributes*: a list of attribute names.

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.
- *rValues*: if *rStatus* is **Success**, a list of sublists, where each sublist corresponds to one object found in Active Directory and is composed of values for that object of the attributes listed in *iAttributes*, in the same order. Otherwise, this value is undefined and MUST NOT be used.

The algorithm MUST perform the following actions to process this event:

- If the value of the **CachedConfigurationNamingContext** ADM element is an empty string, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end.
- If the *iAttributes* list does not contain the attribute name "objectGuid", add it to the list. If the *iAttributes* list does not contain the attribute name "distinguishedName", add it to the list.
- Raise a [Prepare an LDAP Connection \(section 3.1.6.18\)](#) event. The event takes no arguments. If the *rStatus* returned is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end. Otherwise, let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE** ([\[MS-DTYP\]](#) section 2.2.2), which is initialized to the value returned in *rADConnection*.
- Let *BaseDN* be a distinguished name composed depending on the class name in *iObjectClass* according to these rules:
 - If *iObjectClass* is "mSMQConfiguration", "mSMQQueue", or "user", *BaseDN* MUST be set to the value of the **CachedConfigurationNamingContext** ADM element.
 - If *iObjectClass* is "mSMQEnterpriseSettings", *BaseDN* MUST be set to a string formed by concatenating "CN=Services," and the value of the **CachedConfigurationNamingContext** ADM element.
 - If *iObjectClass* is "site", *BaseDN* MUST be set to a string formed by concatenating "CN=Sites," and the value of the **CachedConfigurationNamingContext** ADM element.
 - If *iObjectClass* is "mSMQSiteLink", *BaseDN* MUST be set to a string formed by concatenating "CN=MsmqServices,CN=Services," and the value of the **CachedConfigurationNamingContext** ADM element.
- Construct an **LDAPMessage** ([\[RFC2251\]](#) section 4.1):
 - messageID = set as described in [\[RFC2251\]](#) section 4.1.1.1.
 - protocolOp = searchRequest
 - controls = none
 - baseObject = *BaseDN*
 - scope = baseObject

- *derefAliases* = *neverDerefAliases*
- *sizeLimit* = 0
- *timeLimit* = 0
- *typesOnly* = FALSE
- *filter* = "(objectClass=*)"
- *attributes* = a one-element list consisting of the string "objectClass"
- Perform the Performing an LDAP Operation on an ADConnection ([\[MS-ADTS\]](#) section 7.6.1.6) task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section [2.2.6](#).
 - Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.
 - Processing MUST end.
- Construct an **LDAPMessage**:
 - *messageID* = set as described in [RFC2251] section 4.1.1.1.
 - *protocolOp* = *searchRequest*
 - *controls* = none
 - *baseObject* = *BaseDN*
 - *scope* = *wholeSubtree*
 - *derefAliases* = *neverDerefAliases*
 - *sizeLimit* = 0
 - *timeLimit* = 120
 - *typesOnly* = FALSE
 - *filter* = *iFilter*
 - *attributes* = *iAttributes*
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:

- *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
- Raise a Shut Down an LDAP Connection event.
- Processing MUST end.
- *TaskOutputResultMessages* contains a list of attributes for each object in the directory matching the search filter. For each such list returned, perform the following steps:
 - Let *Sublist* be a list of values, initialized to be empty.
 - For each attribute name in *iAttributes*, find the value for that attribute in the results list and add it to *Sublist*. If there is no matching value in the results, add an empty entry to *Sublist*.
 - Add *Sublist* to *rValues*.
- Raise a Shut Down an LDAP Connection event.
- Processing MUST end.

3.1.6.16 Set Object Properties Using LDAP

This event MUST be generated with the following arguments:

- *iPath*: the distinguished name of the object; optional if *iGuid* is supplied
- *iGuid*: the value of the objectGuid attribute of the object; optional if *iPath* is supplied
- *iAttributes*: a list of name-value pairs

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.

The algorithm MUST perform the following actions to process this event:

- If the value of the **CachedConfigurationNamingContext** ADM element is an empty string, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end.
- Raise a [Prepare an LDAP Connection \(section 3.1.6.18\)](#) event. The event takes no arguments. If the *rStatus* returned is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end. Otherwise, let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE** ([MS-DTYP] section 2.2.2), which is initialized to the value returned in *rADConnection*.
- If *iGuid* is supplied, perform the following steps:
 - Raise a [Find Object By GUID Using LDAP \(section 3.1.6.19\)](#) event with the following arguments:
 - *iGuid* = *iGuid*
 - *iADConnection* = *DirectoryServerConnection*
 - If the value of *rStatus* returned by the event is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the value of *rStatus* returned by the event, and processing MUST end.
 - Set *iPath* to the value of *rDN*.

- Construct an **LDAPMessage** ([\[RFC2251\]](#) section 4.1):
 - messageID = set as described in [\[RFC2251\]](#) section 4.1.1.1.
 - protocolOp = searchRequest
 - controls = none
 - baseObject = *iPath*
 - derefAliases = neverDerefAliases
 - sizeLimit = 0
 - timeLimit = 0
 - scope = baseObject
 - typesOnly = FALSE
 - filter = "(objectClass=*)"
 - attributes = a one-element list consisting of the string "objectClass"
- Perform the Performing an LDAP Operation on an ADConnection ([\[MS-ADTS\]](#) section 7.6.1.6) task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [\[RFC2251\]](#) section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section [2.2.6](#).
 - Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.
 - Processing MUST end.
- Construct an **LDAPMessage**:
 - messageID = set as described in [\[RFC2251\]](#) section 4.1.1.1.
 - protocolOp = modifyRequest
 - controls = none
 - baseObject = *iPath*
 - operation = replace
 - modification = *iAttributes*
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step

- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
 - Raise a Shut Down an LDAP Connection event.
 - Processing MUST end.
- Raise a Shut Down an LDAP Connection event.
- Processing MUST end.

3.1.6.17 Set Object Security Using LDAP

This event MUST be generated with the following arguments:

- *iPath*: the distinguished name of the object; optional if *iGuid* is supplied.
- *iGuid*: the value of the objectGuid attribute of the object; optional if *iPath* is supplied.
- *iSecurity*: a **SECURITY_DESCRIPTOR** ([MS-DTYP] section 2.4.6).

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.

The algorithm MUST perform the following actions to process this event:

- If the value of the **CachedConfigurationNamingContext** ADM element is an empty string, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end.
- Raise a [Prepare an LDAP Connection \(section 3.1.6.18\)](#) event. The event takes no arguments. If the *rStatus* returned is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to **DirectoryOperationResult.DirectoryNotConnected**, and processing MUST end. Otherwise, let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE** ([MS-DTYP] section 2.2.2), which is initialized to the value returned in *rADConnection*.
- If *iGuid* is supplied, perform the following steps:
 - Raise a [Find Object By GUID Using LDAP \(section 3.1.6.19\)](#) event with the following arguments:
 - *iGuid* = *iGuid*
 - *iADConnection* = *DirectoryServerConnection*
 - If the value of *rStatus* returned by the event is not **DirectoryOperationResult.Success**, *rStatus* MUST be set to the value of *rStatus* returned by the event, and processing MUST end.
 - Set *iPath* to the value of *rDN*.
- Construct an **LDAPMessage** ([RFC2251] section 4.1):
 - messageID = set as described in [RFC2251] section 4.1.1.1.
 - protocolOp = searchRequest
 - controls = none

- *baseObject* = *iPath*
- *scope* = *baseObject*
- *derefAliases* = *neverDerefAliases*
- *sizeLimit* = 0
- *timeLimit* = 0
- *typesOnly* = FALSE
- *filter* = "(objectClass=*)"
- *attributes* = a one-element list consisting of the string "objectClass"
- Perform the Performing an LDAP Operation on an ADConnection ([\[MS-ADTS\]](#) section 7.6.1.6) task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section [2.2.6](#).
 - Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.
 - Processing MUST end.
- Construct an **LDAPMessage**:
 - *messageID* = set as described in [RFC2251] section 4.1.1.1.
 - *protocolOp* = *searchRequest*
 - *controls* = *none*
 - *baseObject* = *iPath*
 - *scope* = *baseObject*
 - *derefAliases* = *neverDerefAliases*
 - *sizeLimit* = 0
 - *timeLimit* = 0
 - *typesOnly* = FALSE
 - *filter* = "(nTSecurityDescriptor=*)"
 - *attributes* = a one-element list consisting of the string "objectClass"
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step

- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
 - Raise a Shut Down an LDAP Connection event.
 - Processing MUST end.
- Construct an **LDAPMessage**:
 - *messageID* = set as described in [RFC2251] section 4.1.1.1.
 - *protocolOp* = modifyRequest
 - *controls* = none
 - *baseObject* = *iPath*
 - *operation* = replace
 - *modification* = an attribute-value pair consisting of "ntSecurityDescriptor" and *iSecurity*
- Perform the Performing an LDAP Operation on an ADConnection task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
 - Raise a Shut Down an LDAP Connection event.
 - Processing MUST end.
- Raise a Shut Down an LDAP Connection event.
- Processing MUST end.

3.1.6.18 Prepare an LDAP Connection

This event MUST be generated with no arguments.

Return Values:

- *rADConnection*: An **ADCONNECTION_HANDLE** ([MS-DTYP] section 2.2.2) that is connected to an **Active Directory** server and that is ready to perform **LDAP** operations. The value is undefined if *rStatus* is not **DirectoryOperationResult.Success**.
- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation.

The algorithm MUST perform the following actions to process this event:

- If the **CachedLDAPConnection.Handle** (section 3.1.1.6) ADM element attribute is not NULL:
 - Increment the value of the **CachedLDAPConnection.RefCount** ADM element attribute by 1.

- Set *rADConnection* to the value of **CachedLDAPConnection.Handle**, and set *rStatus* to **DirectoryOperationResult.Success**.
- Take no further action.
- Let *DirectoryServerConnection* be a variable of type **ADCONNECTION_HANDLE**.
- Perform the Initializing an ADConnection ([MS-ADTS] section 7.6.1.1) task, specifying the following parameters:
 - *TaskInputTargetName* = NULL
 - *TaskInputPortNumber* = 389
- *DirectoryServerConnection* is set to the *TaskReturnADConnection* result returned by the task.
- Perform the Setting an LDAP Option on an ADConnection ([MS-ADTS] section 7.6.1.2) task, specifying the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputOptionName* = "LDAP_OPT_PROTOCOL_VERSION"
 - *TaskInputOptionValue* = 3
- Perform the Establishing an ADConnection ([MS-ADTS] section 7.6.1.3) task, specifying the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
- If the *TaskReturnStatus* result is not success, as specified in [RFC2251] section 4.1.10, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
- Perform the Performing an LDAP Bind on an ADConnection ([MS-ADTS] section 7.6.1.4) task, specifying the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
- If the *TaskReturnStatus* result is not success, as specified in [RFC2251] section 4.1.10, *rStatus* MUST be set to **DirectoryOperationResult.GenericError**, and processing MUST end.
- *rADConnection* MUST be set to *DirectoryServerConnection*, and *rStatus* MUST be set to **DirectoryOperationResult.Success**.
- Set the **CachedLDAPConnection.Handle** ADM element attribute to the value of *rADConnection*, and set the **CachedLDAPConnection.RefCount** ADM element attribute to 1.

3.1.6.19 Find Object By GUID Using LDAP

This event MUST be generated with the following arguments:

- *iGuid*: the value of the objectGuid attribute of the requested object.
- *iADConnection*: An **ADCONNECTION_HANDLE** ([MS-DTYP] section 2.2.2).

Return Values:

- *rStatus*: A **DirectoryOperationResult** that indicates the result of this directory operation. If this value is not **DirectoryOperationResult.Success**, the **ADConnection** in *iADConnection* is no longer bound and MUST NOT be used to perform further operations, and the value of *rDN* is undefined.

- *rDN*: The **distinguished name** of the object found.

The algorithm MUST perform the following actions to process this event:

- Let *CompactGuid* be a string variable that is initialized by converting the value of *iGuid* to curly braced GUID string form ([MS-DTYP] section 2.3.4.3), then compacting it by removing all characters that are not hexadecimal digits.
- Construct an **LDAPMessage** ([RFC2251] section 4.1):
 - *messageID* = set as described in [RFC2251] section 4.1.1.1.
 - *protocolOp* = *searchRequest*
 - *controls* = none
 - *baseObject* = an empty string
 - *scope* = *wholeSubtree*
 - *derefAliases* = *neverDerefAliases*
 - *sizeLimit* = 0
 - *timeLimit* = 0
 - *typesOnly* = FALSE
 - *filter* = "(objectGuid=" *CompactGuid* ")"
 - *attributes* = a one-element list consisting of the string "distinguishedName"
- Perform the Performing an LDAP Operation on an ADConnection ([MS-ADTS] section 7.6.1.6) task with the following parameters:
 - *TaskInputADConnection* = *DirectoryServerConnection*
 - *TaskInputRequestMessage* = the **LDAPMessage** constructed in the preceding step
- If the value of *TaskReturnStatus* is not success, as defined in [RFC2251] section 4.1.10, the algorithm MUST perform the following steps:
 - *rStatus* MUST be set to a **DirectoryOperationResult** enumeration value according to the conversion rules for *TaskReturnStatus* specified in section 2.2.6.
 - Raise a [Shut Down an LDAP Connection \(section 3.1.6.21\)](#) event.
 - Processing MUST end.
- Extract the value of the distinguishedName attribute from the result message returned in *TaskOutputResultMessages* and set *rDN* to that value.
- Set *rStatus* to **DirectoryOperationResult.Success**.
- Processing MUST end.

3.1.6.20 Data Element Directory Attribute Tables

3.1.6.20.1 QueueManager

The **QueueManager** ADM element is specified in [\[MS-MQDMPR\]](#) section 3.1.1.1. The following table lists only those **QueueManager** ADM element attributes that are stored in the directory.

QueueManager ADM element attributes MUST be stored as attributes of an mSMQConfiguration ([\[MS-ADSC\]](#) section 2.163) object, except for those noted in the following table, which MUST be stored on the computer ([MS-ADSC] section 2.21) object that is the parent of the mSMQConfiguration object in the directory. In addition, any **QueueManager** ADM element instance for which one or more of the **RoutingServer**, **DirectoryServer**, or **SupportingServer** ADM attributes is TRUE MUST have an mSMQSettings ([MS-ADSC] section 2.167) object in the directory, the attributes of which MUST have the same values as certain attributes of the mSMQConfiguration object. The **distinguished names** for these three objects (mSMQConfiguration, computer, and mSMQSettings) are specified in section [2.2.1](#).

QueueManager ADM element attribute	Stored on computer object?
Identifier	no
ComputerName	no
QueueManagerVersion	no
OperatingSystemType	no
QualifiedComputerName	yes
CreateTime	no
ModifyTime	no
DirectoryServerType	no
RemoteAccessServer	no
QueueManagerQuota	no
JournalQuota	no
ForeignSystem	no
FullPath	no
SiteIdentifierList	no
ConnectedNetworkIdentifierList	This ADM element attribute is not supported in Active Directory -based environments. Any attempt to operate on it results in DirectoryOperationResult.GenericError .
OutRoutingServerIdentifierList	no
InRoutingServerIdentifierList	no
RoutingServer	no
DirectoryServer	no
SupportingServer	no
PublicEncryptionKeyList	no
PublicSigningKeyList	yes
Security	no

QueueManager ADM element attribute	Stored on computer object?
Clustered	yes
OperatingSystemVersion	yes

3.1.6.20.2 Queue

The **Queue** ADM element is specified in [\[MS-MQDMPR\]](#) section 3.1.1.2. The following table lists only those **Queue** ADM element attributes that are stored in the directory.

Queue ADM elements MUST be stored in the directory if the value of the **QueueType** ADM attribute is **Public**. **Queue** ADM elements MUST NOT be stored in the directory if the value of the **QueueType** ADM attribute is not **Public**.

Queue ADM element attribute
Identifier
Label
CreateTime
ModifyTime
Type
Pathname
QualifiedPathname
Journaling
Quota
JournalQuota
Authentication
PrivacyLevel
Transactional
MulticastAddress
Security
BasePriority
FullPath
DirectoryPath

3.1.6.20.3 Enterprise

The **Enterprise** ADM element is specified in [\[MS-MQDMPR\]](#) section 3.1.1.6. The following table lists only those **Enterprise** ADM element attributes that are stored in the directory.

Enterprise ADM element attribute
Identifier
Name
WeakenedSecurity
NonLDAPCapableQueueManagerNotification
DefaultTimeToLive
OldDirectory
Security

3.1.6.20.4 Site

The **Site** ADM element is specified in [\[MS-MQDMPR\]](#) section 3.1.1.7. The following table lists only those **Site** ADM element attributes that are stored in the directory.

Site ADM element attribute	Notes
Identifier	
Name	
PrimarySiteController	This ADM element attribute is not supported in Active Directory -based environments. Any attempt to operate on it results in DirectoryOperationResult.GenericError .
IntraSiteReplicationInterval	
InterSiteReplicationInterval	
FullPath	
ForeignSite	
PublicSigningKeyList	This ADM element attribute is not supported in Active Directory-based environments. Any attempt to operate on it results in DirectoryOperationResult.GenericError .
MigratedFromMsmq10	
Security	

3.1.6.20.5 RoutingLink

The **RoutingLink** ADM element is specified in [\[MS-MQDMPR\]](#) section 3.1.1.8. The following table lists only those **RoutingLink** ADM attributes that are stored in the directory.

RoutingLink ADM element attribute
Identifier
Description

RoutingLink ADM element attribute
FullPath
ActualCost
Site1Identifier
Site2Identifier
SiteGateIdentifierList

3.1.6.20.6 User

The **User** ADM element is specified in [\[MS-MQDMPR\]](#) section 3.1.1.15 and extended in [User Data Element \(section 3.1.1.4\)](#). The following table lists only those **User** ADM element attributes that are stored in the directory.

User ADM element attribute
Identifier
SecurityIdentifier
CertificateDigestList
Certificates
FullPath

3.1.6.21 Shut Down an LDAP Connection

This event MUST be generated with no arguments.

Return Values:

- None.

The algorithm MUST perform the following actions to process this event:

- Decrement the value of the **CachedLDAPConnection.RefCount** (section [3.1.1.6](#)) ADM element attribute by 1.
- If the value of the **CachedLDAPConnection.RefCount** ADM element attribute is greater than zero, take no further action.
- Let *DirectoryServerConnection* be an **ADCONNECTION_HANDLE** ([\[MS-DTYP\]](#) section 2.2.2) that is initialized to the value of the **CachedLDAPConnection.Handle** ADM element attribute.
- Set the **CachedLDAPConnection.Handle** ADM element attribute to NULL.
- Perform the Performing an LDAP Unbind on an ADConnection ([\[MS-ADTS\]](#) section 7.6.1.5) task with the following parameter:
 - *TaskInputADConnection* = *DirectoryServerConnection*

4 Algorithm Examples

None.

5 Security

5.1 Security Considerations for Implementers

Active Directory requires authentication to access the directory via **LDAP**. This authentication is performed via SASL, using the GSS-SPNEGO protocol as described in [\[MS-ADTS\] section 5.1.1](#).

Active Directory performs authorization on each access to each object in the directory, as described in [\[MS-ADTS\] section 5.1.3](#). The **discretionary access control list (DACL)**, if any, found in the nTSecurityDescriptor attribute ([\[MS-ADA3\] section 2.37](#)) of the object is used in this process. This algorithm controls that DACL only for the mSMQConfiguration ([\[MS-ADSC\] section 2.163](#)), mSMQQueue ([\[MS-ADSC\] section 2.166](#)), site ([\[MS-ADSC\] section 2.258](#)), and mSMQEnterpriseSettings ([\[MS-ADSC\] section 2.164](#)) objects; for all of the other Active Directory objects listed in [section 2.2.1](#), the defaults supplied by Active Directory are used.

The following sections describe, for each combination of object and operation, the requested access mask that is compared to the granted rights in the DACL evaluation process used by Active Directory and described in [\[MS-ADTS\] section 5.1.3.3.2](#). If the required access is not granted, Active Directory returns a failure.

5.1.1 QueueManager

The directory representation of a **QueueManager** ([\[MS-MQDMPR\] section 3.1.1](#)) ADM element instance can be created, deleted, read, and modified, as described in sections [3.1.6.1.1](#), [3.1.6.2.1](#), [3.1.6.3.2](#), [3.1.6.4.1](#), and [3.1.6.7.1](#). These operations always act on an mSMQConfiguration ([\[MS-ADSC\] section 2.163](#)) **Active Directory** object and can also act on computer ([\[MS-ADSC\] section 2.21](#)) and mSMQSettings ([\[MS-ADSC\] section 2.167](#)) objects. The default security is discussed in [section 3.1.6.1.1.8](#).

To create the directory representation of a **QueueManager** ADM element instance requires at least **RIGHT_DS_CREATE_CHILD** ([\[MS-ADTS\] section 5.1.3.2](#)) access on the parent computer object. If the **PublicSigningKeyList** ADM attribute is present, as described in [section 3.1.6.1.1.3](#), **RIGHT_GENERIC_WRITE** ([\[MS-ADTS\] section 5.1.3.2](#)) access is also required on the parent object. If one or more associated mSMQSettings objects are required to be created, as described in [section 3.1.6.1.1.3](#), **RIGHT_DS_CREATE_CHILD** access is required on the parent objects of the mSMQSettings objects, which are of class server ([\[MS-ADSC\] 2.250](#)).

To delete the directory representation of a **QueueManager** ADM element instance requires **RIGHT_DS_CREATE_CHILD** access on the parent object and **RIGHT_DELETE** ([\[MS-ADTS\] section 5.1.3.2](#)) access on the mSMQConfiguration object itself. If one or more associated mSMQSettings objects exist, as described in [section 3.1.6.2.1.3](#), **RIGHT_DS_DELETE_CHILD** ([\[MS-ADTS\] section 5.1.3.2](#)) access is required on the parent objects of the mSMQSettings objects, and **RIGHT_DELETE** access on the mSMQSettings objects themselves.

To read the attributes of a **QueueManager** ADM element instance requires **RIGHT_GENERIC_READ** ([\[MS-ADTS\] section 5.1.3.2](#)) access on the object. If any of the **QualifiedComputerName**, **PublicSigningKeyList**, **OperationSystemVersion**, or **Clustered** ADM attributes is specified, **RIGHT_GENERIC_READ** access on the parent object is also required.

To modify the directory representation of a **QueueManager** ADM element instance requires **RIGHT_GENERIC_WRITE** access on the object. If the **PublicSigningKeyList** ADM attribute is to be modified, **RIGHT_GENERIC_WRITE** access on the parent object is also required. As a result of this modification, one or more associated mSMQSettings objects can also be created, modified, or deleted, as described in [section 3.1.6.7.1.9](#). To create mSMQSettings objects, **RIGHT_DS_CREATE_CHILD** access is required on the parent objects of the mSMQSettings objects. To modify, **RIGHT_GENERIC_WRITE** access is required on the mSMQSettings objects. To delete,

RIGHT_DS_DELETE_CHILD access is required on the parent objects of the mSMQSettings objects, and RIGHT_DELETE access on the mSMQSettings objects themselves.

5.1.2 Queue

The directory representation of a **Queue** ([MS-MQDMPR] section 3.1.1.2) ADM element instance can be created, deleted, read, and modified, as described in sections [3.1.6.1.2](#), [3.1.6.2.2](#), [3.1.6.3.3](#), [3.1.6.4.2](#), and [3.1.6.7.2](#). These operations act on an mSMQQueue ([MS-ADSC] section 2.166) **Active Directory** object. The default security is discussed in [MS-MQDMPR] section 3.1.7.1.3.1.

To create the directory representation of a **Queue** ADM element instance requires RIGHT_DS_CREATE_CHILD ([MS-ADTS] section 5.1.3.2) access on the parent mSMQConfiguration ([MS-ADSC] section 2.163) object.

To delete the directory representation of a **Queue** ADM element instance requires RIGHT_DS_DELETE_CHILD ([MS-ADTS] section 5.1.3.2) access on the parent object and RIGHT_DELETE ([MS-ADTS] section 5.1.3.2) access on the mSMQQueue object itself.

To read the attributes of a **Queue** ADM element instance requires RIGHT_GENERIC_READ ([MS-ADTS] section 5.1.3.2) access on the object.

To modify the directory representation of a **Queue** ADM element instance requires RIGHT_GENERIC_WRITE ([MS-ADTS] section 5.1.3.2) access on the object.

5.1.3 Enterprise

The directory representation of an **Enterprise** ([MS-MQDMPR] section 3.1.1.6) ADM element instance can be read and modified, as described in sections [3.1.6.3.4](#), [3.1.6.4.3](#), and [3.1.6.7.3](#). These operations act on an mSMQEnterpriseSettings ([MS-ADSC] section 2.164) **Active Directory** object.

To read the attributes of an **Enterprise** ADM element instance requires RIGHT_GENERIC_READ ([MS-ADTS] section 5.1.3.2) access on the object.

To modify the directory representation of an **Enterprise** ADM element instance requires RIGHT_GENERIC_WRITE ([MS-ADTS] section 5.1.3.2) access on the object.

5.1.4 Site

The directory representation of a **Site** ([MS-MQDMPR] section 3.1.1.7) ADM element instance can be created, deleted, read, and modified, as described in sections [3.1.6.1.3](#), [3.1.6.2.3](#), [3.1.6.3.5](#), [3.1.6.4.4](#), and [3.1.6.7.4](#). These operations act on a site ([MS-ADSC] section 2.258) **Active Directory** object.

To create the directory representation of a **Site** ADM element instance requires RIGHT_DS_CREATE_CHILD ([MS-ADTS] section 5.1.3.2) access on the parent object, which is of class sitesContainer ([MS-ADSC] section 2.261) and has a fixed distinguished name of "CN=Sites,CN=Configuration,<root>".

To delete the directory representation of a **Site** ADM element instance requires RIGHT_DS_DELETE_CHILD ([MS-ADTS] section 5.1.3.2) access on the parent object and RIGHT_DELETE ([MS-ADTS] section 5.1.3.2) access on the site object itself.

To read the attributes of a **Site** ADM element instance requires RIGHT_GENERIC_READ ([MS-ADTS] section 5.1.3.2) access on the object.

To modify the directory representation of a **Site** ADM element instance requires RIGHT_GENERIC_WRITE ([MS-ADTS] section 5.1.3.2) access on the object.

5.1.5 RoutingLink

The directory representation of a **RoutingLink** ([\[MS-MQDMPR\]](#) section 3.1.1.8) ADM element instance can be created, deleted, read, and modified, as described in sections [3.1.6.1.4](#), [3.1.6.2.4](#), [3.1.6.3.6](#), [3.1.6.4.5](#), and [3.1.6.7.5](#). These operations act on an mSMQSiteLink ([\[MS-ADSC\]](#) section 2.168) Active Directory object.

To create the directory representation of a **RoutingLink** ADM element instance requires RIGHT_DS_CREATE_CHILD ([\[MS-ADTS\]](#) section 5.1.3.2) access on the parent mSMQEnterpriseSettings ([\[MS-ADSC\]](#) section 2.164) object.

To delete the directory representation of a **RoutingLink** ADM element instance requires RIGHT_DS_DELETE_CHILD ([\[MS-ADTS\]](#) section 5.1.3.2) access on the mSMQEnterpriseSettings object and RIGHT_DELETE ([\[MS-ADTS\]](#) section 5.1.3.2) access on the mSMQSiteLink object itself.

To read the attributes of a **RoutingLink** ADM element instance requires RIGHT_GENERIC_READ ([\[MS-ADTS\]](#) section 5.1.3.2) access on the object.

To modify the directory representation of a **RoutingLink** ADM element instance requires RIGHT_GENERIC_WRITE ([\[MS-ADTS\]](#) section 5.1.3.2) access on the object.

5.1.6 User

The directory representation of a **User** ([\[MS-MQDMPR\]](#) section 3.1.1.15) ADM element instance can be read and modified, as described in sections [3.1.6.3.7](#), [3.1.6.4.6](#), and [3.1.6.7.6](#). These operations act on a user ([\[MS-ADSC\]](#) section 2.269) **Active Directory** object.

To read the attributes of a **User** ADM element instance requires RIGHT_GENERIC_READ ([\[MS-ADTS\]](#) section 5.1.3.2) access on the object.

To modify the directory representation of a **User** ADM element instance requires RIGHT_GENERIC_WRITE ([\[MS-ADTS\]](#) section 5.1.3.2) access on the object.

5.1.7 Queue Alias

A [queue alias \(section 2.3\)](#) can be read, as described in section [3.1.6.8](#). This operation acts on an mSMQ-Custom-Recipient Active Directory object and requires RIGHT_GENERIC_READ ([\[MS-ADTS\]](#) section 5.1.3.2) access on the object.

5.1.8 Distribution List

A distribution list can be read, as described in section [3.1.6.9](#). This operation acts on a group ([\[MS-ADSC\]](#) section 2.55) object and requires RIGHT_GENERIC_READ ([\[MS-ADTS\]](#) section 5.1.3.2) access on the object.

5.2 Index of Security Parameters

None.

6 Appendix A: Product Behavior

The information in this specification is applicable to the following Microsoft products or supplemental software. References to product versions include updates to those products.

- Windows 2000 operating system
- Windows XP operating system
- Windows Server 2003 operating system
- Windows Server 2003 R2 operating system
- Windows Vista operating system
- Windows Server 2008 operating system
- Windows 7 operating system
- Windows Server 2008 R2 operating system
- Windows 8 operating system
- Windows Server 2012 operating system
- Windows 8.1 operating system
- Windows Server 2012 R2 operating system
- Windows 10 operating system
- Windows Server 2016 operating system
- Windows Server operating system
- Windows Server 2019 operating system
- Windows Server 2022 operating system
- Windows 11 operating system
- Windows Server 2025 operating system

Exceptions, if any, are noted in this section. If an update version, service pack or Knowledge Base (KB) number appears with a product name, the behavior changed in that update. The new behavior also applies to subsequent updates unless otherwise specified. If a product edition appears with the product version, behavior is different in that product edition.

Unless otherwise specified, any statement of optional behavior in this specification that is prescribed using the terms "SHOULD" or "SHOULD NOT" implies product behavior in accordance with the SHOULD or SHOULD NOT prescription. Unless otherwise specified, the term "MAY" implies that the product does not follow the prescription.

[<1> Section 2.2.2](#): The directory attributes mSMQDigestsMig ([\[MS-ADA2\]](#) section 2.554), mSMQSignCertificatesMig ([\[MS-ADA2\]](#) section 2.588), mSMQSiteGatesMig ([\[MS-ADA2\]](#) section 2.594), and mSMQMigrated ([\[MS-ADA2\]](#) section 2.568) are used only on Windows 2000.

[<2> Section 2.2.2](#): The directory attributes mSMQDigestsMig ([\[MS-ADA2\]](#) section 2.554), mSMQSignCertificatesMig ([\[MS-ADA2\]](#) section 2.588), mSMQSiteGatesMig ([\[MS-ADA2\]](#) section 2.594), and mSMQMigrated ([\[MS-ADA2\]](#) section 2.568) are used only on Windows 2000.

<3> [Section 2.2.2](#): The directory attributes mSMQDigestsMig ([MS-ADA2] section 2.554), mSMQSignCertificatesMig ([MS-ADA2] section 2.588), mSMQSiteGatesMig ([MS-ADA2] section 2.594), and mSMQMigrated ([MS-ADA2] section 2.568) are used only on Windows 2000.

<4> [Section 2.2.2](#): The directory attributes mSMQDigestsMig ([MS-ADA2] section 2.554), mSMQSignCertificatesMig ([MS-ADA2] section 2.588), mSMQSiteGatesMig ([MS-ADA2] section 2.594), and mSMQMigrated ([MS-ADA2] section 2.568) are used only on Windows 2000.

<5> [Section 2.2.2](#): The directory attributes mSMQDigestsMig ([MS-ADA2] section 2.554), mSMQSignCertificatesMig ([MS-ADA2] section 2.588), mSMQSiteGatesMig ([MS-ADA2] section 2.594), and mSMQMigrated ([MS-ADA2] section 2.568) are used only on Windows 2000.

<6> [Section 2.2.2](#): The directory attributes mSMQDigestsMig ([MS-ADA2] section 2.554), mSMQSignCertificatesMig ([MS-ADA2] section 2.588), mSMQSiteGatesMig ([MS-ADA2] section 2.594), and mSMQMigrated ([MS-ADA2] section 2.568) are used only on Windows 2000.

<7> [Section 2.2.3](#): For Windows 2000, Windows Server 2003, Windows Server 2008, and Windows Server 2008 R2, an mSMQEnterpriseSettings object is created as part of the creation of each new rootDomainNamingContext.

<8> [Section 2.3](#): Windows 2000 and Windows XP do not use the queue alias object. All other Windows implementations of this protocol use the Windows API function **MQADsPathToFormatName** to translate a directory service path of a queue alias into its associated format name.

<9> [Section 3.1.6.1.1.9](#): The directory attributes mSMQSignCertificatesMig ([MS-ADA2] section 2.588) and mSMQDigestsMig ([MS-ADA2] section 2.555) are set only on Windows 2000.

<10> [Section 3.1.6.11.1](#): Default values are used by all Windows implementations except for Windows 2000. However, the default value of the **QueueManagerQuota** ADM element for Windows XP is 0xFFFFFFFF.

<11> [Section 3.1.6.11.1](#): All Windows implementations except for Windows 2000 use an empty **Unicode** string as the default value.

<12> [Section 3.1.6.11.1](#): The default value used by all Windows implementations except for Windows 2000 is an empty list.

<13> [Section 3.1.6.11.3](#): This is a table of the default values used by all Windows implementations except for Windows 2000.

Enterprise ADM element attribute	Default value
WeakenedSecurity	See section 3.1.6.11.3.2 to convert a mSMQNameStyle value of 0x02.
NonLDAPCapableQueueManagerNotification	"Y"
DefaultTimeToLive	7776000 (Windows XP) 345600 (All Windows implementations except for Windows 2000 and Windows XP.)
OldDirectory	TRUE

<14> [Section 3.1.6.11.3.2](#): All Windows implementations except for Windows 2000 use an internal default that is equivalent to an mSMQNameStyle value of 0x00.

<15> [Section 3.1.6.11.4](#): This is a table of the default values used by all Windows implementations except for Windows 2000.

Site ADM element attribute	Default value
IntraSiteReplicationInterval	2
InterSiteReplicationInterval	10
Foreign	FALSE

<16> [Section 3.1.6.11.5](#): This is a table of the default values used by all Windows implementations except for Windows 2000.

mSMQSiteLink ADM element attribute	Default value
Description	empty string
SiteGateIdentifierList	empty list

<17> [Section 3.1.6.11.6](#): This is a table of the default values used by all Windows implementations except for Windows 2000.

User ADM element attribute	Default value
Certificates	empty
CertificateDigestList	empty list

7 Change Tracking

This section identifies changes that were made to this document since the last release. Changes are classified as Major, Minor, or None.

The revision class **Major** means that the technical content in the document was significantly revised. Major changes affect protocol interoperability or implementation. Examples of major changes are:

- A document revision that incorporates changes to interoperability requirements.
- A document revision that captures changes to protocol functionality.

The revision class **Minor** means that the meaning of the technical content was clarified. Minor changes do not affect protocol interoperability or implementation. Examples of minor changes are updates to clarify ambiguity at the sentence, paragraph, or table level.

The revision class **None** means that no new technical changes were introduced. Minor editorial and formatting changes may have been made, but the relevant technical content is identical to the last released version.

The changes made to this document are listed in the following table. For more information, please contact dochelp@microsoft.com.

Section	Description	Revision class
6 Appendix A: Product Behavior	Added Windows Server 2025 to the list of applicable products.	Major

8 Index

A

Abstract data model

data elements

[CachedConfigurationNamingContext](#) 29

[CachedLDAPConnection](#) 29

[externally defined](#) 28

[ReadDirectoryIteratorState](#) 28

[ReadDirectoryIteratorStateCollection](#) 29

[user](#) 29

[overview](#) 28

[Algorithm - overview](#) 28

[Applicability](#) 12

[Attributes of directory objects](#) 14

C

[Capability negotiation](#) 12

[Change tracking](#) 134

[Common data types](#) 13

directory objects

[attributes](#) 14

[LDAP distinguished names](#) 13

[hash string calculation from queue name](#) 16

[mSMQEnterpriseSettings object](#) 16

[unused active directory properties](#) 16

D

Data model - abstract

data elements

[CachedConfigurationNamingContext](#) 29

[CachedLDAPConnection](#) 29

[externally defined](#) 28

[ReadDirectoryIteratorState](#) 28

[ReadDirectoryIteratorStateCollection](#) 29

[user](#) 29

[overview](#) 28

Data types

[common - overview](#) 13

Data types - common

directory objects

[attributes](#) 14

[LDAP distinguished names](#) 13

[hash string calculation from queue name](#) 16

[mSMQEnterpriseSettings object](#) 16

[unused active directory properties](#) 16

Directory

objects

[attributes](#) 14

[LDAP distinguished names](#) 13

[properties - unused - active](#) 16

[service schema elements](#) 25

[Directory service schema elements](#) 25

E

[Elements - directory service schema](#) 25

[Examples](#) 127

F

[Fields - vendor-extensible](#) 12

G

[Glossary](#) 9

H

[Hash string calculation from queue name](#) 16

I

[Implementer - security considerations](#) 128

[Index of security parameters](#) 130

[Informative references](#) 11

[Initialization](#) 29

[Introduction](#) 9

L

[LDAP distinguished names of directory objects](#) 13

Local events

create

[ADM element From LDAP values](#) 93

[directory object](#) 31

[LDAP attribute list](#) 90

[object using LDAP](#) 107

delete

[directory object](#) 45

[object using LDAP](#) 109

[find object by GUID using LDAP](#) 121

[get object properties using LDAP](#) 112

LDAP connection

[prepare](#) 120

[shut down](#) 126

read

[directory](#) 49

[directory begin](#) 56

[directory end](#) 70

[directory next](#) 69

resolve

[distribution list](#) 88

[queue alias](#) 88

[search using LDAP](#) 113

set object

[properties using LDAP](#) 116

[security using LDAP](#) 118

[write directory](#) 70

M

[Message processing](#) 30

Messages

[common data types](#) 13

directory objects

[attributes](#) 14

[LDAP distinguished names](#) 13

[hash string calculation from queue name](#) 16

[mSMQEnterpriseSettings object](#) 16

[unused active directory properties](#) 16

[queue alias](#) 25
[transport](#) 13
[mSMQEnterpriseSettings object](#) 16

N

[Normative references](#) 10

O

[Objects - mSMQEnterpriseSettings](#) 16
[Overview \(synopsis\)](#) 11

P

[Parameters - security index](#) 130
[Preconditions](#) 11
[Prerequisites](#) 11
[Product behavior](#) 131

Q

Queue
[alias](#) 25
[name - hash string calculation](#) 16

R

[References](#) 10
[informative](#) 11
[normative](#) 10
[Relationship to other protocols](#) 11

S

[Schema elements - directory service](#) 25
Security
[implementer considerations](#) 128
[parameter index](#) 130
[Sequencing rules](#) 30
[Standards assignments](#) 12

T

[Timer events](#) 30
[Timers](#) 29
[Tracking changes](#) 134
[Transport](#) 13

U

[Unused active directory properties](#) 16

V

[Vendor-extensible fields](#) 12
[Versioning](#) 12